Vector Bundles and K-Theory

— *EYNTKA* Series —

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Contents

Part I

Topological K-Theory

1

Vector Bundles

1.0.1 Topological Convention We are interested in topological space with "geometric" properties. Hence, we shall limit ourselves to paracompact topological space. If we are working with manifolds, they shall be Hausdorff. When working with schemes, we shall not assume Hausdorffness (for obvious reasons[a])
Many of the result covered apply to paracompact topological spaces (every open cover has a locally finite refinement). We require the paracompact condition so that many of the constructions we shall want to have will work, namely we have good behaviour of sheaves and sufficient conditions for homotopy theory and differential topology.

[a] see [1] why only trivial schemes are Hausdorff

1.1 Fiber Bundles, Principal Bundles, Vector Bundles

Definition 1.1.1: Fiber Bundle

Let B be a paracompact topological space. Then a quadruple (B, E, π, F) where E is a topological space called the *total space*, $\pi : E \to B$ is a continuous surjection called the *bundle projection*, and B is called the *base space*, and F is the fiber is called *fiber bundle* if:

1. (regular fibers) each $\pi^{-1}(b)$ for $b \in B$ is isomorphic to F, or in particular
$$\pi^{-1}(b) \cong b \times F$$
2. (local trivialization)For each $b \in B$, there exists an open set $U \subseteq B$ containing b such that the following diagram commutes
$$\begin{tikzcd} \pi^{-1}(U) \arrow[r, "\sim"] \arrow[d, "\pi"'] & U \times F \arrow[dl, "\pi_1"] \\ U & \end{tikzcd}$$

When clear from context, the fiber bundle is usually denoted as E.

Definition 1.1.2: Vector Bundle

If $F = V$ is a vector space and each fiber-morphism in the local trivialization is a linear isomorphism, then is called a *vector bundle* (over B). If the vector space is real or complex, the vector bundle can be called a *real vector bundle complex vector bundle*.

Definition 1.1.3: Principal Bundle

If $F = G$ is a topological and $P \to X$ is a fiber bundle over X group where there is a free and transitive continuous right group action $P \times G \to P$ that preserves the fibers of P, then it is called a *principal G-bundle* (or *principal bundle* if G is clear from context.

Each fiber is called a *G-torsor* (which is a space that where the action $G \curvearrowright F_x$ has trivial stabilizer, hence "is" the group G but does not intrinsically have a notion of identity element)

Note that these conditions make each fiber homeomorphic to G. As a consequence the Orbit space P/G is homeomorphic to X.

Example 1.1.4: Fiber Bundle

1. Given a topological B and fiber F, then $E = B \times F$ is the *trivial F-fiber bundle* or simply

the *trivial bundle*. Such a bundle has a *global trivialization*, namely by letting $U = B$:

$$\begin{tikzcd} B \times F \arrow[r, "\mathrm{id}"] \arrow[d, "\pi"'] & B \times F \arrow[dl, "\pi_1"] \\ B & \end{tikzcd}$$

2. If M is a n-manifold, then there are various interesting vector bundles defined on it:
   (a) Tangent bundle TM
   (b) cotangent bundle T^*M
   (c) differential forms $\Omega^k(M)$ for $k \leq n$

   If M has more structure, ex. if is a Riemann manifold, a Khäler manifold, has a complex manifold, etc, then we can define more vector bundles that capture the extra structure (ex. Normal bundles, holomorphic bundles, spin bundles, flat bundles, etc.)

3. Let $p : E \to B$ be a fiber bundles and $A \subseteq B$. Then $p : p^{-1}(A) \to A$ is certainly a fiber bundle. This is called the *restriction of E over A*. The same goes through for vector-bundles.

4. Sub-bundle shave compliments: if $E \to X$ is a bundle (on a paracompact topological space) and F is a subbundle, then there exists a bundle $F^\perp$ such that $E = F \oplus F^\perp$.

5. the Tensor product is well-defined: Given a vector bundle $E \to X$ and $F \to X$, there exits a vector-bundle $E \otimes F \to X$ whose fibers over x are $E_x \otimes F_x$. From this, we get our usual more general bundles such as the *determinant bundle* $\det(E) = \Lambda^n E$ (given E is n-dimensional).

6. Let $E = I \times \mathbb{R}/\sim$ where $(0, t \sim (1, -t)$. Then the projection map $I \times \mathbb{R} \to I$ produces a quotient $p : E \to S^1$ known as the Möbius bundle

7. Let TS^n be the tangent bundle of S^n. Then notice that this can be taken as a sub-bundle of $T\mathbb{R}^{n+1}$.

8. Another important bundle on S^n is the normal bundle which is a line bundle that that is where we take the one dimensional vector space panned by the line perpendicular to each point $p \in S^n$. We shall see that this bundle is isomorphic to the trivial bundle later. If NS^n is the normal bundle, we shall see that $TS^n \oplus NS^n$ is a trivial bundle.

9. Let $\mathbb{R}\mathrm{P}^n$ and $\mathbb{C}\mathrm{P}^n$ be real and complex projective space respectively. Then there is a *canonical line bundle* $p : E \to \mathbb{R}\mathrm{P}^n$ and $p : E \to \mathbb{C}\mathrm{P}^n$ where E is the subspace of $\mathbb{R}\mathrm{P}^n \times \mathbb{R}^{n+1}$ consisting of pairs (ℓ, v) where $v \in \ell)$ and $p(\ell, v)\ell$. Note that the canonical line bundle over $\mathbb{R}\mathrm{P}^n$ has an orthogonal compliment

$$E^\perp = \left\{ (\ell, v) \in \mathbb{R}\mathrm{P}^n \times \mathbb{R}^{n+1} \;\middle|\; v \perp \ell \right\}$$

10. An important bundle for the future is the infinite real/complex projective space $\mathbb{R}\mathrm{P}^\infty$ and $\mathbb{C}\mathrm{P}^\infty$ taking by $\varinjlim_n \mathbb{R}\mathrm{P}^n$ and $\varinjlim_n \mathbb{C}\mathrm{P}^n$.

11. The Grassmann manifold $G_k(\mathbb{R}^n)$ was shown in [14] to be the space of all k-dimensional planes through the origin of $\mathbb{R}^n$.

Definition 1.1.5: Morphism of Fiber Bundles

If E_1, E_2 are two vector bundles, $\varphi : E_1 \to E_2$ is a vector-bundle morphism is a continuous map where this diagram commutes:

$$\begin{tikzcd} E_1 \arrow[rr, "\varphi"] \arrow[dr] & & E_2 \arrow[dl] \\ & B & \end{tikzcd}$$

and it respects local trivialization; this includes a linear homomorphism on each fiber. An isomorphism would have φ be a homeomorphism and is a linear isomorphism on each fiber.

For each dimension n, let $\mathrm{Vect}^n_k(B)$ be the isomorphism classes of vector bundles over B (with field k).

For a topological group G, let $b_G(B)$ be the isomorphism classes of principal G-bundles over B.

Often, if $k = \mathbb{R}$, we shall just write $\mathrm{Vect}^n(B)$. Note that $\mathrm{Vect}^n_k(B)$ and $b_G(B)$ are contravariant functors by applying the pullback map, for example:

$$f : X \to Y \qquad \Rightarrow \qquad f^* : b_G(Y) \to b_G(X)$$

Lemma 1.1.6: Morphism is Fiber-local

Let $p_1 : E_1 \to B, p_2 : E_2 \to B$ be two vector bundles over B and $\varphi : E_1 \to E_2$ a continuous. Then φ is an isomorphism if and only if it takes each fiber $p_1^{-1}(b)$ to the corresponding fiber $p_2^{-1}(b)$ via a linear isomorphism.

***Proof*:**
exercise.

Example 1.1.7: Morphism of Bundles

We can see that the normal bundle NS^n as a subspace of $\mathbb{R}^1$ is isomorphic to $S^n \times \mathbb{R}$ via the map:

$$(x, tx) \to (x, t)$$

We shall later see that only S^1, S^3, S^7 have trivial line bundles (this shall correspond to the existence of division algebra's over $\mathbb{R}$ existing only for dimension $2, 4, 8$, see ref:HERE). For S^1, the isomorphism is simple enough as we are comfortable with complex analysis:

$$\varphi : TS^1 \to S^1 \times \mathbb{R} \qquad (e^{i\theta}, ite^{i\theta}) \mapsto (e^{i\theta}, t)$$

Another important example is that the Möbius bundle is isomorphic to the canonical line bundle over $\mathbb{RP}^1$ (note that $\mathbb{RP}^1 \cong S^1$).

Definition 1.1.8: Section

Let $p : E \to B$ be a fiber bundle. Then a *section* is a continuous map $s : B \to E$ such that $p \circ s = \text{id}$.

We shall focus on sections of vector bundles. We immediately get the following

Definition 1.1.9: Zero Section

Let $p : E \to B$ be a vector bundle. Then the *zero section* is the section $z : B \to E$ given pointwise by $z(p) = 0 \in p^{-1}(b)$.

As each vector-bundle morphism on fibers is linear, the zero section is taken to the zero section. This can be taken advantage of: for example the Möbius bundle is not isomorphic to $S^1 \times \mathbb{R}$ since the compliment of the zero section is connected in the Möbius bundle but not for the product bundle. The other type of section is a *non-vanishing section* where no elements of the section are zero. Famously, by the Hairy-ball theorem (or Hedgehog theorem), there does not exist a section of TS^2 (i.e a vector-field) that is everywhere nonzero (see [9]). We shall see in ref:HERE that S^n has a nonvanishing vector field if and only if n is odd, and so we shall see that the tangent bundle of S^n cannot be isomorphic to the trivial bundle if $n > 0$ is even. The existence of zero's shall be really important, for example it shall be key in how to interpret the Euler characteristic (see ref:HERE).

Lemma 1.1.10: Characterizing Trivial Bundles using Sections

Let $p : E \to B$ be an n-dimensional vector bundle. Then p is trivial if and only if it has n sections $s - 1, ..., s_n$ such that $(s_1(b), ..., s_n(b)$ are linearly independent on each fiber $p^{-1}(b)$.

***Proof*:**
A trivial n-dimensional vector bundle certainly respects these properties. Conversely, given such n section $s_1, ..., s_n$, define:

$$h : B \times \mathbb{R}^n \to E \qquad h(b, t_1, ..., t_n) \to \sum_i t_i s_i(b)$$

this is certainly a linear isomorphism on each fiber, hence by lemma 1.1.6 it is an isomorphism, completing the proof.

Note that another way of characterizing a trivial real bundle is to say there is a projection map $E \to \mathbb{R}^n$ that is a linear isomorphism on each fiber.

Using this lemma, we can show that S^3 has a trivial line bundle by relying on the Quaternions. Recall that any quaternion $z \in \mathbb{H}$ is given by:

$$z = x_1 + ix - 2 = jx_3 = kx_4$$
$$i^2 = j^2 = k^2 = -1,\ ij = k,\ jk = i, ki = j, ji = -k, kj = -i, ik = -j$$

By identifying $\mathbb{H}$ with $\mathbb{R}^4$ via the coordinates (x_1, x_2, x_3, x_4), then the unit sphere S^3 can be given by the three sections of the tangent bundle:

$$\begin{array}{lcl} z \mapsto iz & \text{or} & (x_1, x_2, x_3, x_4) \mapsto (-x_2, x_1, -x_4, x_3) \\ z \mapsto jz & \text{or} & (x_1, x_2, x_3, x_4) \mapsto (-x_3, x_4, x_1, -x_2) \\ z \mapsto kz & \text{or} & (x_1, x_2, x_3, x_4) \mapsto (-x_4, -x_3, x_2, x_1) \end{array}$$

The key idea is that the quaternions $1, i, j, k$ form the standard orthonormal basis for $\mathbb{R}^3$, multiplying by an arbitrary unit quaternion $z \in S^3$ gets another orthonormal basis z, iz, jz, kz, and since $|zw| = |z| \cdot |w|$ (where $|-|$ is the usual norm in $\mathbb{R}^3$, we get that multiplication by a quaternion in the unit sphere S^3 is an isometry of $\mathbb{H}$. A similar process can be done for the Cayley Octonions by thinking of $\mathbb{R}^8$ as $\mathbb{H} \times \mathbb{H}$ and multiplying the Octonions by:

$$(z_1, z_2)(w_1, w_2) = (z_1 w_2 - \overline{w_2} z_2, z_2 \overline{w_1} + w_2 z_1)$$

This also defines a division algebra, and also satisfies $|zw| = |z||w|$, giving 7 orthogonal tangent vectors fields on S^7 that are nowhere vanishing, and hence S^7 is also trivial. We shall show in ref:HERE that these are the only ones.

Definition 1.1.11: Direct Sum of Vector Bundles

Let $p_1 : E_1 \to B$, $p_2 : E_2 \to B$ be vector bundles over B. Then the *direct sum* or *coproduct* (historically called *Whitney sum*) of E_1, E_2 is:

$$E_1 \oplus E_2 = \{(v_1, v_2) \in E_1 \times E_2 \mid p_1(v_1) = p_2(v_2)\}$$

Certainly the fibers of the direct sum is the direct sum of the fibers. An interesting way of thinking of the direct sum is by taking the product of the two vector bundles (including the base space)

$$E_1 \times E_2 \to B \times B$$

Then the restriction no the product to the diagonal $\{(b, b) \in B \times B)\}$ is $E_1 \times E_2$. Note that just like for vector spaces, the (categorical) product and coproduct are isomorphic, however the notation $E_1 \times E_2$ is sometimes be used in literature to also mean the multiplication of the base-space, a convention we shall void.

Definition 1.1.12: Stably Trivial

Let E be a vector bundle. Then E is *stably trivial* if there exists a vector bundle F such that $E \oplus F$ is the trivial bundle.

We saw that the tangent bundle S^n is stably trivial, with NS^n being the compliment. Similarly, $E \oplus E^\perp$ where E is the canonical line bundle with its orthogonal compliment is isomorphic to $\mathbb{R}\mathrm{P}^n \times \mathbb{R}^{n+1}$ via

$$(\ell, v, w) \mapsto (\ell, v + w)$$

In the $n = 1$ case, $E^\perp$ is isomorphic to E by rotating each vector in the plane by $\pi/2$ degrees. As mention earlier, the canonical line bundle E in the $n = 1$ case is isomorphic to the Möbius bundle. Hence, the direct sum of the Möbius bundle with itself is the trivial bundle! Take a moment to think about this one as the “untwisting” is not an isometry and will have self-intersection.

Example 1.1.13: Stable Triviality of $\mathbb{R}\mathrm{P}^n$

By the above discussion, let $S^n \times \mathbb{R}^{n+1} \cong TS^n \oplus NS^n$. Define the equivalence (i.e. quotient by)

$$(x, v) \sim (-x, -v)$$

on both sides of the isomorphism. Then when applied to TS^n, we get $T\mathbb{R}\mathrm{P}^n$ (see [14], or even [1]). This is not preserved by the isomorphism (general quotient usually doesn't), however we do get the closest next thing: what we will show is a special case of the *Direct Sum decomposition for compact space* that will show that $T\mathbb{R}\mathrm{P}^n$ is a direct sum of nontrivial line bundles. We will show this by simply directly computing the two sides.

In NS^n, the identification $(x, v) \sim (-x, -v)$ is explicitly given by $(x, tx) \sim (-x, t(-x))$. This yields the product bundle $\mathbb{R}\mathrm{P}^n \times \mathbb{R}$ as $x \mapsto (-x, -x)$ is well-defined in the quotient. Now the identification $(x, v) \sim (-x, -v)$ in $S^n \times \mathbb{R}^{n+1}$ respects the coordinate macros of $\mathbb{R}^{n+1}$, so the quotient is the direct sum of $n + 1$ copies of line bundles E over $\mathbb{R}\mathrm{P}^n$ given by $(x, t) \sim (-x, -t)$ in $S^n \times \mathbb{R}$. It is now easy to see that E is the canonical line bundle over $\mathbb{R}\mathrm{P}^n$: identifying $S^n \times \mathbb{R}$ via the isomorphism $(x, t) \to (x, tx)$, we have $(-x, -t) \mapsto ((-x, (-t)(-x)) = (-x, tx)$ and so we have the gluing in NS^n that is the canonical line bundle over $\mathbb{R}\mathrm{P}^n$.

Hence, the direct sum of the tangent bundle $T\mathbb{R}\mathrm{P}^n$ with a trivial line bundle is isomorphic to $n+1$ copies of the canonical bundle over $\mathbb{R}\mathrm{P}^n$. When $n = 3$, $T\mathbb{R}\mathrm{P}^3$ is trivial (as shown earlier), and so we get four copies of the canonical line bundle over $\mathbb{R}\mathrm{P}^3$ is trivial (and similarly for the canonical line bundle over $\mathbb{R}\mathrm{P}^7$). We shall fully explore these properties in section ref:HERE.

As we are working with vector bundles, we can define an inner product:

$$\langle -, - \rangle E \oplus E \to \mathbb{R}$$

where E is a vector bundle over B. As we are over a paracompact space, we always have such an inner product

Proposition 1.1.14: Existence of Inner Product

Let $p : E \to B$ be a (real) vector bundle (over a paracompact space). Then there exists an inner product $\langle -, - \rangle$ on E

the complex inner product (also named Hermitian inner product) can be defined similarly.

***Proof*:**

This is the same proof as for the existence of a Riemann metric on a differentiable manifold. Let $\{U_\alpha\}$ be an open cover of B given by local trivializations. These can then pull-back the standard inner product on $\mathbb{R}^n$, and then by setting:

$$\langle v, w \rangle = \sum_{\beta} \varphi_\beta p(v) \langle v, w \rangle_{\alpha(\beta)}$$

where $\{\varphi_\beta\}$ is a partition of unity subordinate to $\{U_\alpha\}$ with support of φ_β contained in $U_{\alpha(\beta)}$.

Vector bundles are nice enough so satisfy the complementary property:

Proposition 1.1.15: Vector Bundles Satisfy Complimentary Property

Let $E \to B$ be a vector bundle (over a paracompact space B) and let $E_0 \subseteq E$ be a vector subbundle. Then there exists a vector subbundle $E_0^{\perp} \subseteq E$ such that

$$E_0 \oplus E_0^{\perp} \cong E$$

***Proof*:**
Let E have dimension n. Choose an inner product on E. Let $E_0^{\perp}$ be the subspace of E which in each fiber is the orthogonal to vectors in E_0. First, the natural projection $E_0^{\perp} \to B$ is a vector bundle. If this is so, then $E_0 \oplus E_0^{\perp}$ is isomorphic to E via the map $(v, w) \mapsto v + w$ by lemma 1.1.10.

To see that $E_0^{\perp}$ satisfies the local triviality condition, note that we may assume E is the product $B \times \mathbb{R}^n$ as the question is local in B. As E_0 is a vector bundle, it has some $m \leq n$ independent local sections $b \mapsto (b, s_i(b))$ near each point $b_0 \in B$. This set of m independent local sections of E_0 can be enlarged to a set of n independent local sections $b \mapsto (b, s_i(b))$ of E by choosing $s_{m+1}, \ldots, s_n$ first in the fiber $p^{-1}(b_0)$, then taking the same vectors for all nearby fibers, since if $s_1, \ldots, s_m, s_{m+1}, \ldots, s_n$ are independent at b_0, they will be independent near b by continuity of the determinant function. Applying the Gram-Schmidt orthogonalization process to $s_1, \ldots, s_m, s_{m+1}, \ldots, s_n$ in each fiber we obtain new sections s_i'. Note that each s_i' is continuous as they were obtain via the explicit formulas for the Gram-Schmidt process which is certainly a continuous process, and the first m of them are a basis for E_0 in each fiber. The sections s_i' allow us to define a local trivialization $h : p^{-1}(U) \to U \times \mathbb{R}^n$ with $h(b, s_i'(b))$ equal to the i^{th} standard basis vector of $\mathbb{R}^n$. Now, this h carries E_0 to $U \times \mathbb{R}^m$ and $E_0^{\perp}$ to $U \times \mathbb{R}^{n-m}$, so $h|_{E_0^{\perp}}$ is a local trivialization of $E_0^{\perp}$, as we sought to show.

When $E_0 = E$, then this proof also shows that local trivializations are also isometries. When B is compact, then we get that vector bundles are "projective"

Proposition 1.1.16: Vector Bundle Over Compact Are Projective

Let $E \to B$ be a vector bundle over a compact Hausdorff space B. Then there exists a vector bundle $E' \to B$ such that $E \oplus E'$ is the trivial bundle

Note this fails for non-compact spaces, we shall show this for $\mathbb{RP}^n$ in example 3.2.5(3).

***Proof*:**
Suppose first this holds, so E is a subbundle of a trivial bundle $B \times \mathbb{R}^N$. Composing the inclusion of E into this product with the projection of the product onto $\mathbb{R}^N$ yields a map $E \to \mathbb{R}^N$ that is a linear injection on each fiber. We shall show such a map exists by first constructing a map $E \to \mathbb{R}^N$ that is a linear injection on each fiber, then showing that this gives an embedding of E in $B \times \mathbb{R}^N$ as a direct summand.

To that end, each point $x \in B$ has an open neighborhood U_x over which E is trivial. By Urysohn's Lemma there is a map $\varphi_x : B \to [0, 1]$ that is nonzero at x and has support contained in U_x. Letting x vary, the sets $\varphi_x^{-1}(0, 1]$ form an open cover of B. By compactness, there exists finite subcover. Relabel the corresponding U_x's and φ_x's as U_i and φ_i. Define $g_i : E \to \mathbb{R}^n$ by

$g_i(v) = \varphi_i(p(v))(\pi_i h_i(v))$ where p is the projection $E \to B$ and $\pi_i h_i$ is the composition of a local trivialization $h_i : p^{-1}(U_i) \to U_i \times \mathbb{R}^n$ with the projection π_i to $\mathbb{R}^n$. Then g_i is a linear injection on each fiber over $\varphi_i^{-1}(0,1]$, so making the various g_i's the coordinates of a map $g : E \to \mathbb{R}^N$ with $\mathbb{R}^N$ a product of copies of $\mathbb{R}^n$, then g is a linear injection on each fiber.

The map g is the second coordinate of a map $f : E \to B \times \mathbb{R}^N$ with first coordinate p. The image of f is a subbundle of the product $B \times \mathbb{R}^N$ since projection of $\mathbb{R}^N$ onto the ith $\mathbb{R}^n$ factor gives the second coordinate of a local trivialization over $\varphi_i^{-1}(0,1]$. Thus, E isomorphic to a subbundle of $B \times \mathbb{R}^N$ so by the preceding proposition there is a complementary subbundle E' with $E \oplus E'$ isomorphic to $B \times \mathbb{R}^N$, as we sought to show.

Next, we define the tensor product.

Definition 1.1.17: Tensor Product of Vector Bundles

Let $p_1 : E_1 \to B$, $p_2 : E_2 \to B$ be vector bundles over B. Then *tensor product* is a vector bundle where as the set

$$E_1 \otimes E_2 = \coprod_{x \in B} p_1^{-1}(x) \otimes p_2^{-1}(x)$$

and for the topology, choose an open cover $\{U_\alpha\}$ with local trivializations $h_i : p_i^{-1}(UU) \to U \times \mathbb{R}^{n_i}$, then the topology $\mathcal{T}_U$ on $\pi_1^{-1}(U) \otimes \pi_1^{-1}(U)$ is given by letting the fiberwise tensor product map:

$$h_1 \otimes h_2 : p_1^{-1}(U) \otimes p_2^{-1}(U) \to U \times (\mathbb{R}^{n_1} \times \mathbb{R}^{n_2})$$

be a homeomorphism.

This topology is independent of the choice of local trivialization cover as any other choice is will be obtained by composing with isomorphism of $U \times \mathbb{R}^{n_i}$ of the form:

$$(x, v) \mapsto (x, g_i(x)(v))$$

where $g_i : U \to \mathrm{GL}_{n_i}(\mathbb{R})$. Thus, $h_1 \otimes h_2$ changes by composing with analogous isomorphism of $U \times (\mathbb{R}^{n_1}, \otimes \mathbb{R}^{n_2})$ whose second coordinate $g_1 \otimes g_2$ are continuous maps

$$U \to \mathrm{GL}_{n_1 n_2}(\mathbb{R})$$

If the reader is familiar with the fibered product of schemes, or is comfortable with the more "categorical" approach, then we may also define the tensor product by first defining the vector-bundle as the gluing of $\coprod_\alpha (U_\alpha \times \mathbb{R}^n)$ where on intersections we have the gluing condition:

$$g_{\beta\alpha} : U_\alpha \cap U_\beta \to \mathrm{GL}_n(\mathbb{R})$$

that satisfy the cocycle condition on $U_\alpha \cap U_\beta \cap U_\beta$

$$g_{\gamma\beta} g_{\beta\alpha} = g_{\gamma\alpha}$$

Then the tensor product between $E_1 \to B$ and $E_2 \to B$ is given by choosing an open cover $\{U_\alpha\}$ that is locally trivial for both E_1, E_2 and getting gluing functions $g^i_{\beta\alpha} : U_\alpha \cap U_\beta \to \mathrm{GL}_{n_i}(\mathbb{R})$ for each E_i. Then the tensor bundle $E_1 \otimes E_2$ is the tensor product of the functions $g^1_{\beta\alpha} \otimes g^2_{\beta\alpha}$ assigning to each$x \in U_\alpha \cap U_\beta$ the tensor product of the matrices.

The key is that it is a fiberwise product. Hence, over a fixed base we get that it is commutative, associate, has an identity element (is the trivial bundle), and distributive with direct sum. If we only look at line bundles, then the tensor product of two line bundles is a line bundle. In that case, $\mathrm{Vect}^1(B)$ (the isomorphism classes of line bundles) is an abelian group with respect to the tensor product; the inverse is giving by inverses matrices $g_{\beta\alpha}(x) \in \mathrm{GL}_1(\mathbb{R})$ and the cocycle condition is preserved as 1 by 1 matrices commute. Giving E an inner product, we can re-scale the local trivialization h_α to be isometries, which then the vectors in fivers of E go to vectors in $\mathbb{R}^1$ of the same length. Thus ,all values of the gluing functions $g_{\beta\alpha}$ are ± 1. Then the gluing function for $E \otimes E$ are the squares of these $g_{\beta\alpha}$, and hence are identically 1; thus every element of $\mathrm{Vect}^1(B)$ is its own inverse. This should remind the reader of the cup product in algebraic topology, and indeed in ref:HERE we shall see that $\mathrm{Vect}^1(B)$ is isomorphic to the cohomology group $H^1(B;\mathbb{Z})$ been B is homotopy equivalent to a CW complex.

The same idea goes through for complex vector bundles but with some different behaviour: the gluing functions $g_{\beta\alpha}$ will have norm 1 (not just ± 1, so it is not $E \otimes E$ need not be trivial. We shall see that $\mathrm{Vect}^1_{\mathbb{C}}(B) \cong H^2(B;\mathbb{Z})$ when B is homotopy equivalent to a CW complex. However, is we take the *conjugate bundle* $\overline{E} \to B$ defined in the natural way, then as $z\overline{z} = 1$ when $|z| = 1$, $E \otimes \overline{E}$ is the trivial line bundle.

With the tensor product, we can then also define the symmetric and alternating bundle, which also gives us the determinant bundle.

1.2 Classifying Vector Bundles

Proposition 1.2.1: Pullback of Vector bundle

Let $f : A \to B$ be a continuous map and $p : E \to B$ a vector bundle. Then there exists a vector bundle $p' : E' \to A$ with a map $f' : E' \to E$ taking the fibers of E' over each $a \in A$ isomorphically onto the fibers of a over $f(a)$. The vector bundle E' is unique up to isomorphism.

Thus, we get the pullback map $f^* : \mathrm{Vect}(B) \to \mathrm{Vect}(A)$ mapping isomorphism classes of E to isomorphism classes of $f^*(E) = E'$.

***Proof*:**
Let
$$E' = \{(a, v) \in A \times E \big| f(a) = p(v)\}$$
This subspace of $A \times E$ fits into the following commutative diagram

$$\begin{tikzcd} E' \arrow[r, "f'"] \arrow[d, "p'"'] & E \arrow[d, "p"] \\ A \arrow[r, "f"] & B \end{tikzcd}$$

where $p'(a, v) = a$ and $f'(a, v) = v$. Let Γ_f denote the graph of f in $A \times B$. Then p' factors as the composition $E' \to \Gamma_f \to A$, $(a, v) \mapsto (a, p(v)) = (a, f(a)) \mapsto a$. The first of these two maps is the restriction of the vector bundle $\nVdash \times p : A \times E \to A \times B$ over the graph Γ_f, so it is a vector bundle, and the second map is a homeomorphism, so their composition $p' : E' \to A$ is a vector bundle. The map f' obviously takes the fiber E' over a isomorphically onto the fiber of E over $f(a)$, showing existence.

For uniqueness, construct an isomorphism from an arbitrary E' satisfying the conditions in the proposition to the particular one just constructed by sending $v' \in E'$ to the pair $(p'(v'), f'(v'))$. This map takes each fiber of E' to the corresponding fiber of $f^*(E)$ by a vector space isomorphism, so by lemma 1.1.10 it is an isomorphism of vector bundles, completing the proof.

When restricting to Vect^1, notice the trivial bundle pulls back to the trivial bundle and the tensor product is well-behaved; hence this pullback becomes a group homomorphism.

Example 1.2.2: Vector Bundle Pullbacks

1. Let $A \subseteq B$ so that $\iota : A \hookrightarrow B$. Then the restriction of E over B to A is the pullback $\iota^*(E)$.
2. Let $f : A \to B$ be a constant map: $f(A) = b$. Then
$$f^*(E) = A \times p^{-1}(B)$$
3. The tangent bundle TS^n is the pullback of $T\mathbb{R}\mathrm{P}^n$ via he map $S^n \to \mathbb{R}\mathrm{P}^n$.
4. Consider the möbius bundle $E \to S^1$, and take $f : S^1 \to S^1$ given by $z \mapsto z^2$. By thinking of the Möbius bundle as $S^1 \times \mathbb{R}$ under $(z, t) \sim (-z, -t)$, then $f^*(E)$ is the trivial bundle (think about this for awhile!). If $f(z) = z^n$, then E_n is the trivial bundle if n is even, and the Möbius bundle is n is odd.
5. Let $F(E)$ be the flag bundle of n-dimensions (rest here; [15, p.20]).

Proposition 1.2.3: Properties of Vector-bundle Pullback

Let E be a vector bundle and f, g continuous maps, everything over the appropriate domains. Then:

1. $(fg)^*(E) \cong g^*(f^*(E))$
2. $\mathrm{id}^*(E) \cong E$
3. $f^*(E_1 \oplus E_2) \cong f^*(E_1) \oplus f^*(E_2)$
4. $f^*(E_1 \otimes E_2) \cong f^*(E_1) \otimes f^*(E_2)$

Proof:
exercise

Let us next show that isomorphism classes of vector bundles are invariant under homotopy. We first need a technical lemma:

Lemma 1.2.4: Homotopy Invariant Vector Bundle

Let $E \to X \times I$ be a vector bundle where X is paracompact. Then the restriction over $X \times \{0\}$ and $X \times \{1\}$ is isomorphic.

***Proof*:**
First two smaller proofs:

1. A vector bundle $p : E \to X \times [a, b]$ is trivial if its restrictions over $X \times [a, c]$ and $X \times [c, b]$ are both trivial for some $c \in (a, b)$. Indeed, let these restrictions be $E_1 = p^{-1}(X \times [a, c])$ and $E_2 = p^{-1}(X \times [c, b])$, and let $h_1 : E_1 \to X \times [a, c] \times \mathbb{R}^n$ and $h_2 : E_2 \to X \times [c, b] \times \mathbb{R}^n$ be isomorphisms. These isomorphisms may not agree on $p^{-1}(X \times \{c\})$, but they can be made to agree by replacing h_2 by its composition with the isomorphism $X \times [c, b] \times \mathbb{R}^n \to X \times [c, b] \times \mathbb{R}^n$ which on each slice $X \times \{x\} \times \mathbb{R}^n$ is given by $h_1 h_2^{-1} : X \times \{c\} \times \mathbb{R}^n \to X \times \{c\} \times \mathbb{R}^n$. Once h_1 and h_2 agree on $E_1 \cap E_2$, they define a trivialization of E.

2. For a vector bundle $p : E \to X \times I$, there exists an open cover $\{U_\alpha\}$ of X where each restriction $p^{-1}(U_\alpha \times I) \to U_\alpha \times I$ is trivial. Indeed, for each $x \in X$ find open neighborhoods $U_{x,1}, \ldots, U_{x,k}$ in X and a partition $0 = t_0 < t_1 < \cdots < t_k = 1$ of $[0, 1]$ such that the bundle is trivial over $U_{x,i} \times [t_{i-1}, t_i]$, using compactness of $[0, 1]$. Then by (1) the bundle is trivial over $U_\alpha \times I$ where $U_\alpha = U_{x,1} \cap \cdots \cap U_{x,k}$.

Now for the proposition, by (2), choose an open cover $\{U_\alpha\}$ of X so that E is trivial over each $U_\alpha \times I$. Let's assume for now the simpler case that X is compact Hausdorff. Then a finite number of U_α's cover X. Relabel these as U_i for $i = 1, 2, \ldots, m$. As shown in proposition 1.1.14, there is a corresponding partition of unity by functions φ_i with the support of φ_i contained in U_i. For $i \geq 0$, let $\psi_i = \varphi_1 + \cdots + \varphi_i$, so in particular $\psi_0 = 0$ and $\psi_m = 1$. Let X_i be the graph of ψ_i, the subspace of $X \times I$ consisting of points of the form $(x, \psi_i(x))$, and let $p_i : E_i \to X_i$ be the restriction of the bundle E over X_i. Since E is trivial over $U_i \times I$, the natural projection homeomorphism $X_i \to X_{i-1}$ lifts to a homeomorphism $h_i : E_i \to E_{i-1}$ which is the identity outside $p^{-1}(U_i \times I)$ and which takes each fiber of E_i isomorphically onto the corresponding fiber of E_{i-1}. Explicitly, on points in $p^{-1}(U_i \times I) = U_i \times I \times \mathbb{R}^n$ let $h_i(x, \psi_i(x), v) = (x, \psi_{i-1}(x), v)$. The composition $h = h_1 h_2 \cdots h_m$ is then an isomorphism from the restriction of E over $X \times \{1\}$ to the restriction over $X \times \{0\}$.

In the general case that X is only paracompact, there is a countable cover $\{V_i\}_{i\geq 1}$ of X and a partition of unity $\{\varphi_i\}$ with φ_i supported in V_i, such that each V_i is a disjoint union of open sets each contained in some U_α (see [14]). Then E is trivial over each $V_i \times I$. As before we let $\psi_i = \varphi_1 + \cdots + \varphi_i$ and let $p_i : E_i \to X_i$ be the restriction of E over the graph of ψ_i. Also as before we construct $h_i : E_i \to E_{i-1}$ using the fact that E is trivial over $V_i \times I$. The infinite composition $h = h_1 h_2 \cdots$ is then a well-defined isomorphism from the restriction of E over $X \times \{1\}$ to the restriction over $X \times \{0\}$ since near each point $x \in X$ only finitely many φ_i's are nonzero, so there is a neighborhood of x in which all but finitely many h_i's are the identity.

Theorem 1.2.5: Homotopy Invariance Vector Bundles

Let $p : E \to B$ a vector bundle and $f_0, f_1 : A \to B$ be homotopic maps. Then if A is paracompact,

$$f_0^*(E) \cong f_1^*(E)$$

Note this result holds for fiber bundles more generally.

***Proof*:**
Let $F : A \times I \to B$ be a homotopy from f_0 to f_1. Then the restriction $f^*(E)$ over $A \times \{0\}$ and $A \times \{1\}$ are $f_0^*(E)$ and $f_1^*(E)$, and we get the result by the above lemma.

Corollary 1.2.6: Homotopy For N-vector Bundles

Let $f : A \to B$ be a homotopy equivalence of paracompact spaces. Then

$$f^* : \mathrm{Vect}^n(B) \to \mathrm{Vect}^n(A)$$

is a bijection. In particular, vector bundles over a contractible paracompact base is trivial.

***Proof*:**
Let g be the homotopy inverse of f. Then

$$f^*g^* = \mathrm{id}^* = \mathrm{id} \qquad g^*f^* = \mathrm{id}^* = \mathrm{id}$$

completing the proof.

1.2.1 Classifying Vector Bundles on Spheres

Using the tensor product, we can construct some non-trivial bundles on the sphere S^k. The clutching function is a way of creating a bundle on the sphere S^k by gluing a bundle on the upper and lower hemisphere with a compatibility condition on the "equator subspace" (this is the "clutch"). In particular, let $D^k+_, D^k_-$ be the upper and lower hemisphere so that $D^k_+ \cap D^k_- \cong S^{k-1}$. Given a map $f : S^{k-1} \to \mathrm{GL}_n(\mathbb{R})$, we can define:

$$E_f = \left((D^k_- \times \mathbb{R}^n) \sqcup (D^k_- \times \mathbb{R}^n)\right) / \sim$$
$$\partial D^k_- \times \mathbb{R}^n \ni (x, v) \sim (x, f(x)(v)) \in \partial D^k_+ \times \mathbb{R}^n$$

Then we get a natural n-vector bundle map $f : E_f \to S^k$. Naturally, replacing $\mathbb{R}$ with $\mathbb{C}$ would require $f : S^{k-1} \to \mathrm{GL}_n(\mathbb{C})$.

Example 1.2.7: Clutching Function

1. Let us create TS^2 using a clutching function. Let v_+ be a vector field on D^2_+ given by choosing a tangent vector at the north pole and parallel-transporting it along each meridian circle (see [14] for parallel-transports). Here is a visual on only a quarter of the upper hemisphere:

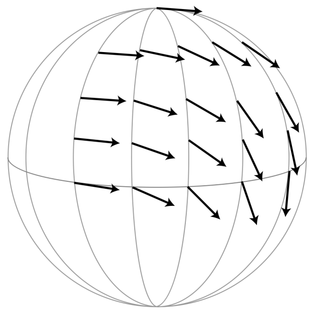

Figure 1.1: v_+ on D^2_+ (from Hatcher [15])

Reflect across the equation to create the vector field v_- on D^2_-. Then take w_+ and w_- to be v_+ and v_- respectively each rotate by $\pi/2$ degree when viewed from outside the sphere. Then these for vector fields give a trivialization of TS^2 over $D^2_\pm$: the clutching function f read off the coordinates of v_-, w_- in the v_+, w_+ coordinate system, in particular$f$ rotates (v_+, w_+) to (v_-, w_-). As we go around the equator S^1 counter-clockwise, the angle will go from 0 to 4π.

2. Can you think of how do do this for $\mathbb{C}\mathrm{P}^1 = S^2$ (hint: we should get a clutching function $f : S^1 \to \mathrm{GL}_1(\mathbb{C})$ where $f(z) = (z^{-1})$.).

The advantage of working with clutching functions is that we get that if f, g are clutching functions for E_f, E_g we have $E_f \cong E_g$ if and only if f, g are homotopic. When $k = \mathbb{C}$, we get a bijection (we shall show what we need to modify for the bijection when our $\mathbb{R}$, particularly it will have to be dealt with that $\mathbb{R} \setminus \{0\}$ is no longer connected)

Proposition 1.2.8: Clutching Function Vector Bundle Correspondence

The map:

$$\Phi : [S^{k-1}, \mathrm{GL}_n(\mathbb{C})] \to \mathrm{Vect}^n_{\mathbb{C}}(S^k)$$

sending homotopy classes of clutching functions $f : S^{k-1} \to \mathrm{GL}_n(\mathbb{C})$ to complex vector bundles E_f is a bijection

***Proof*:**
Let $[X, Y]$ denote the set of homotopy equivalences of functions. If $f \simeq g$, then there is a homotopy $F : S^{k-1} \times I \to \mathrm{GL}_n(\mathbb{C})$. Then the same sort of clutching construction can be used to produce a vector bundle $E_F \to S^K \times I$ that restricts to E_f over $S^k \times \{-\}$ and E_g over $S^k \times \{1\}$, hence $E_f \cong E_g$ by theorem 1.2.5. We thus have a map:

$$\Phi : [S^{k-1}, \mathrm{GL}_n(\mathbb{C})] \to \mathrm{Vect}^n_{\mathbb{C}}(S^k)$$

Let us find an inverse Ψ. Let $p : E \to S^k$ be an n-vector bundle, and let E_+, E_- be its restriction to D^k_+ and D^k_-. $D^k_\pm$ is contractible, by theorem 1.2.5 these bundles are trivial. Choose trivializations $h_\pm : E_\pm \to D^k_\pm \times \mathbb{C}^n$. Then we get that $h_+ h_-^{-1}$ defines a map $S^{k-1} \to \mathrm{GL}_n(\mathbb{C})$ whose homotopy class is by construction

$$\Psi(E) \in [S^{k-1}, \mathrm{GL}_n(\mathbb{C})]$$

Let us make sure $\Psi(E)$ is well-defined, that is the trivialization is unique up to homotopy. First off, any two choices of $h_\pm$ differ by a map $D^k_\pm \to \mathrm{GL}_n(\mathbb{C})$, as $D^k_\pm$ is contractible, these maps are homotopic to a constant map. As $\mathrm{GL}_n(\mathbb{C})$ is path connected, elementary row operations in $\mathrm{GL}_n(\mathbb{C})$ can be realized by a path in $\mathrm{GL}_n(\mathbb{C})$ (insert a factor of $t \in [0,1]$ in front of the scalar multiples). As $\mathbb{C}$ is algebraically closed, $M \in \mathrm{GL}_n(\mathbb{C})$ can be diagonalized. As the set of diagonal matrices in $\mathrm{GL}_n(\mathbb{C})$ is path-connected (since it is homeomorphic to the product of n copies of $\mathbb{C} - \{0\}$), we see that h_+ and h_- is unique up to homotopy. Thus $h_+h_-^{-1} : S^{k-1} \to \mathrm{GL}_n(\mathbb{C})$ is unique up to homotopy, hence $\Psi : \mathrm{Vect}^n_{\mathbb{C}}(S^k) \to [S^{k-1}, \mathrm{GL}_n(\mathbb{C})]$ is well-defined, and by construction it is clear it is the inverse, completing the proof.

This correspondence gives some immediate results. For example, every complex vector bundles over S^1 is trivial since every class $\mathrm{Vect}^n_{\mathbb{C}}(S^1)$ corresponds to $[s^0, \mathrm{GL}_n(\mathbb{C})]$ which as a single element (as $\mathrm{GL}_n(\mathbb{C})$ is path connected). Another important result is that the canonical line bundle $H \to \mathbb{C}\mathrm{P}^1$ satisfies the relation:

$$(H \otimes H) \oplus 1 \cong H \oplus H$$

where 1 is the trivial one-dimensional bundle (see [15, p.24]).

For the real case, we have that $\mathrm{GL}_n(\mathbb{R})$ has two homeomorphic path-connected components given by the pre-image of det. However, everything passes through if we consider *oriented (real) vector bundles* (where all the transition maps have the same sign) as we only need the fact the component is path connect:

Proposition 1.2.9: Clutching Function Vector Bundle Correspondence

The map:

$$\Phi : [S^{k-1}, \mathrm{GL}^+_n(\mathbb{R})] \to \mathrm{Vect}^n_+(S^k)$$

sending homotopy classes of clutching functions $f : S^{k-1} \to \mathrm{GL}^+_n(\mathbb{R})$ to oriented real vector bundles E_f is a bijection.

***Proof*:**
exercise.

From this, let us figure out some information about $\mathrm{Vect}^n_{\mathbb{R}}(S^k)$ for various n, k. To understand this structure better, let us first define a map $\mathrm{Vect}^n_{x_0,\mathbb{R}}(S^k) \to \mathrm{Vect}_{\mathbb{R}}(S^k)$. The set $\mathrm{Vect}^n_{x_0,\mathbb{R}}(S^k)$ is all isomorphism classes with an orientation specified by the point $x_0 \in S^{k-1}$; by our above results we get that $\mathrm{Vect}^n_{x_0,\mathbb{R}}(S^k)$ is in bijective correspondence with the homotopy classes of maps $S^{k-1} \to \mathrm{GL}_n(\mathbb{R})$ taking x_0 to $\mathrm{GL}^+_n(\mathbb{R})$. Then we get the map

$$\mathrm{Vect}^n_{x_0,\mathbb{R}}(S^k) \to \mathrm{Vect}^n_{\mathbb{R}}(S^k)$$

that forgets the orientation over x_0. This maps is certainly surjective. It is 2-to-1 at all points except where a bundle has an automorphism that reverses the orientation of the fiber over x_0; on such points it is 1-to-1.

When $k = 1$ and $n \in \mathbb{N}_{>0}$, there is two homotopy classes $S^0 \to \mathrm{GL}_n(\mathbb{R})$: x_0 goes to $\mathrm{id} \in \mathrm{GL}^+_n(\mathbb{R})$, and then the other point goes to to either the identity or the reflection. When $n = 1$, we get the trivial bundle and the Möbius bundle. When $n > 1$, we get that these bundles all have automorphisms reversing the orientations of the fibers. Hence, $\mathrm{Vect}^n_{\mathbb{R}}(S^1)$ always have two elements for each $n \geq 1$, with the nontrivial bundle being nonorientable.

When $k > 1$, then S^{k-1} is path-connected, and hence $S^{k-1} \to \mathrm{GL}_n(\mathbb{R})$ taking x_0 to $\mathrm{GL}_n^+(\mathbb{R})$ must take all of S^{k-1} to $\mathrm{GL}_n^+(\mathbb{R})$, implying that we get a bijection

$$\mathrm{Vect}_+^n(S^k) \xrightarrow{\sim} \mathrm{Vect}_{x_0,\mathbb{R}}^n(S^k)$$

and so every vector bundle over S^k is orientable and has (exactly) two different orientations. Thus, the map:

$$\mathrm{Vect}_+^n(S^k) \to \mathrm{Vect}^n(S^k)$$

is surjective, being 2-to-1 on bundles not having an orientation-reversing automorphism, and is injective otherwise.

As we are working over paracompact spaces, and have access to an inner product, we get that we can deformation-retract $\mathrm{GL}_n(\mathbb{R})$ to the orthogonal subgroup $O(n)$ where $O(n) \leqslant \mathrm{GL}_n(R)$ where:

$$O(n) = \left\{A \in \mathrm{GL}_n(\mathbb{R}) \;\middle|\; AA^T = I\right\}$$

The deformation retract is easy by keeping track the Gram-Schmidt process and applying "paths" to each process (as has been done earlier). Similarly, $\mathrm{GL}_n^+(\mathbb{R})$ is a deformation retract of the special orthogonal group $SO(n)$ where

$$SO(n) = \{A \in O(n) \mid \det(A) = 1\}$$

In the complex case, we get a deformation retract onto the unitary group $U(n)$ where

$$U(n) = \{A \in \mathrm{GL}_n(\mathbb{C}) \mid AA^* = I\}$$

and can be characterized by having column forming an orthonormal basis.

Now, as we have a deformation retract $SO(n) \hookrightarrow \mathrm{GL}_n^+(\mathbb{R})$ and $U(n) \hookrightarrow \mathrm{GL}_n(\mathbb{C})$ we get natural bijections:

$$[S^{k-1}, SO(n)] \xrightarrow{\sim} [S^{k-1}, \mathrm{GL}_n^+(\mathbb{R})] \qquad \text{and} \qquad [S^{k-1}, U(n)] \xrightarrow{\sim} [S^{k-1}, \mathrm{GL}_n(\mathbb{C})]$$

This can greatly simply classification efforts as $SO(n)$ is a much smaller Lie groups. For example, $SO(2)$ is a circle (orientation-preserving isometric of $\mathbb{R}^2$ are rotations). Letting $k = 2$ and $n = 2$, we get for each integer $n \in \mathbb{Z}$ a map $f_m : S^1 \to S^1$ given by $z \mapsto z^n$ which is a clutching function for an oriented 2-dimensional vector bundle $E_n \to S^2$. Note that E_2 is TS^2 and E_1 is the real vector bundle underlying the canonical complex line bundle over $\mathbb{C}\mathrm{P}^1 = S^2$. As every arbitrary continuous map $f : S^1 \to S^1$ is homotopic to a map $z \mapsto z^n$ with a unique $n \in \mathbb{Z}$ (see [9]), we get:

$$\mathrm{Vect}_+^2(S^2) \cong \mathbb{Z}$$

Note that reversing the orientation of E_n gives E_{-n}, and hence:

$$\mathrm{Vect}_{\mathbb{R}}^2(S^2) \cong \mathbb{N}$$

As $U(1)$ is homeomorphic to S^1, we similarly get:

$$\mathrm{Vect}_{\mathbb{C}}^2(S^2) \cong \mathbb{Z}$$

For $k > 2$, recall that any two maps $S^{k-1} \to S^1$ are homotopic if $k > 2$ (such a map lifts to a covering space $\mathbb{R}$ of S^1 and $\mathbb{R}$ is contractible, see [9]). Hence, every 2-dimensional vector bundle over S^k for $k \geq 3$ is trivial:

$$\mathrm{Vect}_{\mathbb{R}}^2(S^k) \cong \{0\} \qquad k \geq 3$$

The same logic applies to complex vector bundles:

$$\mathrm{Vect}^2_{\mathbb{C}}(S^k) \cong \{0\} \qquad k \geq 3$$

We shall give some more examples after the next proposition. We first point out a couple more important facts. First, for any space X, the set $[X, SO(n)]$ and $[X, U(n)]$ has a natural group structure given by point-wise multiplication. Using this fact, we have that:

$$[S^i, SO(n)] \cong_{\mathrm{Grp}} \pi_i(SO(n)) \qquad \text{and} \qquad [S^i, U(n)] \cong_{\mathrm{Grp}} \pi(U(n))$$

where homotopies preserve base points.

Proposition 1.2.10: Vector bundles and Suspension

et SX denote the suspension of a paracompact space X. Then given a choice of base-point $x \in X$, there is a 1-1 correspondence between the set $\mathrm{Vect}^n(SX)$ of isomorphism classes of n-dimensional (resp. real, complex, or quaternion) vector bundles over SX and the respective set of based homotopy classes of maps

$$[X, O_n]_*, \quad [X, U_n]_*, \quad \text{or} \quad [X, Sp_n]_*$$

from X to the orthogonal group O_n, unitary group U_n, or symplectic group Sp_n.

***Proof*:**
The proof is a complicated proof, the following is a sketch given from [6]. From [9], we know that SX is the union of two contractible cones C_1 and C_2 whose intersection is X. As every vector bundle on the cones C_i is trivial as it is contractible, and every vector bundle on SX is obtained from an isomorphism of trivial bundles over X via the clutching construction (see [6]; it is the Mayer-Vietoris sequence for this suspension construction). Such an isomorphism is given by a homotopy class of maps from X to GL_n, or equivalently, to the appropriate deformation retract (O_n, U_n, or Sp_n) of GL_n. Then the indeterminacy in the resulting map from $[X, GL_n]$ to classes of vector bundles is eliminated by insisting that the base point of X map to $1 \in GL_n$.

Now, from [13] or [9], we know the computations of the homotopy groups classical Lie groups, which we now take advantage of to find the isomorphism classes of vector bundles on spheres

Example 1.2.11: Vector Bundles On Spheres

1. Let $X = S^1$, and recall that $|\pi_0(O_n)| = 2$ for real vector bundles of dimension n for all $n \geq 1$. This corresponds to the trivial bundle and the möbius bundle. The Whitney direct sum of these two shall give all non-trivial real vector bundles on S^1.

   As $|\pi_0(U_n)| = 1$ for all n, all complex vector bundles are trivial!

2. Let $X = S^2$. Then $\pi_1(O_1) = 0$, and so there is only the trivial real line bundle on S^1. As $\pi_1(O_2) = \mathbb{Z}$, there are infinitely many 2-dimensional real line bundles of dimension 2 (indexed by the degree of the clutching map mentioned in the above proof). As $\pi_1(O_n) = \mathbb{Z}/2\mathbb{Z}$ for $n \geq 2$, there is only two real line bundles for dimension $n \geq 3$.

   For complex line bundles on S^3, we get the same phenomena as $\pi_1(U_2) = \mathbb{Z}$

3. Recall from [13] that $\pi_2(G) = 0$ for all compact Lie groups. Thus, all vector bundles on S^3 are trivial!

4. From the above, we can conclude that all (real and complex) line bundles on S^4 trivial. as:

$$\begin{array}{ccc} \pi_3(O_n) = 3 & \pi_3(U_n) = \mathbb{Z} & \pi_3(Sp_n) = \mathbb{Z} \\ \text{for } n \geq 5 & \text{for } n \geq 2 & \text{for } n \geq 1 \end{array}$$

We get the corresponding infinite family of n-dimensional Real/Complex/Quaternion vector bundles.

1.2.2 Universal Bundles

In this section, we shall show that every vector bundles is the pullback of a canonical vector bundle on the Grassmannian G_n over $\mathbb{R}, \mathbb{C}$, or $\mathbb{H}$. As a quick reminder of the Grassmannian, let V be a finite dimensional vector space and let $\mathrm{Grass}_k(V)$ be all k-dimensional subspaces of V; i.e. the *Grassmannian manifold* of all sub-spaces of V . Let's define a topology on $\mathrm{Grass}_n(\mathbb{R}^k)$ by first defining the *Stiefel Manifold* $V_n(\mathbb{R}^k)$ which is the space of orthonormal n-frames in $\mathbb{R}^k$. This is a subspace of the product of $n$copies of the unit sphere S^{k-1}, that is the subspace of orthonormal n-tuples. It is a closed subspace as orthogonality is given by an algebraic equation. Hence, $V_n(\mathbb{R}^k)$ is compact, being the product of spheres. Then there is a natural surjection $V_n(\mathbb{R}^k) \to \mathrm{Grass}_n(\mathbb{R}^k)$ sending n-frames to the subspace it spans. This gives the quotient topology on $G_n(\mathbb{R}^k)$, and it is easy to verify it is also compact. To show its a manifold, see see [14]. The *infinite Grassmannian* Grass_n is the union all $\mathrm{Grass}_n(V)$ as V ranges over all finite dimensional subspace of a fixed infinite dimensional vector space (usually $\mathbb{R}^\infty$, $\mathbb{C}^\infty$, or $\mathbb{H}^\infty$). With a bit of work, we see that it is an infinite-dimensional CW complex (it is the limit of CW complexes respecting the appropriate directed mappings, see 1.2.3). These are rather natural, for example if taking $n = 1$, we get that Grass_1 is $\mathbb{R}\mathrm{P}^\infty$, $\mathbb{C}\mathrm{P}^\infty$, or $\mathbb{H}\mathrm{P}^\infty$ respectively.

Just like for $\mathbb{R}\mathrm{P}^n$, $\mathbb{C}\mathrm{P}^n$, and $\mathbb{H}\mathrm{P}^n$, there a canonical line bundle for Grassmannian: let:

$$E_n(\mathbb{R}^k) = \left\{(\ell, v) \in \mathrm{Grass}_n(\mathbb{R}^k) \times \mathbb{R}^k \;\middle|\; v \in \ell\right\}$$

and similarly for $\mathbb{C}^k, \mathbb{H}^k$. Then the inclusion $\mathbb{R}^k \subseteq \mathbb{R}^{k+1} \subseteq \cdots$ induces inclusions $E_n(\mathbb{R}^k) \subseteq E_n(\mathbb{R}^{k+1}) \subseteq \cdots$, and hence

$$E_n(\mathbb{R}^\infty) = \cup_k E_n(\mathbb{R}^k)$$

where it is given the colimit topology, and similarly for the complex and quaternion case. Then:

Proposition 1.2.12: Grassmannian Bundle is Vector Bundle

The projection $p : E_n(\mathbb{R}^k) \to \mathrm{Grass}_n(\mathbb{R}^k)$ where $p(\ell, v) = \ell$ is a vector bundle for finite and infinite k.

***Proof*:**
By proving the finite case, the infinite case follows by local trivializations working over every $k \in \mathbb{N}$, hence we prove the finite case. For $\ell \in G_n(\mathbb{R}^k)$, let $\pi_\ell : \mathbb{R}^k \to \ell$ be orthogonal projection and let

$$U_\ell = \left\{\ell' \in G_n(\mathbb{R}^k) \;\middle|\; \pi_\ell(\ell') \text{ has dimension } n\right\}.$$

so that $\ell \in U_\ell$. We will show that U_ℓ is open in $G_n(\mathbb{R}^k)$ and that the map $h : p^{-1}(U_\ell) \to U_\ell \times \ell \cong U_\ell \times \mathbb{R}^n$ defined by $h(\ell', v) = (\ell', \pi_\ell(v))$ is a local trivialization of $E_n(\mathbb{R}^k)$.

For U_ℓ to be open is equivalent to its preimage in $V_n(\mathbb{R}^k)$ being open. This preimage consists of orthonormal frames v_i, $1 \leq i \leq n$, such that $\pi_\ell(v_1), \dots, \pi_\ell(v_n)$ are independent. Let A be the matrix of π_ℓ with respect to the standard basis in the domain $\mathbb{R}^k$ and any fixed basis in the range ℓ. The condition on v_i is then that the $n \times n$ matrix with columns Av_i has nonzero determinant. Since the value of this determinant is a continuous function of v_i, it follows that the frames v_i yielding a nonzero determinant form an open set in $V_n(\mathbb{R}^k)$.

Then it's clear that h is a bijection which is a linear isomorphism on each fiber. Furthermore h is continuous since its two coordinate functions are continuous. To see that h^{-1} is continuous, for $\ell' \in U_\ell$ there is a unique invertible linear map $L_{\ell'} : \mathbb{R}^k \to \mathbb{R}^k$ restricting to π_ℓ on ℓ' and the identity on $\ell^\perp = \text{Ker } \pi_\ell$. Then $L_{\ell'}$, regarded as a $k \times k$ matrix, depends continuously on ℓ'. Indeed, we can write L_ℓ as a product AB^{-1} where:

- B sends the standard basis to $v_1, \dots, v_n, v_{n+1}, \dots, v_k$ with $v_1, \dots, v_n$ an orthonormal basis for ℓ' and $v_{n+1}, \dots, v_k$ a fixed basis for $\ell^\perp$.
- A sends the standard basis to $\pi_\ell(v_1), \dots, \pi_\ell(v_n), v_{n+1}, \dots, v_k$.

Both A and B depend continuously on $v_1, \dots, v_n$. Since matrix multiplication and matrix inversion are continuous operations, it follows that the product $L_{\ell'} = AB^{-1}$ depends continuously on $v_1, \dots, v_n$. But since $L_{\ell'}$ depends only on ℓ', not on the basis $v_1, \dots, v_n$ for ℓ', it follows that $L_{\ell'}$ depends continuously on ℓ' since $G_n(\mathbb{R}^k)$ has the quotient topology from $V_n(\mathbb{R}^k)$. We have $h^{-1}(\ell', w) = (\ell', L_{\ell'}^{-1}(w))$ with $L_{\ell'}^{-1}$ depending continuously on ℓ' since matrix inversion is continuous, so h^{-1} is continuous, and so we have a local trivialization, as we sought to show.

Theorem 1.2.13: Classifying Vector Bundles

Let X be a paracompact space. Then $\text{Vect}^n(X)$ is in correspondence with the set of homotopy classes of maps $X \to \text{Grass}_n$, denoted $[X, \text{Grass}_n]$. In other words:

$$\text{Vect}^n(X) \cong [X, \text{Grass}_n]$$

In particular, if $E \to X$ is an n-dimensional vector bundle, it is isomorphic to $f^*(E_n)$ for some $f : X \to \text{Grass}_n$ where E determines f up to homotopy.

***Proof*:**
We shall prove it for the real case, the complex, quaternion, and orientable real case follow similarly. To shorten notation, let G_n denote the real Grassmannian for n-dimensional subspaces.

The key observation is the following: For an n-dimensional vector bundle $p : E \to X$, an isomorphism $E \cong f^*(E_n)$ is equivalent to a map $g : E \to \mathbb{R}^\infty$ that is a linear injection on each fiber. To see this, suppose that we have a map $f : X \to G_n$ and an isomorphism $E \cong f^*(E_n)$.

Then we have a commutative diagram

$$\begin{tikzcd} E \arrow[r, "\cong"] \arrow[dr, "p"'] & f^*(E_n) \arrow[r, "\widetilde{f}"] \arrow[d] & E_n \arrow[r, "\pi"] \arrow[d] & \mathbb{R}^\infty \\ & X \arrow[r, "f"] & G_n & \end{tikzcd}$$

where $\pi(\ell, v) = v$. The composition across the top row is a map $g : E \to \mathbb{R}^\infty$ that is a linear injection on each fiber, since both $\tilde{f}$ and π have this property. Conversely, given a map $g : E \to \mathbb{R}^\infty$ that is a linear injection on each fiber, define $f : X \to G_n$ by letting $f(x)$ be the n-plane $g(p^{-1}(x))$. This clearly yields the above commutative diagram.

To show surjectivity of the map $[X, G_n] \to \text{Vect}^n(X)$, suppose $p : E \to X$ is an n-dimensional vector bundle. Let $\{U_\alpha\}$ be an open cover of X such that E is trivial over each U_α. As X is paracompact, there a countable open cover $\{U_i\}$ of X such that E is trivial over each U_i, and there is a partition of unity $\{\varphi_i\}$ with φ_i supported in U_i. Let $g_i : p^{-1}(U_i) \to \mathbb{R}^n$ be the composition of a trivialization $p^{-1}(U_i) \to U_i \times \mathbb{R}^n$ with projection onto $\mathbb{R}^n$. The map $(\varphi_i p)g_i$, $v \mapsto \varphi_i(p(v))g_i(v)$, extends to a map $E \to \mathbb{R}^n$ that is zero outside $p^{-1}(U_i)$. Near each point of X only finitely many φ_i's are nonzero, and at least one φ_i is nonzero, so these extended $(\varphi_i p)g_i$'s are the coordinates of a map $g : E \to (\mathbb{R}^n)^\infty = \mathbb{R}^\infty$ that is a linear injection on each fiber.

For injectivity, if we have isomorphisms $E \cong f_0^*(E_n)$ and $E \cong f_1^*(E_n)$ for two maps $f_0, f_1 : X \to G_n$, then these give maps $g_0, g_1 : E \to \mathbb{R}^\infty$ that are linear injections on fibers, as in the first paragraph of the proof. Then g_0 and g_1 are homotopic through maps g_t that are linear injections on fibers. If this is so, then f_0 and f_1 will be homotopic via $f_t(x) = g_t(p^{-1}(x))$.

To show this, a homotopy g_t is constructed. The first step is to compose g_0 with the homotopy $L_t : \mathbb{R}^\infty \to \mathbb{R}^\infty$ defined by $L_t(x_1, x_2, \ldots) = (1-t)(x_1, x_2, \ldots) + t(x_1, 0, x_2, 0, \ldots)$. For each t this is a linear map whose kernel is easily computed to be 0, so L_t is injective. Composing the homotopy L_t with g_0 moves the image of g_0 into the odd-numbered coordinates. Similarly we can homotope g_1 into the even-numbered coordinates. Still calling the new g's g_0 and g_1, let $g_t = (1-t)g_0 + tg_1$. This is linear and injective on fibers for each t since g_0 and g_1 are linear and injective on fibers.

Definition 1.2.14: Universal Bundle

The Grassmannians Grass_n are called the *universal bundles*; for $\mathbb{R}, \mathbb{C}, \mathbb{H}$ they are often denoted BO_n, BU_n,and BSp_n respectively.

As we saw we can put a natural structure group on vector bundles (i.e. O_n, U_n or Sp_n depending on whether the vector bundle is a real, complex, or quaternion bundle respectively). Hence, the infinite Grassmannian is called "classifying space" of O_n, U_n or Sp_n. For this reason, the infinite Grassmannian is denoted BO_n, BU_n, or BSp_n respectively.

1.2.3 *Cell-Structure on Grassmannian

HERE

2

K-Theory for Vector Bundles

Would go over the basics, Bott-Periodicity, and some more stuff

2.1 K0-Group

fix later

Let us now take a moment to examine the K-theory on the spaces explored in chapter 1. We shall build-up to an interesting result known as *Bot Periodicity.* First, we can define the homotopy groups of U and O by:

$$\pi_k(U) = \pi_k(\varinjlim_n U(n)) \qquad \pi_k(O) = \pi_k(\varinjlim_n U(O))$$

which is well-defined as the fibrations

$$U(n-1) \hookrightarrow U(n) \to S^{2n-1} \qquad O(n-1) \hookrightarrow O(n) \to S^{n-1}$$

naturally induces the inclusion $\pi_k(U(n-1)) \to \pi_k(U(n))$ and $\pi_k(O(n-1)) \to \pi_k(O(n))$. Looking at these groups, we get the following fascinating result:

Theorem 2.1.1: Bott Periodicity

Let U, O be above groups. Then the homotopy groups of U, O are:

$k \pmod 8$	0	1	2	3	4	5	6	7
$\pi_k(U)$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$
$\pi_k(O)$	$\mathbb{Z}/2$	$\mathbb{Z}/2$	0	$\mathbb{Z}$	0	0	0	$\mathbb{Z}$

In particular, we see that $\pi_k(U)$ is 2-periodic, and $\pi_k(O)$ is 8-periodic.

***Proof*:**
Delayed till ref:HERE

The details of this proof is beyond the scope for this book, but an outline of the idea shall be presented.

2.1.1 K-groups over different fields

From chapter 1 recall the collection of isomorphism classes of vector bundles on a nonempty[1] paracompact space X is denoted $\mathbb{VB}_{\mathbb{R}}(X)$, $\mathbb{VB}_{\mathbb{C}}(X)$, or $\mathbb{VB}_{\mathbb{H}}(X)$ depending on whether it is for real, complex, or quaternion vector bundles. We shall focus on the real and complex case. Certainly under Whitney sums the real and complex vector bundles form an abelian monoid. As we can define the tensor product (the point-wise definition extends naturally, see the beginning of chapter 1 or [9]). Hence, we can define the Grothendeick completions of $\mathbb{VB}_{\mathbb{R}}(X)$ and $\mathbb{VB}_{\mathbb{C}}(X)$ and get:

$$KO(X) \qquad KU(X)$$

give commutative rings with identity being the trivial line bundles. If it is clear over which field we are, we may write $K(X)$. We have omitted $KSp(X)$ as $\mathbb{H}^m \otimes_{\mathbb{H}} \mathbb{H}^n \ncong \mathbb{H}^{mn}$. However,

$$\mathbb{H}^m \otimes_{\mathbb{R}} \mathbb{R}^n \cong \mathbb{H}^{mn}$$

and so it has a natural $\mathbb{R}$-module structure.

Some immediate results we can get form algebraic topology

1. if $X = \{*\}$ is the one point space (or more generally a contractible space as vector bundle isomorphism classes are defined up to homotopy), then we are working with the usual K-theory and so:
$$KO(*) = KU(*) = \mathbb{Z}$$
2. If $f : X \to Y$ is continuous, the induced line bundle map $E \mapsto f^*(E)$ gives a morphism of monoids and semi-rings
$$f^* : \mathbf{VB}(Y) \to \mathbf{VB}(X)$$
which then induces
$$f^* : K(Y) \to K(X)$$
Thus, as there is a universal map $f : X \to *$, there is a universal map from $K(*) = \mathbb{Z} \to K(X)$. As we are assuming $X \neq \emptyset$, we may also define a map $* \to X$ which can be compose with the above universal map and which splits, and hence we further get that $\mathbb{Z}$ is in fact a direct summand of $K(X)$
3. The direct sum of trivial bundles will form a cofinal object when X is compact (namely since any map $X \to \mathbb{N}$ can only take finitely many values as the has to be finitely many components, and the subbundle of a bundle always has an orthogonal compliment and so can be decomposed into 1-dimensional trivial bundles, but then $[X, \mathbb{N}]$ and $\mathbb{N}$ would both be cofinal). From this, we can just like in section 6.3 have the subrings:
$$\mathbb{Z} \subseteq [X, \mathbb{Z}] \subseteq K(X)$$

[1] If it is empty, the vector bundle is always trivial, and it will be a bit of a headache to mention it constantly

and the "splitting" map:
$$\dim : K(X) \to [X, \mathbb{Z}]$$
which gives us:
$$K(X) \cong \tilde{K}(X) \oplus [X, \mathbb{Z}]$$
where $\tilde{K}(X)$ is the kernel of the dimension map.

Let us now take a look at the classifying spaces BO and BU and their K_0-group.

Theorem 2.1.2: Stabilization Theorem of Classifying Spaces

Let X be either a compact space or a finite-dimensional connected CW complex. Then there is an isomorphism $\widetilde{K}(X) \cong \varinjlim \mathbf{VB}(X) \cong \varinjlim [X, \mathrm{Grass}_n]$. In particular,
$$\widetilde{KO}(X) \cong \varinjlim [X, BO_n], \quad \widetilde{KU}(X) \cong \varinjlim [X, BU_n],$$
and
$$\widetilde{KSp}(X) \cong \varinjlim [X, BSp_n].$$

***Proof*:**
Since the monoid of (componentwise) trivial vector bundles is cofinal in $\mathbf{VB}(X)$, we shall show that every element of $\widetilde{K}(X)$ is represented by an element
$$[E] - n$$
of some $\mathbf{VB}_n(X)$ by lemma 6.1.3.Now, if $[E] - n = [F] - n$, then $E \oplus T^n \cong F \oplus T^n$ in some $\mathbf{VB}_{n+\ell}(X)$. But that E is stably free for large enough n, and so we must have
$$\widetilde{K}(X) \cong \varinjlim \mathbf{VB}_n(X)$$
As this is the case for each O, U, Sp, we get our desired result.

2.1.2 Bott Periodicity

Let us return to the setting of paracompact spaces X wit structure groups O_n, U_n, or SP_n respectively. The vector bundles are given by the classifying spaces BO, BU and BSp which can be realised as the natural vector bundle on the limit of Grammarians $\mathrm{Grass}_n(V)$ for finite dimensional V of $\mathbb{R}^\infty$, $\mathbb{C}^\infty$ or $\mathbb{H}^\infty$ (depending on the type of vector bundle). Then Bott periodicity can be sated as:
$$BU \times \mathbb{Z} \cong \Omega^2(BU \times \mathbb{Z}) \qquad BO \times \mathbb{Z} \cong \Omega^8(BO \times \mathbb{Z})$$
where Ω^n is the n-dimensional loop space (i.e. The space not up to homotopy). In Bott's original proof, he uses spaces known as "symmetric spaces" which are compact Riemann manifolds such that for every point $p \in M$, there is an isometry ι_p that fixes p and reverses the geodesics through p (hence a symmetric space). The sphere gives rise to such a space, and by taking homotopies we get U, O, Sp. Here, Bott looks at a new spaces M^ν where $\nu = (P, Q, h)$ is a triple of two points in

M and h a homotopy class of curves connecting P, Q, and M^ν is the set of all geodesics of minimal length which join P and Q that are in h. Then he showed that:

$$\pi_k(M^\nu) = \pi_{k+1}(M)$$

For appropriate dimension considerations (see Bott's paper for more details). From this, he managed conclude:

$$\pi_k(U) \cong \pi_{k+1}(BU) \cong \pi_{k+2}(U)$$

and by manipulating U, O, and Sp, we also get:

$$\pi_k(O) \cong \pi_{k+1}(BO) \cong \pi_{k+2}(U/O) \cong \pi_{k+3}(Sp/U) \cong \pi_{k+4}(Sp)$$
$$\pi_{k+5}(BSp) \cong \pi_{k+6}(U/Sp) \cong \pi_{k+7}(O/U) \cong \pi_{k+8}(O)$$

which would give our desired periodicity! The details are beyond this book, but it should be enlightening to see how this theory results are rather simple to calculate consider how in general it is very difficult to calculate homotopy groups, for example here are the known homotopy groups for spheres:

image lost

In fact, we can see the results of Bott periodicity in this graph with the stability of the results for as n get's large for $\pi_{n+k}(S^n)$ where $k \geq 2$, and how this result it much harder for the rest of the cases!

3

Characteristic Classes

In this chapter, we shall explore the relation between vector bundles and cohomology rings, in particular there shall be many vector bundles that correspond to certain generators of cohomology rings, sometimes even generating them. The images of these vector bundles into cohomology rings shall be called *characteristic classes*, or sometimes more rarely *cohomology classes*. As cohomology rings represent some non trivial twisting, so do vector bundles. This connection between "twisting" or some obstruction to triviality shall be made precise in this section by showing how many linearly independent sections can exist for certain characteristic classes.

This section shall cover 4 important characteristic classes;

1. Stiefel-Whitney classes: $w_i(E) \in H^i(B; \mathbb{Z}/2\mathbb{Z})$ for real vector bundles $E \to B$
2. Chern classes: $c_i(E) \in H^{2i}(B; \mathbb{Z})$ for complex vector bundles $E \to B$
3. Pontryagin classes: $p_i(E) \in H^{4i}(B; \mathbb{Z})$ for oriented real vector bundles $E \to B$
4. Euler class: $e(E) \in H^n(B; \mathbb{Z})$ for an oriented n-dimensional real vector bundle $E \to B$ (note there is only one)

We shall explore each of these characteristic classes throughout this chapter. The idea behind all of these classes is to measure how far away a vector bundle $E \to B$ is from being trivial. By theorem 1.2.5, if $f : B \to \mathrm{Grass}_n$ is nullhomotopic, $f : E \to B$ is trivial. Measuring the failure of this map being nullhomotopic would then use the tools of homotopy and cohomology theory.

3.1 Definition and Intuitions

As (co)homology classes can be defined over general groups $H^i(X; G)$, it is worth-while being a little more general in our definition:

Definition 3.1.1: Characteristic Classes

Let G be a topological group, and $b_G(-)$ be the contravariant functor associating to each topological space$B$ the set (sometimes group, sometimes ring)[a] $b_G(B)$ and f to the pullback f^*. Then a *characteristic class* is a natural transformation

$$c : b_G \Rightarrow H^*$$

In particular, a characteristic class c associated to each principal G-bundle $P \to X$ to an element $c(P) \in H^*(X)$ where

$$c(f^*P) = f^*c(P)$$

where the f^* on the left hand side is the pullback map and the f^* on the right hand side is the image of the class of P (i.e. $c(P)$) by the induced map in cohomology. For a specific H^i, we shall denote the characteristic class as c_i.

[a] depending on the choice of G

For the remainder of this section, we shall focus on the case of $G = V$ is a vector space, either real or complex, and stick to the convention that $\text{Vect}^n(B)$ means $\text{Vect}^n_{\mathbb{R}}(B)$. Then we have that $\text{Vect}(B)$ is associated to $[B, \text{Grass}_n]$,In particular, if $f : B \to \text{Grass}_n$ is a classifying map and $E = f^*(E_n)$, then

$$c_i(E) = c_i(f^*(E_n)) = f^*(c_i(E_n))$$

Thus, if f induces the trivial map on G-cohomology, then the characteristic classes $c_i(E)$ are trivial.

Let us now thing of the specific classes:

Definition 3.1.2: Stiefel-Whitney Classes

If $G = \mathbb{Z}/2\mathbb{Z}$ and $b_G(B) = \text{Vect}^n(B)$ for some $n \in \mathbb{N}_{>0}$, then the characteristic class is called the *Stiefel-Whitney Class*, and will be denoted $w(E)$ (or $w_i(E)$ for the $H^i(B; \mathbb{Z}/2\mathbb{Z})$ cohomology)

These classes shall be of interest to us as they shall generate the cohomology ring $H^*(\text{Grass}_n; \mathbb{Z}/2\mathbb{Z})$, in case:

$$H^*(\text{Grass}_n; \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}_2[w_1(E_n), ..., w_n(E_n)]$$

I fact we shall prove in ref:HERE. This is not the case if $\mathbb{Z}$-coefficients are used, we will need the Pontryagin class to characterize these:

Definition 3.1.3: Pontryagin Classes

If $G = \mathbb{Z}$ and $b_G(B) = \text{Vect}^n(B)$ for some $n \in \mathbb{N}_{>0}$, then the characteristic class is called the *Pontryagin-Whitney Class*, and will be denoted $w(E)$ (or $w_i(E)$ for the $H^{4i}(B; \mathbb{Z})$ cohomology)

As hinted in the above, the values of interest are in H^{4i} as these are the only classes with nonzero values. We shall show that f induces the trivial map in $\mathbb{Z}$ coefficient if the Stiefel-Whitney and Pontryagin classes of E are all zero. If E is orientable, then we can take $f : E \to \widetilde{\text{Grass}}_n$ where $\widetilde{\text{Grass}}_n$ has the orientation structure defined earlier. Then we get:

Definition 3.1.4: Euler Class

If $G = \mathbb{Z}$ and $b_G(B) = \widetilde{\mathrm{Vect}^n}(B)$ of orientable real vector bundles for some $n \in \mathbb{N}_{>0}$, then the characteristic class is called the *Euler Class*, and will be denoted $e(E)$.

We shall show that f induces the trivial map for an oriented real vector bundle if the Stiefel-Whitney, Pontryagin, and Euler classes are trivial.

Finally, as the complex Grassmannian has a different structure do to how eliminating 0 from $\mathbb{C}$ leaves a path-connected component, we shall have the following:

Definition 3.1.5: Chern Class

If $G = \mathbb{Z}$ and $b_G(B) = \mathrm{Vect}^n_{\mathbb{C}}(B)$ of complex bundles for some $n \in \mathbb{N}_{>0}$, then the characteristic class is called the *Chern Class*, and will be denoted $c(E)$ (or $c_i(E)$ for the $H^{2i}(B;\mathbb{Z})$ cohomology)

We shall show that

$$H^*(\mathrm{Grass}_n(\mathbb{C});\mathbb{Z}) \cong \mathbb{Z}[c_1, ..., c_n]$$

where $c_i = c_i(\mathrm{Grass}_n(\mathbb{C}))$

Note how in many of the above we required to use many types of characteristic classes to determine triviality. This is because trivial characteristic classes are necessary but not sufficient conditions to determine triviality. The simplest non-trivial example if S^5, which we established in section 2.1.2 is non-trivial, but it shall have trivial characteristic classes for all but the Euler class. This is because the first three characteristic classes have the property of being *stably invariant* meaning that direct sums with trivial bundles does not change the characteristic class. As we stall, all sphere are stably trivial, meaning they direct sum to the trivial bundle, and hence the first 3 classes cannot distinguish odd dimensional spheres.

Let us now think about for a moment what characteristic classes are detecting. The first “obstruction” it detects is orientability. If $E \to B$ is a vector bundle where B is path-connected, then we may define a map:

$$\pi_1(B) \to \mathbb{Z}/2\mathbb{Z}$$

where it assigns 0 or 1 to each loop according to whether orientation of fibers is preserved or reversed (as one goes around the loop). As $\mathbb{Z}/2\mathbb{Z}$ is abelian, the homomorphism factors through the abelianization $H_1(B)$ of $\pi_1(B)$, and the homomorphisms $H_1(B) \to \mathbb{Z}/2\mathbb{Z}$ are elements of $H^1(B;\mathbb{Z}/2\mathbb{Z})$. From this,we have elements of $H^1(B;\mathbb{Z}/2\mathbb{Z})$ associated to E which when E is orientable. This shall be the interpretation of the first Stiefel-Whitney class $w_1(E)$.

To generalize to higher Stiefel-Whitney classes, let us re-interpret this result when B is a CW complex. Then E is orientable if and only if the restriction of the 1-skeleton B^1 of B is orientable, as all lops in B can be deformed to lie in B^1. Then a vector bundle over a 1-dimensional CW complex is trivial if and only if it is orientable, so we see that $w_1(E)$ measures whether E is trivial over the 1-skeleton B^1. Notice that the fibers can be higher dimensional, namely on each loop we are looking at vector bundles on S^1. Now, we can ask the following: given E is trivial over the 1-skeleton, can we extend it to be trivial over the 2-skeleton. Assuming E is trivial over B^1 and its fibers are n-dimensional, we can choose n orthonormal sections over B^1 and ask whether these sections extend to orthonormal sections over each 2-cell. As each two cell is given by the characteristic map $D^2 \to B$, we can pull back E over the disk D^2, where the sections over B^1 pullback to sections over ∂D^2. As D^2 is contractible, this bundle is trivial. Choosing a trivialization, the sections over ∂D^2 determine a map:

$$\partial D^2 \to O(n)$$

Choosing a trivialization that has the same orientation as the trivialization of E over B^1, we have the image is in $SO(n)$. Such a map is exactly an element of $\pi_1(SO(n))$, hence we have that each 2-cell is identified with an element of $\pi_1(SO(n))$. Then B^1 extends to orthonormal sections over B^2 if and only if this element of $\pi_1(SO(n))$ is zero for each 2-cell.

This groups is also relatively easy to compute : when $n = 1$ it is 0, when $n = 2$ it is $\mathbb{Z}$ (as $SO(2)$ is homeomorphic to S^1). For $n \geq 3$, it is $\mathbb{Z}/2\mathbb{Z}$ (exercise). Thus, the inclusion $SO(2) \hookrightarrow SO(n)$ induces a surjection $\pi_1(SO(2)) \twoheadrightarrow \pi_1(SO(n))$, so the generator of $\pi_1(SO(n))$ can be represented by the loop given by rotating the plane through 2π degrees, fixing the orthogonal compliment of the plane. From this, we can define a function which assigns each 2-cell (regarded as a 2-dimensional cellular cochain) with coefficients in $\pi - 1(SO(n))$. This shall be a cellular cocycle, and so an element of $H^2(B; \pi_1(SO(n)))$. This element shall be called $w_2(E)$, and will not depend on the choice of sections over B^1. We can thus see that E is trivial over B^2 if and only if $w_2(E)$ is zero in $H^2(B; \pi_1(SO(n)))$. Note that if $n = 1$, clearly $w_2(E) = 0$. If $n = 2$, then $w_2(E) \in H^2(B; \mathbb{Z})$ and is equal to the Euler class $e(E)$ which shall be covered alter. If $n \geq 3$, then $w_2(E)$ is in $H^2(B; \mathbb{Z}/2\mathbb{Z})$.

The natural next question to ask if if E is trivial over B^2, can it be extended to be trivial over B^3 which again is given by looking at elements of $\pi_2(SO(n))$. Fortunately for us, $\pi_2(SO(n)) = 0$ for all $n \in \mathbb{N}_{>0}$ and so the sections automatically extend over B^3. For extended to 4-cells, we have that $\pi_3(SO(n))$ is relatively simple to compute (see [15, p.75]). For higher dimensional extension B^k for $k \geq 5$, things become a good deal more complicated as the groups $\pi_{k-1}(SO(n))$ become increasingly difficult to compute. We know they are finitely generated abelian groups whose free part is fairly easy to compute, but the torsion part is complicated[1]. In the range of $n > k$, then Bott periodicity gives the value of $\pi_{k-1}(SO(n))$. For $k = 4i$, we get:

$$\pi_{4i-1}(SO(n)) \cong \mathbb{Z}$$

which is why the Pontryagin class $p_i(E) \in H^{4i}(B; \mathbb{Z})$, and from the above we see it detects the non-triviality of bundles with obstruction:

$$w_{4i}(E) \in H^{4i}(B; \pi_{4i-1}(SO(n)))$$

Given the dimension $n > 4i$, we get that the other nontrivial stable groups are:

$$\pi_{k-1}(SO(n)) \cong \mathbb{Z}/2\mathbb{Z} \qquad k = 8i + 1, 8i + 2$$

However, the obstructions $w_k(E)$ in these cases shall not be detected by Stiefel-Whitney or Pontryagin classes except when $i = 0$. (see ref:HERE).

Another we of measuring triviality will lead to the Euler class. Consider an n-dimensional vector bundle $E \to B$. Let's say we are looking for an obstruction to the existence of n orthonormal sections. Let us first consider the case where we are looking for an obstruction of there being 1 orthonormal section, or more particularly a section of unit vectors. As before we can proceed by induction over the skeleta of B. Assuming we have section over B^{k-1}, to extend over the k-cell we can pull back E

[1] Interestingly, this contrasts with elliptic curves over rationals $E(\mathbb{Q})$ for which it is relatively easy to compute the torsion but a million dollar problem (literally) to compute the free part

as before to the trivial bundle over D^k via the characteristic map so that we are look at the unit vector section ∂D^k. But this time we were not looking for orientability, this is *exactly* the what we tried to accomplish. Hence, if $k < n$, such an extension always is possible, so the first (possibly) nontrivial obstruction occurs when $k = n$ where

$$\pi_{k-1}(S^{n-1}) \cong \mathbb{Z}$$

Note that to identify $\mathbb{Z}$ to $\pi_{n-1}(S^{n-1})$ in a consistent way, it is *necessary* be that E is orientable. Thus class is the Euler class and shall be denoted $e(E)$. Dropping the orientability condition, we must work with $\mathbb{Z}/2\mathbb{Z}$ coefficients and we get the Stiefel-Whitney class $w_n(E) \in H^n(B; \mathbb{Z}/2\mathbb{Z})$.

3.2 Stiefel-Whitney and Chern Classes

The intuition presented above is computationally difficult. The approach we shall take here will be require more algebraic topology, but will be equivalent (see 3.4 to for the comparisons). As usual, we shall assume the base space is paracompact. The following two results will be needed for the proof of the existence of the characteristic classes:

Theorem 3.2.1: Leray-Hirsch Theorem

Let R be a commutative ring and $p : E \to B$ be a fiber bundle. In [9] it was shown $H^*(E; R)$ is a $H^*(B; R)$-module where

$$\alpha\beta = p^*(\alpha) \smile \beta$$

with $\alpha \in H^*(B)$ and $\beta \in H^*(E)$. If

1. for each fiber F the inclusion $F \hookrightarrow E$ induces a surjection on $H^*(-; R)$
2. $H^n(F; R)$ is a free R-module of finite rank for each n.

Then $H^*(E; R)$ is a free $H^*(B; R)$-module. A basis for the $H^*(B; R)$-module $H^*(E; R)$ can be obtained by choosing any set of elements in $H^*(E; R)$ that map to a basis for $H^*(F; R)$ under the map induced by the inclusion.

***Proof*:**
see [9].

Proposition 3.2.2: Splitting Principle

Let $\pi : E \to B$ be a vector bundle. Then there exists a space $F(E)$ and a map $p : F(E) \to B$ such that the pullback $p^*(E) \to F(E)$ splits as a direct sum of line bundles, and

$$p^* : H^*(B; \mathbb{Z}/2\mathbb{Z}) \hookrightarrow H^*(F(E); \mathbb{Z}/2\mathbb{Z})$$

is injective

***Proof*:**
Consider the pullback $P(\pi)^*(E)$ of E via the map $P(\pi) : P(E) \to B$ where $P(E)$ is the space of lines through the origin in all fibers of E. This pullback contains a natural one-dimensional subbundle

$$L = \{(\ell, v) \in P(E) \times E \mid v \in \ell\}$$

An inner product on E pulls back to an inner product on the pullback bundle, and thus a splitting of the pullback as a sum $L \oplus L^\perp$ with the orthogonal bundle $L^\perp$ having dimension one less than E. Then by the Leray-Hirsch theorem to $P(E) \to B$, $H^*(P(E); \mathbb{Z}/2\mathbb{Z})$ is the free $H^*(B; \mathbb{Z}/2\mathbb{Z})$-module with basis $1, x, \dots, x^{n-1}$ and in particular the induced map $H^*(B; \mathbb{Z}/2\mathbb{Z}) \to H^*(P(E); \mathbb{Z}/2\mathbb{Z})$ is injective since one of the basis elements is 1.

This construction can be repeated with $L^\perp \to P(E)$ in place of $E \to B$, until we get the splitting, completing the proof.

Theorem 3.2.3: Existence of Stiefel-Whitney Classes

Let $E \to B$ be a real vector bundle. Then there exists a sequence of functions $w_1, w_2, \ldots$ where $w_i(E) \in H^i(B; \mathbb{Z}/2\mathbb{Z})$ that depends only on the isomorphism type of E such that

1. $w_i(f^*(E) = f^*(w_i(E))$ for a pullback $f^*(E)$
2. $w(E_1 \oplus E_2) = w(E_1) \smile w(E_2)$ where $w = 1 + w_1 + w_2 + \cdots \in H^*(B; \mathbb{Z}/2\mathbb{Z})$
3. $w_i(E) = 0$ if $i > \dim E$
4. Given $E \to \mathbb{R}\mathrm{P}^\infty$ is the canonical line bundle, $w_1(E)$ is a generator of $H^1(\mathbb{R}\mathrm{P}^\infty; \mathbb{Z}/2\mathbb{Z})$.

The Stiefel-Whitney class w is called the *total Stiefel-Whitney class.*

Note that by (3), w only has finitely many nonzero terms, showing it is well-defined in $H^i(B; \mathbb{Z}/2\mathbb{Z})$. The $\oplus$ to $\smile$ is shorthand for the Whitney sum formula:

$$w_n(E_1 \oplus E_2) = \sum_{i+j=n} w_i(E_1) \smile E_j(E_2)$$

where $w_0 = 1$. Property (4) is a non triviality condition, as we can just say $w_i(E) = 0$ for all $i > 0$.

As proving the result for Chern classes is ostensibly the same, we shall in fact do both:

Theorem 3.2.4: Existence of Chern Classes

Let $E \to B$ be a complex vector bundle. Then there exists a sequence of functions $c_1, c_2, \ldots$ where $c_i(E) \in H^{2i}(B; \mathbb{Z})$ that depends only on the isomorphism type of E such that

1. $c_i(f^*(E) = f^*(c_i(E))$ for a pullback $f^*(E)$
2. $c(E_1 \oplus E_2) = c(E_1) \smile c(E_2)$ where $c = 1 + c_1 + c_2 + \cdots \in H^*(B; \mathbb{Z})$
3. $c_i(E) = 0$ if $i > \dim E$
4. Given $E \to \mathbb{C}\mathrm{P}^\infty$ is the canonical line bundle, $c_1(E)$ is a generator of $H^1(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z})$ specified in advance.

The extra condition for (4) is a non triviality *and* a normalization condition , as each function c_i can be replaced by $k^i c_i$ for a fixed integer k which gives new functions satisfying (1)-(3).

***Proof*:**
Associated to a vector bundle $\pi : E \to B$ with fiber $\mathbb{R}^n$ is the projective bundle $P(\pi) : P(E) \to B$, where $P(E)$ is the space of all lines through the origin in all the fibers of E, and $P(\pi)$ is the natural projection sending each line in $\pi^{-1}(b)$ to $b \in B$. Topologize $P(E)$ as a quotient of the complement of the zero section of E, that is the quotient obtained by factoring out scalar multiplication in each fiber. Over a neighborhood $U \subseteq B$ where E is a product $U \times \mathbb{R}^n$, the aforementioned quotient is $U \times \mathbb{RP}^{n-1}$, so $P(E)$ is a fiber bundle over B with fiber $\mathbb{RP}^{n-1}$.

To construct the classes, we use Leray-Hirsch theorem (theorem 3.2.1) for cohomology with $\mathbb{Z}/2\mathbb{Z}$ coefficients to this bundle $P(E) \to B$. To that end, there must be classes $x_i \in H^i(P(E);\mathbb{Z}/2\mathbb{Z})$ restricting to generators of $H^i(\mathbb{RP}^{n-1};\mathbb{Z}/2\mathbb{Z})$ in each fiber $\mathbb{RP}^{n-1}$ for $i = 0, \dots, n-1$. Recall from the proof of theorem 1.2.13 that there is a map $g : E \to \mathbb{R}^\infty$ that is a linear injection on each fiber. Projectivizing the map g by deleting zero vectors and factoring out scalar multiplication produces a map

$$P(g) : P(E) \to \mathbb{RP}^\infty$$

Let α be a generator of $H^1(\mathbb{RP}^\infty;\mathbb{Z}/2\mathbb{Z})$ and let $x = P(g)^*(\alpha) \in H^1(P(E);\mathbb{Z}/2\mathbb{Z})$. Then the powers x^i for $i = 0, \dots, n-1$ are the desired classes x_i since a linear injection $\mathbb{R}^n \to \mathbb{R}^\infty$ induces an embedding $\mathbb{RP}^{n-1} \hookrightarrow \mathbb{RP}^\infty$ for which α pulls back to a generator of $H^1(\mathbb{RP}^{n-1};\mathbb{Z}/2\mathbb{Z})$, hence x^i pulls back to a generator of $H^i(\mathbb{RP}^{n-1};\mathbb{Z}/2\mathbb{Z})$. Note that any two linear injections $\mathbb{R}^n \to \mathbb{R}^\infty$ are homotopic, so the induced embeddings $\mathbb{RP}^{n-1} \hookrightarrow \mathbb{RP}^\infty$ of different fibers of $P(E)$ are all homotopic. Again by theorem 1.2.13, any two choices of g are homotopic through maps that are linear injections on fibers, so the classes x^i are independent of the choice of g.

Then Leray-Hirsch theorem (theorem 3.2.1) says that $H^*(P(E);\mathbb{Z}/2\mathbb{Z})$ is a free $H^*(B;\mathbb{Z}/2\mathbb{Z})$-module with basis $1, x, \dots, x^{n-1}$. Consequently, x^n can be expressed uniquely as a linear combination of these basis elements with coefficients in $H^*(B;\mathbb{Z}/2\mathbb{Z})$. Thus there is a unique relation of the form

$$x^n + w_1(E)x^{n-1} + \cdots + w_n(E) \cdot 1 = 0$$

for certain classes $w_i(E) \in H^i(B;\mathbb{Z}/2\mathbb{Z})$ where $w_i(E)x^i$ means $P(\pi)^*(w_i(E)) \smile x^i$, by the definition of the $H^*(B;\mathbb{Z}/2\mathbb{Z})$-module structure on $H^*(P(E);\mathbb{Z}/2\mathbb{Z})$. For completeness, define $w_i(E) = 0$ for $i > n$ and $w_0(E) = 1$. We must show these satisfy the desired properties.

For property (1), consider a pullback $f^*(E) = E'$, fitting into the diagram

$$\begin{aligned} P(\tilde{f})^*(\sum_i P(\pi)^*(w_i(E)) \smile x(E)^{n-i}) &= \sum_i P(\tilde{f})^* P(\pi)^*(w_i(E)) \smile P(\tilde{f})^*(x(E)^{n-i}) \\ &= \sum_i P(\pi')^* f^*(w_i(E)) \smile x(E')^{n-i} \end{aligned}$$

If $g : E \to \mathbb{R}^\infty$ is a linear injection on fibers then so is $g\tilde{f}$. It follows that $P(\tilde{f})^*$ takes the canonical class $x = x(E)$ for $P(E)$ to the canonical class $x(E')$ for $P(E')$. Then the relation $x(E)^n + w_1(E)x(E)^{n-1} + \cdots + w_n(E) \cdot 1 = 0$ defining $w_i(E)$ pulls back to the relation $x(E')^n + f^*(w_1(E))x(E')^{n-1} + \cdots + f^*(w_n(E)) \cdot 1 = 0$ defining $w_i(E')$. By the uniqueness of this relation, $w_i(E') = f^*(w_i(E))$.

For property (2), the inclusions of E_1 and E_2 into $E_1 \oplus E_2$ give inclusions of $P(E_1)$ and $P(E_2)$ into $P(E_1 \oplus E_2)$ with $P(E_1) \cap P(E_2) = \emptyset$. Let $U_1 = P(E_1 \oplus E_2) - P(E_1)$ and $U_2 = P(E_1 \oplus E_2) - P(E_2)$.

These are open sets in $P(E_1 \oplus E_2)$ that deformation-retract onto $P(E_2)$ and $P(E_1)$, respectively. A map $g : E_1 \oplus E_2 \to \mathbb{R}^\infty$ which is a linear injection on fibers restricts to such a map on E_1 and E_2, so the canonical class $x \in H^1(P(E_1 \oplus E_2); \mathbb{Z}/2\mathbb{Z})$ for $E_1 \oplus E_2$ restricts to the canonical classes for E_1 and E_2. If E_1 and E_2 have dimensions m and n, consider the classes $\omega_1 = \sum_j w_j(E_1)x^{m-j}$ and $\omega_2 = \sum_j w_j(E_2)x^{n-j}$ in $H^*(P(E_1 \oplus E_2); \mathbb{Z}/2\mathbb{Z})$, with cup product

$$\omega_1\omega_2 = \sum_j \left[\sum_{r+s=j} w_r(E_1)w_s(E_2)\right] x^{m+n-j}$$

By the definition of the classes $w_j(E_1)$, the class ω_1 restricts to zero in $H^m(P(E_1); \mathbb{Z}/2\mathbb{Z})$, hence ω_1 pulls back to a class in the relative group

$$H^m(P(E_1 \oplus E_2), P(E_1); \mathbb{Z}/2\mathbb{Z}) \cong H^m(P(E_1 \oplus E_2), U_2; \mathbb{Z}/2\mathbb{Z})$$

and similarly for ω_2. The following commutative diagram, with $\mathbb{Z}/2\mathbb{Z}$ coefficients understood, then shows that $\omega_1\omega_2 = 0$:

$$\begin{array}{ccc} H^m(P(E_1 \oplus E_2), U_2) \times H^n(P(E_1 \oplus E_2), U_1) & \xrightarrow{\smile} & H^{m+n}(P(E_1 \oplus E_2), U_1 \cup U_2) = 0 \\ \downarrow & & \downarrow \\ H^m(P(E_1 \oplus E_2)) \times H^n(P(E_1 \oplus E_2)) & \xrightarrow{\smile} & H^{m+n}(P(E_1 \oplus E_2)) \end{array}$$

Thus the defining relation for the Stiefel-Whitney classes of $E_1 \oplus E_2$ is

$$\omega_1\omega_2 = \sum_j \left[\sum_{r+s=j} w_r(E_1)w_s(E_2)\right] x^{m+n-j} = 0$$

and so $w_j(E_1 \oplus E_2) = \sum_{r+s=j} w_r(E_1)w_s(E_2)$.

For property (3), it holds by definition.

For property (4), recall that the canonical line bundle is $E = \{(\ell, v) \in \mathbb{RP}^\infty \times \mathbb{R}^\infty \big| v \in \ell\}$. The map $P(\pi)$ in this case is the identity. The map $g : E \to \mathbb{R}^\infty$ which is a linear injection on fibers can be taken to be $g(\ell, v) = v$. So $P(g)$ is also the identity, hence $x(E)$ is a generator of $H^1(\mathbb{RP}^\infty; \mathbb{Z}/2\mathbb{Z})$. The defining relation $x(E) + w_1(E) \cdot 1 = 0$ then says that $w_1(E)$ is a generator of $H^1(\mathbb{RP}^\infty; \mathbb{Z}/2\mathbb{Z})$.

The proof of uniqueness of the classes w_i comes from the *splitting principle*. Looking at the construction in proposition 3.2.2, $L^\perp$ consists of pairs $(\ell, v) \in P(E) \times E$ with $v \perp \ell$. At the next stage we form $P(L^\perp)$, whose points are pairs (ℓ, ℓ') where ℓ and ℓ' are orthogonal lines in E. Continuing in this way, we see that the final base space $F(E)$ is the space of all orthogonal splittings $\ell_1 \oplus \cdots \oplus \ell_n$ of fibers of E as sums of lines, and the vector bundle over $F(E)$ consists of all n-tuples of vectors in these lines. Alternatively, $F(E)$ can be described as the space of all chains $V_1 \subset \cdots \subset V_n$ of linear subspaces of fibers of E with $\dim V_i = i$. Such chains are called flags, and $F(E) \to B$ is the flag bundle associated to E. Note that the description of points of $F(E)$ as flags does not depend on a choice of inner product in E.

Finally, Property (4) determines $w_1(E)$ for the canonical line bundle $E \to \mathbb{RP}^\infty$. Property (3) then determines all the w_i's for this bundle. Since the canonical line bundle is the universal line bundle, property (1) therefore determines the classes w_i for all line bundles. Property (2) extends this to sums of line bundles, and finally the splitting principle implies that the w_i's are determined for all bundles.

For complex vector bundles we can use the same proof, but with $\mathbb{Z}$ coefficients since $H^*(\mathbb{CP}^\infty;\mathbb{Z}) \cong \mathbb{Z}[\alpha]$, with α now two-dimensional. The defining relation for the $c_i(E)$'s is modified to be

$$x^n - c_1(E)x^{n-1} + \cdots + (-1)^n c_n(E) \cdot 1 = 0$$

with alternating signs. This is equivalent to changing the sign of α, so it does not affect the proofs of properties (1)-(3), but it has the advantage that the canonical line bundle $E \to \mathbb{CP}^\infty$ has $c_1(E) = \alpha$ ratheR than $-\alpha$, since the defining relation in this case is $x(E) - c_1(E) \cdot 1 = 0$ and $x(E) = \alpha$, completing the proof.

Note that for property (4), the canonical line bundle over $\mathbb{RP}^1$ could have been used instead of $\mathbb{RP}^\infty$ as $\mathbb{RP}^1 \hookrightarrow \mathbb{RP}^\infty$ induces an isomorphism

$$H^1(\mathbb{RP}^\infty;\mathbb{Z}/2\mathbb{Z}) \cong H^1(\mathbb{RP}^1;\mathbb{Z}/2\mathbb{Z})$$

and similarly for Chern classes.

Example 3.2.5: Stiefel-Whitney and Chern Classes

1. By property (1), the trivial bundle $B \times \mathbb{R}^n$ has

$$w_i(E) = 0 \qquad \forall i > 0$$

as the product is the pullback of the bundle over a point.

2. By property (b), the direct sum of a bundle with a product bundle does not change its Stiefel-Whitney classes; this property is called *stable*. Using this, we see that the tangent bundle TS^n to S^n is stably trivial, and hence

$$w_i(TS^n) = 0 \qquad \forall i > 0$$

More generally, as :

$$(1 + w_1 + \cdots)(1 + w_1' + \cdots) = 1 + (w_1 + w_1') + (w_2 + w_1 w_2' + w_2') + \cdots$$

if we have $w(E_1)$ and $w(E_1 \oplus E_2)$, we can determine $w(E_2)$ as:

$$\begin{aligned} w_1 + w_1' &= a_1 \\ w_2 + w_1 w_1' + w_2' &= a_2 \\ &\vdots \\ \sum_i w_{n-1} w_i' &= a_n \end{aligned}$$

If $E_1 \oplus E_2$ is the trivial bundle, $a_i = 0$ for all $i > 0$, so $w(E_1)$ determines $w(E_2)$ uniquely by the above formulas. The same applies to Chern classes.

3. We can prove there is no bundle $E \to \mathbb{RP}^\infty$ whose sum with the line bundle $E_1(\mathbb{R}^\infty)$ is trivial. For

$$w(E_1(R^\infty)) = 1 + \omega$$

where ω is a generator of $H^1(\mathbb{RP}^\infty; \mathbb{Z}/2\mathbb{Z})$. Then we have (as we are using $\mathbb{Z}/2\mathbb{Z}$ coefficients):

$$w(E) = (1+\omega)^{-1} = 1 + \omega + \omega^2 + \cdots$$

Thus we have nonzero $w_i(E) = \omega^i$ for all i, but that contradicts $w_i(E) = 0$ for all $i > \dim E$. We finally showed the necessity for compactness for proposition 1.1.16.

4. (Hatcher construct an n-dimensional vector bundle $E \to B$ where $w_i(E)$ is nonzero for each $i \leq E$, [15, p.82]).

Proposition 3.2.6: Relating Chern and Stiefel-Whitney Classes

Let $E \to B$ be a complex n-dimensional vector bundle, and regard it as a $2n$-dimensional real vector bundle. Then

$$w_{2i+1}(E) = 0$$

and $w_{2i}(E)$ is the image of $c_i(E)$ under the coefficient homomorphism:

$$H^{2i}(B; \mathbb{Z}) \to H^{2i}(B; \mathbb{Z}/2\mathbb{Z})$$

***Proof*:**
The bundle E has two projectivizations $\mathbb{RP}(E)$ and $\mathbb{CP}(E)$, consisting of all the real and all the complex lines in fibers of E, respectively. There is a natural projection $p : \mathbb{RP}(E) \to \mathbb{CP}(E)$ sending each real line to the complex line containing it, since a real line is all the real scalar multiples of any nonzero vector in it and a complex line is all the complex scalar multiples. This projection p fits into the following commutative diagram

$$\begin{tikzcd}
\mathbb{RP}^{2n-1} \arrow[r] \arrow[d] & \mathbb{RP}(E) \arrow[r, "\mathbb{RP}(g)"] \arrow[d, "p"] & \mathbb{RP}^\infty \arrow[d] \\
\mathbb{CP}^{n-1} \arrow[r] & \mathbb{CP}(E) \arrow[r, "\mathbb{CP}(g)"] & \mathbb{CP}^\infty
\end{tikzcd}$$

where the left column is the restriction of p to a fiber of E and the maps $\mathbb{RP}(g)$ and $\mathbb{CP}(g)$ are obtained by projectivizing, over $\mathbb{R}$ and $\mathbb{C}$, a map $g : E \to \mathbb{C}^\infty$ which is a $\mathbb{C}$-linear injection on fibers. It is easy to see that all three vertical maps in this diagram are fiber bundles with fiber $\mathbb{RP}^1$, the real lines in a complex line. As the Leray-Hirsch theorem (theorem 3.2.1) applies to the bundle $\mathbb{RP}^\infty \to \mathbb{CP}^\infty$, with $\mathbb{Z}/2\mathbb{Z}$ coefficients, so if β is the standard generator of $H^2(\mathbb{CP}^\infty; \mathbb{Z})$, the $(\mathbb{Z}/2\mathbb{Z})$-reduction $\bar{\beta} \in H^2(\mathbb{CP}^\infty; \mathbb{Z}/2\mathbb{Z})$ pulls back to a generator of $H^2(\mathbb{RP}^\infty; \mathbb{Z}/2\mathbb{Z})$, namely the square α^2 of the generator $\alpha \in H^1(\mathbb{RP}^\infty; \mathbb{Z}/2\mathbb{Z})$. Hence the $(\mathbb{Z}/2\mathbb{Z})$-reduction $\bar{x}_\mathbb{C}(E) = \mathbb{CP}(g)^*(\bar{\beta}) \in H^2(\mathbb{CP}(E); \mathbb{Z}/2\mathbb{Z})$ of the basic class $x_\mathbb{C}(E) = \mathbb{CP}(g)^*(\beta)$ pulls back to the square of the basic class $x_\mathbb{R}(E) = \mathbb{RP}(g)^*(\alpha) \in H^1(\mathbb{RP}(E); \mathbb{Z}/2\mathbb{Z})$. Consequently the $(\mathbb{Z}/2\mathbb{Z})$-reduction of the defining relation for the Chern classes of E, which is

$$\bar{x}_\mathbb{C}(E)^n + \bar{c}_1(E)\bar{x}_\mathbb{C}(E)^{n-1} + \cdots + \bar{c}_n(E) \cdot 1 = 0$$

pulls back to the relation

$$x_\mathbb{R}(E)^{2n} + \bar{c}_1(E)x_\mathbb{R}(E)^{2n-2} + \cdots + \bar{c}_n(E) \cdot 1 = 0$$

which is the defining relation for the Stiefel-Whitney classes of E. This means that $w_{2i+1}(E) = 0$ and $w_{2i}(E) = \bar{c}_i(E)$, as we sought to show.

3.2.1 Cohomology of Grassmannian

here

3.3 Euler and Pontryagin Classes

here
(Pontryagin numbers, a value deduced form Pontryagin classes, characterizes cobordism classes!!!)

3.4 Characteristic Classes as Obstructions

Part II

C*-Algebras and K-Theory

(The course barely covered these, I'm putting them here for the moment)

4

Basics of C^* algebras

C^* algebras where one of the heavily studied objects of K-theory, so a word must be said about them.

Definition 4.0.1: C^*-Algebra

A C^*-algebra A is an algebra over $\mathbb{C}$ with norm $a \mapsto \|a\|$ and involution $a \mapsto a^*$ such that A is complete with respect to the norm, $\|ab\| \leq \|a\|\|b\|$, and $\|a^*a\| = \|a\|^2$ for all $a, b \in A$

A C^*-algebra is called *unit*l if it has a multiplicative identity. It is called *separable* if it contains a countable dense subset.

Given the axioms, it is easy to check that $\|a\| = \|a^*\|$.

Definition 4.0.2: *-Homomorphism

Let A, B be C^*-algebras. Then a map $\varphi : A \to B$ is called a *-homomorphism if it is a ring homomorhpism that satisfies $\varphi(a^*) = \varphi(a)^*$. If A, B are unital and $\varphi(1_A) = 1_B$, then φ is called *unital* (or *unital preserving*)

A sub-C^*-algebra of A is a non-empty subset $B \subseteq A$ if it is closed under the 4 operations of A (addition, multiplication, adjoint, and scalar multiplication), as well as being a closed subset of A (so that it remains complete under the norm). If we simply have a sub *-algebra of a C^*-algebra, then taking the norm closure gives us a sub C^*-algebra (since the 4 operations are continuous with respect to $\| - \|$). Similar natural elaborations occure for generating C^*-algebra from a subset $S \subseteq A$. Quotients and products of C^*-algebras naturally follow the discussion from Banach Spaces (see Functional Analysis notes for details).

An important characterization of C^*-algebras is their "geometric" dual

Theorem 4.0.3: Gelfand-Naimark Duality

Let A be a C^*-algebra. Then there exists a Hilbert space H and an isometric *-homomorphsim $\varphi : A \to B(H)$. Thus every C^*-algebra is isomorphic to a sub C^*-algebra of $B(H)$.If A is separable, then H can be chosen to be a separable Hilbert Space

***Proof*:**
see notes on functional Analysis, Gelfand Duality ([8]). The high-level idea is that we may view A is a vector-space equipped with a suitable inner prodcut that comes frome the positive linear functionals on A, which then we may complete. The rest is similar to Yonneda's Embedding theorem (recall the special case of a ring R being embedded in $B(R) = \mathrm{End}_{\mathbf{Ab}}(R)$)

As mentioned, we shall consider C^*-algebras to not necessarily be unial. However, we may universally assocaite a unique untial C^*-algebra $\tilde{A}$ to a non-unital one A that contains A, with the characterization that $\tilde{A}/ \cong \mathbb{C}$.

Proposition 4.0.4: Unitization

Let A be a C^*-algebra. Then there exists a unital C^*-algebra $\tilde{A}$ such that $\tilde{A}/A \cong \mathbb{C}$. Furthermore, unitization if functorial.

***Proof*:**
Let

$$\tilde{A} = A \oplus \mathbb{C}$$

with operation

$$(a, \alpha) \cdot (b, \beta) = (ab + \beta + \alpha b, \alpha\beta) \qquad (a, \alpha) = (a^*, \overline{\alpha})$$

and addition/scalar-multiplication is done coordinate wise. Then it is enough now to show that $(0, 1)$ is the unit in $\overline{A}$. The rest of the properties now easily follow.

If A is already unital, then we get $\tilde{A} \cong A \oplus \mathbb{C}$ satisfies this property.

Every C^*-algebra naturally induces a new C^*-algebra $M_n(A)$ known as its *matrix algebra*. Each element of $M_n(A)$ is an $n \times n$ matrix whose elements are in A, addition, scalar multiplication, and involution is done term-wise, multiplication is matrix multiplication, and norm is the max of all the norms on the entries. With this norm, a function $f : X \to M_n(A)$ is continuous if and only if each component map $f_{ij} : X \to A$ is continuous. Naturally, if $\varphi : A \to B$ is a C^*-homomorphism, then $\varphi_n : M_n(A) \to M_n(B)$ given component wise operation is a C^*-homomorphism, and $M_n(-)$ is functorial.

Defining semi-group via Projections

First, we shall say two *elements* $a, b \in X$ are homotopic if there is a continuous path $[0, 1] \to X$ where $\gamma(0) = a$ and $\gamma(1) = b$ (this comes from the idea that X can be embedded in a natural hilbert space $L^2(X)$)

Definition 4.0.5: Projections And Equivalences

Let A be a C^*-algebra. The set of all projection of A shall be denoted $\mathcal{P}(A)$ (set of all elements such that $p = p^2 = p^*$). We shall define two equivalence relations on $\mathcal{P}(A)$:

1. (*von Neumann equivalence*) $p \sim q$ if there exists $\in A$ such that $p = v^*v$ anfd $q = vv^*$
2. (*unitary equivalence*) $p \sim_u q$ if there exists a unitary element $u \in \mathcal{U}(\tilde{A})$ where $q = upu^*$.

Definition 4.0.6: Set of projections

Let

$$\mathcal{P}(A) = \mathcal{P}(M_n(A)) \qquad \mathcal{P}_\infty(A) = \bigcup_n^\infty \mathcal{P}_n(A)$$

Then define $\sim_0$ on $\mathcal{P}(A)$ as follows

$$p \sim_0 q \qquad \iff \qquad v \in M_{m,n}(A), p = v^*v \text{ and } q = vv^*$$

now define $\oplus$ on $\mathcal{P}_\infty(A)$ via:

$$p \oplus q = \operatorname{diag}(p, q) = \begin{pmatrix} p & 0 \\ 0 & q \end{pmatrix}$$

Then it is not hard to see this forms a semi-group structure

Proposition 4.0.7: Semigroup Of Projections

Let p, q, r, p', q' be projection in $\mathcal{P}_\infty(A)$ for some C^*-algebra A. Then:

1. $p \sim_0 p \oplus 0_n$ for every n
2. $p \sim_0 p'$ and $q \sim_0 q'$, then $p \oplus q \sim_0 p' \oplus q'$
3. $p \oplus q \sim_0 q \oplus p$
4. if p, q are projections in $\mathcal{P}_n(A)$ such that $pq = 0$, then $p+q$ i a projection and $p+q \sim_0 p \oplus q$
5. $(p \oplus q) \oplus r = p \oplus (q \oplus r)$

***Proof*:**
all relatively simple, will leave for now

Definition 4.0.8: The Semigroup for C^*-algebras

Let $(\mathcal{P}(A), \sim_0, \oplus)$ be as in the above. Then

$$\mathcal{D}(A) = \mathcal{P}_\infty(A)/ \sim_0$$

forms a semi-group where

$$[p]_\mathcal{D} + [q]_\mathcal{D} = [p \oplus q]_\mathcal{D}$$

Naturally, by the Grothendieck construction, this abelian group can be turned into a group. We shall use this in section 4.1 to define the K_0 group.

4.1 K_0 of C^*-algebra

Definition 4.1.1: K_0 of C^*-algebra

Let A be a unital C^* algebra and $(\mathcal{D}(A), +)$ be the semi group defined in definition 4.0.8. Then the group $K_0(A)$ is the Grothendeick construction applied to this semi-group.

All the usual properties of the K_0-group apply: functoriality, transfer maps, it's connection to H_0 and so forth. Note that even if A is not-unital, we can define the K_0 group in the above way, but we shall loose exactness in some important cases, for example:

$$0 \to C_0(X \setminus \{x_0\}) \to C(X) \to \mathbb{C} \to 0$$

is a split exact sequence if X is a connected Compact Hausdorff space, but applying $K_0(-)$ can loose exactness (for example if $X = S^2$).

4.2 K1 of C^*-algebra

As this course covers C^*-algebras, we must add a word on the K-theory of C^*-algebras.

Part III

Algebraic K-Theory

All ring swill be associative with $1 \neq 0$. An R-module will be a right R-module unless specified. We do this so that a homomorphism $R^n \to R^m$ may be thought of as an $m \times n$ matrix with entries in R acting on column in R^n.

The field of K-theory emerged from the work of Alexander Grothendeick in trying to find invariants on Compact Hausdroff spaces and Schemes. The original idea tried to generalize natural transformations such as the determinant, and lead to a series of groups known as the K-groups which give important invariants that we shall explore.

5

Projective modules

If R is a field or a divisor ring, every R-module is a vector-space which is algebraically characterized up to dimension. More generally, a free R-module is a module that is R-module isomorphic to $R^{\oplus I}$ for same indexing set I. Any R-module homomorphism between two free modules is based on where chosen basis of the domain is represented by the chosen basis of the codomain; namely it can be represented by a matrix (or the generalization thereof for infinite dimension). These are the simplest modules that can be put over a ring. A natural question would be to see what are the possible "next weakest" conditions we can put, and what rings R "force" this condition. We shall explore a few of these conditions, one important one shall be for a module to be *projective* which we shall show how this could be thought of as being *locally free*. Measuring how this structure differs from being free will be the beginnings of the study of the K_0-group in the next chapter.

5.1 IBP, Stably Free, and Projective Modules

If R is a field or a divisor ring, then $R^n \cong R^m$ implies $n = m$. If R is commutative this is still the case (consider $R/\mathfrak{m}$ by a maximal ideal), group rings (for finite groups), and any (left)-Noetherian ring satisfies this property as well (using a version of Nakayama's lemma can prove this, see [2]). This property is *not* true for general rings:

Example 5.1.1: Failure of Invariant Basis Number
let R be a commutative ring and let M be the $\mathbb{N} \times \mathbb{N}$ matrices where each column has finitely many nonzero entries, this is sometimes denoted $CFM_{\mathbb{N}}(R)$. Then $CFM_{\mathbb{N}}(R) \cong CFM_{\mathbb{N}}(R)^2$ via the map:

$$\varphi : CFM_{\mathbb{N}}(R) \to CFM_{\mathbb{N}}(R)^2 \qquad M \mapsto (\text{odd columns of } M,\ \text{even columns of } M)$$

Another example the reader could work through is $\mathrm{End}_k(k^{\oplus \mathbb{N}})$.

Definition 5.1.2: Invariant Basis Property

Let R be a ring. Then we say that R satisfies the (right) *invariant basis property* (IBP) if

$$R^n \cong R^m \qquad \Rightarrow \qquad n = m$$

If R satisfies IBP, then the rank of a free R-module M is an invariant, independent of the choice of basis.

Proposition 5.1.3: Necessary Condition For Ibp

Let R be a ring. Then R satisfies IBP if there exists an injective ring map $f : R \to F$ from R to a field (or division ring) F

***Proof*:**
Given such a map, notice that any basis for a free R-module M is the basis for the vector-space $V = M \otimes_R F$, and V is independent of choice of basis, and hence any two bases must have the same cardinality.

Kernel's of a surjective R-module homomorphisms $\varphi : R^n \to R^m$ need not be free, however they must be stably free:

Definition 5.1.4: Stably Free

Let P be an R-module. Then P is called *stably free* of rank $n - m$ if there exists n, m such that

$$P \oplus R^m \cong R^n$$

If R satisfies IBP, then the rank of a stably free module is independent of the choice of n and m.

Certainly, the kernel stably free as R^m is free as we must have a split sequence (see [2]). Not all R-modules are stably free (or more interestingly, not all stably free modules are free). The notion of stably free seems very similar to *projective*, however we have not specified the condition for the 2nd summand in the definition of projective modules and hence it is a slightly weaker condition. We shall get to projective modules shortly.

Let us take a moment to consider the case where an R-module P is stably free of rank 1 , that is $P \oplus R \cong R^n$. In this case, we may ask when can we "cancel" the R on each side, a fruitful discussion considering the K-theory we are building up to. In this case, let σ be a row vector of the right hand side, and call it the *unimodular row.* We get the following interesting set of equivalences:

1. σ is unimodular
2. $R^n \cong P \oplus R$ where $P = \ker(\pi)$ where $\pi : R^n \to R$ (given by σ)
3. $R = r_1 R + \cdots + r_n R$
4. $1 = r_1 s_1 + \cdots + r_n s_n$ for appropriate $s_i \in R$

Note that if P where free in $R^n \cong P \oplus R$, then a basis of P would give a new basis for R^n, which corresponds to an invertible matrix (say g) whose first row is the unimodular row $\sigma : R^n \to R$ corresponding to P. From this we get the following citation:

Proposition 5.1.5: Free and Unimodular

Let P be an R-module that stably free of rank 1. Then P is free if and only if the corresponding unimodulr row ma ybe completed to an invertible matrix.

***Proof*:**
combine the above discussion.

If R is commutative, then every unimodular row of length 2 may be completed: if $r_1s_1 + r_2s_2 = 1$, then the invertible matrix is:

$$\begin{pmatrix} r_1 & r_2 \\ -s_2 & s_1 \end{pmatrix}$$

And so, if$R^2 \cong R \oplus P$, then $P \cong R$. We shall get stronger versions of this result later (note here though that commutativity was important, this fails for unimodular row of length to wit ha non-commutative ring).

In dimension 3, we can already find counter example:

Example 5.1.6: Stably Free But Not Free

Take

$$R = \frac{\mathbb{R}[x,y,z]}{(x^2+y^2+z^2-1)}$$

Note that ever element $(f,g,h) \in R^3$ gives a vector-field in $\mathbb{R}^3$. Let σ be the unimodular row $\sigma = (x,y,z)$. Then $1s$ is the vector-field pointing radially outward. If P were free, a basis of P would yield two tangent vector fields on S^2 which are linearly independent at every point of S^2 (since together with σ, they span the tangent space of 3-space at each point). But this contradicts the infamous Hairy-sphere theorem[a].

[a]Or more humorously *Hairy ball theorem*

Let us now return to the discussion of projective modules.

Definition 5.1.7: Projective Module

Let P be an R-module. Then P is *projective* if there exists an R-module Q such that $P \oplus Q$ is a free R-module.

See [2] for details on the equivalence and examples, on important one is that there is always a lift in the following commutative diagram:

$$\begin{tikzcd} & P \arrow[dl, dashed, "\exists!"'] \arrow[d, "g"] & \\ M \arrow[r, "f"] & N \arrow[r, "0"] & 0 \end{tikzcd}$$

where f is surjective. Naturally, free and stably free module are projective. A more nuanced example is that all $M_n(k)$-modules (with $n > 1$) are projective but not free (which can be deduced using Artin-Wederburn). A more canonical set of examples comes from vector bundles. Take $R = C^0(X)$ to be the set of continuous real valued function on a compact topological space X, and let $\Psi : E \to X$ be a vector-bundle. Then the set $\Gamma(E)$ of continuous sections naturally forms a projective R-module. Let us show this (see [1] for the algebraic geometry perspective of the equivalence between vector bundle and locally free sheaves). The trivial bundle is $\mathbb{R}^n \times X \to X$ and we get $\Gamma(\mathbb{R}^n \times X) = R^n$. Now assume $\Psi : E \to X$ is a non-trivial vector-bundles (up to vector-bundle isomorphism). Then if $\Gamma(E)$ were free, there would be sections $\{s_1, ..., s_n\}$ that would for ma basis $f : T^n \to E$ which would give an isomorphism $\Gamma(T^n) = \Gamma(E)$. As the kernel and cokernel of bundles of f have no nonzero sections, the must banish ,and so f would be an isomorphism, and so $\Psi : E \to X$ would in fact be trivial.

If X is compact, then the famous *Swan's Theorem* gives an equivalence between the category of projective modules over $C^0(X)$ and the category of vector bundles over X, showing that vector bundles are the geometric interpretation of projective modules, which motivates similar discussions over sheaves of module for a scheme.

Proposition 5.1.8: Finitely Generated Projective Modules and Idempotents

Let P be a finitely generated projective R-module. Then P corresponds to an idempotent element in $M_n(R)$ for appropriate n.

***Proof*:**
If P is finitely generated, then $P \oplus Q \cong R^n$, meaning there is an idempotent element in R which can be associated to an element of $M_n(R)$. To find this element, we can take the natural projection-inclusion

$$R^n \to P \to R^n$$

This map is an idempotent in $e \in M_n(R)$ (when representing this composition as a matrix under a choice of basis). Labeling this composition e, we see that $e(R^n) = P$.

If P is a projective R-module over a PID, then it is in fact free (as given by the structure theorem for modules over PID[1]). Another famous class of modules where all finitely generated projectives are free comes from Quillen and Suslin stating that finitely generated modules over a polynomial ring (or a Laurent polynomial ring) over a PID is free[2]. This then gives the result mentioned earlier about group rings over finite abelian groups, for $\mathbb{Z}[G]$ is the Laurent polynomial ring $\mathbb{Z}[x, x^{-1}, ..., z, z^{-1}]$. On the other hand, if G is *not* abelian (but has the conditions of being torsion-free and nilpotent), then any projective $\mathbb{Z}[G]$-module P is stably free but not free, in particular that:

$$P \oplus \mathbb{Z}[G] \cong (\mathbb{Z}[G])^2$$

As of writing this book, it is unknown of the condition “nilpotent” can be dropped and produce a similar (or the same) result.

Another important class of finitely generated projective R-module that are always free are those over *local rings*. For informative purposes, the result can in fact be extended to infinitely generated

[1] Note this even works for infinitely generated projective modules, something briefly outlined in some exercises in [2]
[2] This result appears in [1] where it is used to show that all locally free $\mathcal{O}_X$-modules on affine space must be free

projective R-modules over local rings (Kaplansky gave such a proof). The following is given as a reminder:

Proposition 5.1.9: Projective Over Local Ring, then Free

Let R be a local ring. Then every finitely generated projective Rmodule is free, where

$$P \cong R^n \qquad p = \dim_{R/\mathfrak{m}}(P/\mathfrak{m}P)$$

***Proof*:**
see [5, chapter 20.3]

Proposition 5.1.10: Free At Stalks

Let R be a commutative ring, $\mathfrak{p}$ a prime ideal, and P a finitely generated projective R-module. Then the localization $P_{\mathfrak{p}}$ is isomorphic to $(R_{\mathfrak{p}})^n$ for some $n \geq 0$. Furthermore, there exists some $s \in R \setminus \mathfrak{p}$ where the localization away from s is free:

$$P\left[\frac{1}{s}\right] \cong R\left[\frac{1}{s}\right]^n$$

And we further get that $P_{\mathfrak{p}'} \cong (R_{\mathfrak{p}'})^n$ for every prime ideal $\mathfrak{p}'$ of R not containing S

***Proof*:**
As P is projective, let $P \oplus Q = R^m$. Then $P_{\mathfrak{p}} \oplus Q_{\mathfrak{p}} = (R_{\mathfrak{p}})^m$, hence $P_{\mathfrak{p}}$ is a finitely generated projective $R_{\mathfrak{p}}$-module. As $R_{\mathfrak{p}}$ is local, $P_{\mathfrak{p}}$ is in fact free. Now, every element of $P_{\mathfrak{p}}$ is of the form p/s, for $p \in P$ and $s \in R \setminus \mathfrak{p}$. By choosing a basis for $P_{\mathfrak{p}}$, we may clear the denominators to get an R-module homomorphism $\varphi : R^n \to P$ that become an isomorphism when localizing at $\mathfrak{p}$.

Now, as $\operatorname{coker}(\varphi)$ is a finitely generated R-module which vanishes upon localization, it must be annihilated by some $s \in R \setminus \mathfrak{p}$. Given this s, we get a surjective map:

$$\varphi\left[\frac{1}{s}\right] : \left(R\left[\frac{1}{s}\right]\right)^n \to P\left[\frac{1}{s}\right]$$

As $P\left[\frac{1}{s}\right]$ is projective (as localization preserves projectiveness), $\left(R\left[\frac{1}{s}\right]\right)^n$ is isomorphic to the direct sum of $P\left[\frac{1}{s}\right]$ and a finitely generated $R\left[\frac{1}{s}\right]$-module M (where $M_{\mathfrak{p}} = 0$). As M is annihilated by some $t \in R \setminus \mathfrak{p}$, we get:

$$\varphi\left[\frac{1}{st}\right] : \left(R\left[\frac{1}{st}\right]\right)^n \xrightarrow{\cong} P\left[\frac{1}{st}\right]$$

completing the proof.

We should next define the "vector bundle" structure that we can put on modules. To that end, we will have to assume some algebraic geometry, nothing more than some knowledge on spectra (see [5, chapter 21.7] or [1, chapter 2.1]). Let $\operatorname{Spec} R$ be the collection of prime ideals of R, and define the

Zariski topology with basic open sets given by:

$$D(s) = \{\mathfrak{p} \in \operatorname{Spec} R \mid s \notin \mathfrak{p}\} \cong \operatorname{Spec} R\left[\frac{1}{s}\right]$$

Definition 5.1.11: Rank At Prime Ideal

Let R be a commutative ring and M be a finitely generated R-module. Then the *rank* at $\mathfrak{p}$ is:

$$\operatorname{rank}_{\mathfrak{p}}(M) = \dim_{\kappa(\mathfrak{p})} M \otimes_R \kappa(\mathfrak{p})$$

Note that $M_{\mathfrak{p}}/\mathfrak{p}M_{\mathfrak{p}} \cong \kappa(\mathfrak{p})^{\operatorname{rank}_{\mathfrak{p}}(M)}$, hence $\operatorname{rank}_{\mathfrak{p}}(M)$ is the minimal number of generators for $M_{\mathfrak{p}}$. If P is a finitely generated projective R-module, then $\operatorname{rank}(P) : \mathfrak{p} \mapsto \operatorname{rank}_{\mathfrak{p}}(P)$ is a continuous function between the $\operatorname{Spec} R$ with the Zariski topology and $\mathbb{N}$ with the discrete topology.

If $\operatorname{Spec} R$ is connected, then $\operatorname{rank}(P) : \operatorname{Spec} R \to \mathbb{N}$ must always be a constant function, hence every finitely generated projective R-module has constant rank. In the simplest case, R is an integral domain, then the rank is constant with:

$$\operatorname{rank}(P) = \dim_{\operatorname{Frac}(R)}(P \otimes_R \operatorname{Frac}(R))$$

Note that if P is of constant rank, then it is finitely generated[3]. If M is not projective, then $\operatorname{rank}(M)$ need not be continuous: consider $R = \mathbb{Z}$ with $M = \mathbb{Z}/p\mathbb{Z}$.

We conclude this section with an important observation about the "lack" of importance of higher dimensions. For the following, recall that M, M' are stably isomorphic if $M \oplus R^m \cong M' \oplus R^m$ for some $m \geq 0$.

Theorem 5.1.12: Base Serre Cancellation

Let R be a commutative Noetherian ring with $\operatorname{kdim}(R) = d$, and let P be a projective R-module of constant rank $n > d$. Then:

1. $P \cong P_0 \oplus R^{n-d}$ for some projective R-module P_0 of constant rank d
2. If P is stably isomorphic to P', then $P \cong P'$
3. For all M, M', if $P \oplus M$ is stably isomorphic to M', then

$$P \oplus M \cong M'$$

The idea is that any extra structure when considering higher dimension vector bundles only adds "trivial" information, making the information from vector-bundles ties to the local information of the base space.

***Proof*:**
see H. Bass, Algebraic K-theory, Benjamin.

[3] Prove it for the case of rank 1, then for the rank r case show that $\Lambda^r(P)$ is finitely generated

5.1.1 *Milnor Square

The following material is going to be more useful in section 6.5 when looking over Mayer-Vietoris sequences, however it appears when studying the Picard group of a ring. The idea is that the Milnor square gives us a way of gluing together different parts of a ring akin to the Mayer-Vietoris sequence from algebraic topology (see [9]). Recall from algebraic topology that this construction was express done to be computationally rather simple. We "algebras" this technique by looking at a particularly nice way to "patch" two rings in such a way that will allow for some simple computations of various invariance (ex. Picard group, K-groups,etc)

Let R be a ring, $I \subseteq R$ an ideal, $\varphi : R \to S$ a ring homomorphism that maps I isomorphically to an ideal in S, which we shall also label I. Then R is the pullback of S and R/I in the following diagram:

$$\begin{tikzcd} R \arrow[r] \arrow[d] & S \arrow[d] \\ R/I \arrow[r] & S/I \end{tikzcd}$$

This square is called the *Milnor square*. In algebraic geometry, this represents R as the fibered product of S and R/I over S/I.

We can patch modules given a Milnor square. Let M_1 be an S-module and M_2 be an R/I-module. The Milnor square defines the natural S/I-module isomorphism:

$$g : M_2 \otimes_{R/I} S/I \cong M_1/IM_1$$

Then M can be thought of as a triple (M_1, g, M_2) and is a subset of $M_1 \times M_2$ given by the kernel of the R-module homomorphism:

$$M_1 \times M_2 \to M_1/IM_1 \qquad (m_1, m_2) \mapsto \overline{m}_1 - g(\overline{f}(m_2))$$

Example 5.1.13: Milnor Patching A Module

Take S^n and $(R/I)^n$ with $1 \in \mathrm{GL}_n(S/I)$. Consider $\overline{f} : R/I \to S/I$ Then:

$$\left\{ (\vec{s}, \vec{r+I}) \;\middle|\; \overline{\vec{s}} = \overline{f}(r+I) \right\}$$

Hence, we have patched together two vector-spaces of the same dimension over different rings that can glue over R/I, or geometrically we have two trivial bundles over R/I and S and we glued those bundles to create a bundle over R.

The most important properties of Milnor patching that we shall use is summarized in the following:

Theorem 5.1.14: Milnor patching

Let

$$\begin{array}{ccc} R & \longrightarrow & S \\ \downarrow & & \downarrow \\ R/I & \longrightarrow & S/I \end{array}$$

be a commutative diagram where $I \subseteq R$ is identified with its image in S and also represented as I (i.e. a Milnor square). Then:

1. If P is obtained by patching together a finitely generated projective S-module P_1 and a finitely generated projective R/I-module P_2, then P is a finitely generated projective R-module.
2. $P \otimes_R S \cong P_1$ and $P/IP \cong P_2$.
3. Every finitely generated projective R-module arises in this way.
4. If P is obtained by patching free modules along $g \in GL_n(S/I)$ and Q is obtained by patching free modules along g^{-1}, then $P \oplus Q \cong R^{2n}$.

***Proof*:**

1. Let's start with a Milnor square, which is a pullback diagram:

$$\begin{array}{ccc} R & \to & S \\ \downarrow & & \downarrow \\ R/I & \to & S/I \end{array}$$

Suppose we have a finitely generated projective S-module P_1 and a finitely generated projective R/I-module P_2, along with an isomorphism $g : P_1 \otimes_S S/I \cong P_2/IP_2$. Let

$$P = \{(x, y) \in P_1 \times P_2 \big| \overline{x} = g(\overline{f}(y))\}$$

To show P is finitely generated, Since P_1 and P_2 is a finitely generated, there exists a surjection $S^n \to P_1$ for some n and a surjection $(R/I)^m \to P_2$ for some m. Then it is certainly the case that we can construct a surjection $R^{m+n} \to P$, showing that P is finitely generated.

For showing it is projective, since P_1 is projective over S, there exists an S-module Q_1 such that $P_1 \oplus Q_1 \cong S^n$ for some n. Similarly, there exists an R/I-module Q_2 such that $P_2 \oplus Q_2 \cong (R/I)^m$ for some m.

The isomorphism $g : P_1 \otimes_S S/I \cong P_2/IP_2$ extends $\tilde{g} : S^n \otimes_S S/I \cong (R/I)^m/I(R/I)^m$, namely it is the isomorphism induced by taking $Q_1 \otimes_S S/I$ and Q_2/IQ_2. Now, define F by:

$$F = \{(x, y) \in S^n \times (R/I)^m \big| \overline{x} = g(\overline{f}(y))\}$$

By the properties of the Milnor square, F is a free R-module. The inclusion maps $P_1 \hookrightarrow S^n$ and $P_2 \hookrightarrow (R/I)^m$ induce an inclusion $P \hookrightarrow F$. Moreover, the projections $S^n \to P_1$ and $(R/I)^m \to P_2$ induce a projection $F \to P$ such that the composition $P \to F \to P$ is the identity on P. But then P is projective,

2. This follows by our above construction

3. If M is any R-module, the Milnor square gives a natural map from M to the R-module M' by patching $M_1 = M \otimes_R S$ and $M_2 = M \otimes_R (R/I) = M/IM$ along the natural isomorphism

$$(M/IM) \otimes_{R/I} (S/I) \cong M \otimes_R (S/I) \cong (M \otimes_R S)/I(M \otimes_R S)$$

Now, tensor M with $0 \to R \to (R/I) \oplus S \to S/I \to 0$. This gives an exact sequence (recall that tensoring is right-exact, but not necessarily left-exact).

$$\mathrm{Tor}_1^R(M, S/I) \to M \to M' \to 0$$

so in general M' is just a quotient of M. However, if M is projective, the Tor term is zero and $M \cong M'$. Thus every projective R-module may be obtained by patching.

5.2 Picard Group of a Commutative Ring

Let us now tackle the connection between projective modules and vector bundles more closely. To fully understand the interplay between the algebraic and the geometric, the following theorem must be quoted:

Theorem 5.2.1: Serre-Swan's Theorem

1. **(Serre's Theorem)** Let R be a commutative Noetherian ring. Then the category of finitely generated projective R-modules is equivalent to the category of algebraic vector bundles[a] on $\mathrm{Spec}\, R$

2. **(Swan's Theorem)** Let X be a compact Hausdorff space. Then the category of finitely generated projective R-modules over the continuous functions $C(X)$ is equivalent to the category of finite-rank vector bundles on X, the equivalence being established by sending vector bundles its module of continuous section.

[a] locally free sheaves of structure sheaf-modules of constant finite rank

***Proof*:**
See Serre's paper and Swan's paper.

Due to these connections, we have the following definition:

Definition 5.2.2: Algebraic Line Bundle

Let R be a commutative ring. Then an *algebraic line bundle* L over R is a finitely generated projective R-module of constant rank 1.

In this section, we shall just say *line bundle*. As line bundles are locally 1 dimensional, it is easy to verify that the tensor product of line bundles is again a line bundle (hint: $\dim(V \otimes W) = \dim(V) \cdot \dim(W)$). Furthermore, it should be immediate to the current reader that $L \otimes_R R \cong L$ for any line bundle (even projective module). Then $\otimes_R$ forms a commutative monoid. If we show there is an inverse, we have a group, and indeed:

Lemma 5.2.3: Inverse Line Bundle

Let L be a line bundle and $L^\vee = \mathrm{Hom}_R(L, R)$. Then

$$L \otimes_R L^\vee \cong R$$

***Proof*:**
First, $L^\vee$ is line bundle as $\mathrm{rank}(L) = \mathrm{rank}(L^\vee)$. Next, take:

$$f \otimes_R \ell \mapsto f(\ell)$$

If $L = R$, this is certainly an isomorphism. Hence in the general case, localizing at $\mathfrak{p}$ we get:

$$(L^\vee \otimes_R L)_\mathfrak{p} \cong L^\vee_\mathfrak{p} \otimes_{R_\mathfrak{p}} L_\mathfrak{p} \to R_\mathfrak{p}$$

which is then certainly an isomorphism. As being an isomorphism is a local property, we get the global isomorphism, completing the proof.

Definition 5.2.4: Picard Group

Let R be a commutative ring. Then the *picard group* of R is the set of isomorphism classes of line bundles on R with the operation being the tensor product $\otimes_R$ and inverses $L^{-1} = L^\vee$.

Proposition 5.2.5: Picard Group Homomorphism

Let $R \to S$ be a ring homomorphism between commutative rings. Then $\mathrm{Pic}\,(R) \to \mathrm{Pic}\,(S)$ is a group homomorphism mapping $L \mapsto L \otimes_R S$

***Proof*:**
Certainly $L \otimes_R S$ is a line bundle over S. Then it is an exercise in algebra to see that:

$$(L \otimes_R M) \otimes_R S \cong (L \otimes_R S) \otimes_R (M \otimes_R S)$$

showing that $\otimes_R S$ is a group-homomorphism, as we sought to show.

We prove the following representation of line bundles for later:

Lemma 5.2.6: Representation of Line Bundle

Let L be a line bundle over R. Then

$$\mathrm{End}_R(L) \cong R$$

***Proof*:**
Multiplication by elements of r give a map $R \to \mathrm{End}_R(L)$. This map is locally an isomorphism, and hence an isomorphism, completing the proof.

Example 5.2.7: Picard Groups of Commutative Rings

1. Let $R = k$. Then as k is a field, every projective k-module is free. But then any projective k module of rank 1 is a dimension 1 vector-space, namely it is isomorphic to k. Hence:
$$\mathrm{Pic}\,(k) = 0$$
2. Similarly, as every module over a PID is free, if $R = \mathbb{Z}$, then:
$$\mathrm{Pic}\,(\mathbb{Z}) = 0$$
3. Let R be a Dedekind domain (usually a number ring). We shall heavily be relying on the algebra presented in [5, chapter 22.6]. First, recall that each ideal $I \subseteq R$ is a projective R-module, and it is certainly locally of rank 1 (every ideal is generated by at most 2 elements, and localizing at the appropriate prime makes the ideal principal). Next, recall that any torsion-free R-module of rank n is isomorphic to:
$$M \cong R^{n-1} \oplus I$$
for an ideal I that can be replaced by $J = (a)I$ for $a \in R$. As all projective modules are torsion free, and we are looking at rank 1 modules, we see that the line bundles on R are in correspondence to the ideals of R. As principal ideals are isomorphic to R, we have that:
$$\mathrm{Pic}\,(R) \cong \mathrm{Cl}\,(R)$$
4. (If you know some divisor theory) Building off the work from the last point, If X is a nonsingular affine variety over an algebraically closed field $\overline{k}$ then $\mathrm{Pic}\,(k[X])$ (the Picard group of its coordinate ring) is isomorphic to the divisor class group of X.
5. it was proven by Quillen and Suslin that every projective module over polynomial ring is free. Thus, the picard group of $k[x_1, ..., x_n]$ is always trivial:
$$\mathrm{Pic}\,(k[x_1, ..., x_n]) = 0$$
6. (foreshadowing for elliptic curve theory). An elliptic curve is a smooth algebraic curve (over an algebraically closed field $\overline{k}$) whose genus is 1[a]. We shall see that there is a natural subgroup of the Picard group, $\mathrm{Pic}^{\,0}$, and that if E is an elliptic curve associated to $R = \overline{k}[x, y, z]/I$, then:
$$\mathrm{Pic}^{\,0}(R) \cong R$$
Which is a rather interesting idempotence! See cite:AlgGeo for more details.

[a]Every nonsingular curve maps onto a compact Riemann surface of genus g

We shall next define a natural notion known as the *determinant line bundle*. Its key feature of interest to us is that all finitely generated projective R-modules are determined by their rank and determinant. First, recall the following properties of the exterior product:

Proposition 5.2.8: Properties of The Exterior Product

Let R be a commutative ring and M an R-module. Then:

1. $\Lambda^k(R^n)$ is a free R-module of rank $\binom{n}{k}$ generated by $e_{i_1} \wedge \cdots \wedge e_{i_n}$ where $i_1 \leq \cdots \leq i_n$. Note that $\Lambda^n(R^n) \cong R$
2. If $R \to S$ is a ring map, there is an isomorphism:
$$\left(\Lambda^k M\right) \otimes_R S \cong \Lambda^k(M \otimes_R S)$$
3. There is a natural isomorphism:
$$\bigwedge^k (P \oplus Q) \cong \bigoplus_{i=0}^{k} (\bigwedge^i P) \otimes (\bigwedge^{k-i} Q)$$
4. If P is projective, $\Lambda^k(P)$ is projective (as it is locally free).

***Proof*:**
exercise, or see [5, chapter 15.4]

Recall that $\Lambda^n(M)$ for a n-rank R-module is generated by a single element, and by moving all the coefficients to the front we get the determinant. This motivates the following terminology:

Definition 5.2.9: Determinant Bundle

Let P be a finitely generated projective R-module of constant rank n. Then $\Lambda^n(P)$ is called the *determinant line bundle*, and is denoted $\det(L)$.

If P is not constant rank, we define it component-wise. If $g : P \to P$ is an endomorphism, $\det(-)$ is a functor $\deg(g) : \det(P) \to \det(P)$, and by lemma 5.2.6 we can think of $\deg(g) \in R$.

We shall state, but not prove, the following useful classification result:

Proposition 5.2.10: Classification of Projective Module Over 1-dimensional Ring

Let R be a commutative Noetherian 1-dimensional ring. Then all finitely generated projective R-modules are completely classified by their rank and determinant. In particular, every finitely generated projective R-module P of rank ≥ 1 is isomorphic to $L \oplus R^f$, where $L = \det(P)$ and $f = \text{rank}(P) - 1$.

***Proof*:**
hint: Show that R is the only stably free-line bundle.

Let us next focus on line bundles over integral domains. We shall see that all line bundles on integral domains are given by *invertible ideal*:

Proposition 5.2.11: Classification of Projective Modules over Integral Domains

Let R be a commutative integral domain. L is a line bundle if and only if it is (isomorphic to) an invertible ideal. If I and J are fractional ideals and I is invertible, then $I \otimes_R J \cong IJ$. Furthermore, there is an exact sequence of abelian groups:

$$1 \to R^{\times} \to F^{\times} \xrightarrow{\text{Div}} J_K(R) \to \text{Pic}\,(R) \to 0.$$

where $J_K(R)$ is the ideal group.

***Proof*:**
If I and J are invertible ideals such that $IJ \subseteq R$, then we can interpret elements of J as homomorphisms $I \to R$. If $IJ = R$, then we can find $x_i \in I$ and $y_i \in J$ so that $x_1y_1 + \cdots + x_ny_n = 1$. The $\{x_i\}$ assemble to give a map $R^n \to I$ and the $\{y_i\}$ assemble to give a map $I \to R^n$. The composite $I \to R^n \to I$ is the identity, because it sends $r \in I$ to $\sum x_iy_ir = r$. Thus I is a summand of R^n, i.e., I is a finitely generated projective module.

As R is an integral domain and $I \subseteq F$, $\text{rank}(I)$ is the constant $\dim_F(I \otimes_R F) = \dim_F(F) = 1$. Hence I is a line bundle.

This construction gives a set map $J_K(R) \to \text{Pic}\,(R)$; to show that it is a group homomorphism, it suffices to show that $I \otimes_R J \cong IJ$ for invertible ideals. Suppose that I is a submodule of F which is also a line bundle over R. As I is projective, $I \otimes_R -$ is an exact functor. Thus if J is an R-submodule of F, then $I \otimes_R J$ is a submodule of $I \otimes_R F$. The map $I \otimes_R F \to F$ given by multiplication in F is an isomorphism because I is locally free and F is a field. Therefore the composite

$$I \otimes_R J \subseteq I \otimes_R F \xrightarrow{\text{multiply}} F$$

sends $\sum x_i \otimes y_i$ to $\sum x_iy_i$. Hence $I \otimes_R J$ is isomorphic to its image $IJ \subseteq F$. This proves the third assertion.

The kernel of $J_K(R) \to \text{Pic}\,(R)$ is the set of invertible ideals I having an isomorphism $I \cong R$. If $f \in I$ corresponds to $1 \in R$ under such an isomorphism, then $I = fR = \text{div}(f)$. This proves exactness of the sequence at $J_K(R)$.

Clearly the units $R^{\times}$ of R inject into $F^{\times}$. If $f \in F^{\times}$, then $fR = R$ if and only if $f \in R$ and f is in no proper ideal, i.e., if and only if $f \in R^{\times}$. This proves exactness at $R^{\times}$ and $F^{\times}$.

Finally, we have to show that every line bundle L is isomorphic to an invertible ideal of R. Since $\text{rank}(L) = 1$, there is an isomorphism $L \otimes_R F \cong F$. This gives an injection $L \cong L \otimes_R R \subseteq L \otimes_R F \cong F$, i.e., an isomorphism of L with an R-submodule I of F. Since L is finitely generated, I is a fractional ideal. Choosing an isomorphism $\tilde{L} \cong J$, Lemma 3.1 yields

$$R \cong L \otimes_R \tilde{L} \cong I \otimes_R J \cong IJ.$$

Hence $IJ = fR$ for some $f \in F^{\times}$, and $I(f^{-1}J) = R$, so I is invertible.

There is further discussion different interesting divisor groups, class groups, and their relation to the Picard group, but we leave that material to cite:algGeo.

6

K_0 Group

We saw the Picard group allowed us to treat the collection of (isomorphism classes of) line bundles of constant rank one as a group under $\otimes$. We shall now be looking at a similar situation with the direct sum.

6.1 Algebraic Build-up

Let M be an abelian monoid. Then we universally augment it into an abelian group, $M^{-1}M$ through the usual free-quotient construction. In particular, take $\mathcal{F}(M \times M)/\left([m+n]-[m]-[n]\right)$. This is called the *Grothendieck Completion.* A *Cancellation monoid* is a monoid where $m+a=n+a$ if and only if $m=n$.

Proposition 6.1.1: Properties of Monoid Completion

Let M be a monoid, $M^{-1}M$ be its completion. Then:

1. Every element of $M^{-1}M$ can be written as $[m]-[n]$ for $m,n \in M$
2. For each $[m],[n] \in M^{-1}M$, $[m]=[n]$ if and only if there exists a p such that $m+p=n+p$.
3. The map $M \times M \to M^{-1}M$ where $(m,n) \mapsto [m]-[n]$ is surjective
4. M injects into $M^{-1}M$ if and only if M is a cancellation monoid.
5. The Grothendieck Completion is a functor that is left adjoint to the forgetful functor from the category of abelian group to the category of abelian monoids.

***Proof*:**
these are all simple exercises, but you can take a look at [5, chatper 15].

If we augment a monoid with a 2nd operation that satisfies the ring axioms, then we get a *semiring*. The semiring $\mathbb{N}$ is the initial semiring in this category. Then the Grothendeick functor can also be seen as a functor form semirings to rings.

When we shall be studying $K_0(R)$ groups, we shall find that the free R-modules play an important role. These elements are important enough so that we consider the following special type of subgroup of a monoid

Definition 6.1.2: Cofinal Monoid

Let M be a monoid. Then $L \subseteq M$ is called *cofinal* if for every $m \in M$, there exists a $m' \in M$ such that $m + m' \in L$.

The terminology *cofinal* comes from order theory, were a subset $A \subseteq B$ of an preorder $(B, \leq)$ is cofinal if for every $b \in B$, there is an $a \in A$ such that $b \leq a$.

Lemma 6.1.3: Cofinal Monoids

Let $L \subseteq m$ be a cofinal monoid. Then:

1. $L^{-1}L \subseteq M^{-1}M$ is a subgroup
2. Every element $[m] \in M^{-1}M$ can be written as $[m'] - [\ell]$ for $m' \in M$ and $\ell \in L$.
3. If $[m] = [m'] \in M^{-1}M$, then $m + \ell = m' + \ell$ for some $\ell \in L$.

***Proof*:**
this is a matter of checking definitions

Example 6.1.4: Grothendeick Completion

1. some good examples in the book!

6.2 K_0 of a ring

Definition 6.2.1: First K-ring: $K_0(R)$

Let R be a [not necessarily commutative] ring, $P(R)$ the collection of (isomorphism classes of) finitely generated projective R-modules. Then combining it with $\oplus$ and appending an identity element 0, we get an abelian monoid. It's Grothendeick completion is the group $K_0(R)$.

When R is commutative, then along with $\otimes_R$, this forms a ring.

We restrict to finitely generated modules simply because the usual behaviour of infinitely generated modules trivializes the structure, for example if P is any finitely generated projective R-module then, then if we had to consider the infinitely generated module R^∞ we get:

$$P \oplus R^\infty \cong R^\infty$$

Therefore $[P] = 0$ and $K_0(R) = 0$, and so we naturally stick to the finitely generated case.

If R is commutative, then $K_0(R)$ is commutative with identity $1 = [R]$. This is indeed the identity, for $\oplus$ and $\otimes$ distribute, and $P \otimes_R R \cong P$. The canonical trivial example is when $R = k$ (or any division ring). Then $K_0(k) \cong \mathbb{Z}$. Similarity, if R is a local ring or a PID, then $K_0(R) \cong \mathbb{Z}$ (as these are sufficient conditions for a projective module to be free). We shall give non-trivial example after we have developed some more theory to be able to identify which properties lead to structural results for $K_0(R)$.

It should be noted that $K_0(-)$ is a functor from **Ring** to **Ab** or from **CRing** to **CRing**, namely if $\varphi : R \to S$ is a ring homomorphism, then $\otimes_R S : P(R) \to P(S)$ is a monoid map which induces the map $K_0(R) \to K_0(S)$. If R, S are commutative, then $\otimes_R S : P(R) \to P(S)$ is a semiring map as:

$$(P \otimes_R Q) \otimes_R S \cong (P \otimes_R S) \otimes_R (Q \otimes_R S)$$

and hence we get a ring homomorphism. This shall become handy when trying to translate properties from known $K_0(R)$ groups to one's we are trying to study.

In $K_0(R)$, the free-modules play a particularity important role. First of all, they are *cofinal* in $P(R)$ (see definition 6.1.2). Then every element of $K_0(R)$ can be written as:

$$[P] - [R^n]$$

Furthermore, by lemma 6.1.3, we have $[P] = [Q]$ if and only if:

$$P \oplus R^n \cong Q \oplus R^m$$

that is, P, Q are stably isomorphic, an $[P] = [R^n]$ if P is stably free. If R satisfies the IBP, then $L \subseteq P(R)$ is isomorphic to $\mathbb{N}$. If R respect IBP, we get the following useful characterization of trivial K_0 groups:

Lemma 6.2.2: Ibp and Trivial K_0

Let $\mathbb{N} \to P(R)$ be the monoid map sending $n \to R^n$ inducing the group homomorphism $\mathbb{Z} \to K_0(R)$. Then:

1. this map is injective if and only if R satisfies IBP
2. If R satisfies IBP, then

$$K_0(R) \cong \mathbb{Z} \text{ if and only if every finitely generated projective } R\text{-module is stably free}$$

***Proof*:**

1. This is immediate by IBP
2. $P \oplus R^n \cong R^m$, and so each $[P] = [R^m]$ for some n.

Example 6.2.3: K_0 of some rings

1. As mentioned before, fields/division rings, PIDs, and local rings all have $K_0(R) = \mathbb{Z}$
2. If $R = R_1 \times R_2$, then by [5, chatper 23] we have the equivalence of categories:
$$(R \times S)\textbf{-Mod} \cong R\textbf{-Mod} \times S\textbf{-Mod}$$
Thus we get $P(R) = P(R_1) \times P(R_2)$, and its completion gives $K_0(R) = K_0(R_1) \times K_0(R_2)$. We may thus compute $K_0(R)$ component-wise.
3. Let $R = M_n(F)$ Then R is a simple ring, and so every $M_n(F)$-module is projective. Recall that an $M_n(F)$-module is some direct sums of F, and the *length* of an $M_n(F)$-module is the number of copies. Then the length gives a natural map:
$$K_0(M_n(F)) \cong \mathbb{Z}$$
Note that $M_n(F)$ has length n, and so the subgroup generated by the free modules have index n, so the natural inclusion map $\mathbb{Z} \to K_0(M_n(F))$ does not split!
4. Let R be a semi-simple ring so that by Artin-Wedderburn:
$$R \cong M_{n_1}(D_1) \times \cdots \times M_{n_k}(D_k)$$
Thus, we get:
$$K_0(R) = \prod_i K_0(M_{n_i}(D_i)) \cong \mathbb{Z}^k$$
5. If R is a commutative ring, we have a natural map from R to a field, $R \to R/\mathfrak{m}$ for some maximal ideal $\mathfrak{m}$; call this field F. Then this induces a natural map $K_0(R) \to K(F)$, that is $K_0(R) \to \mathbb{Z}$ which maps $[R]$ to 1. Note that there doesn't always exist, such a map, we shall see an example where $K_0(R) \cong \mathbb{Q}$.
6. (If you know some algebraic geometry) Recall that a sheaf is *flasque* if all restriction maps are surjective. The key interesting property of Flasque sheaves is that they have trivial cohomology, $H^i(X, \mathcal{F}) = 0$ for $i > 0$ and $\mathcal{F}$ a flasque sheaf. We shall link the $K_0(R)$ group to (co)homological considerations, and will find the following definition to be the appropriate purely algebraic counter-part. We say a ring R is *flasque* if there is an R-bimodule M that is finitely generated and projective as a right R-module, such that there is a R-bimodule isomorphism:
$$\varphi : R \oplus M \xrightarrow{\sim} M$$
From this, we get that $K_0(R) = 0$. To see this, note that:
$$P \oplus (P \otimes_R M) \cong P \otimes_R (R \oplus M) \cong (P \otimes_R M)$$
Hence $[P] + [P \otimes_R M] = [P \otimes_R M]$, implying $[P] = [0]$. If R is flasque and the right R-module structure of M is R, then R is said to be an *infinite sum ring*. The isomorphism φ becomes $R^2 \cong R$. An example of such a ring is $\mathrm{End}_R(R^\infty)$.

7. (There is an example on p.76 of the K-theory textbook I'm following on Von Neumann regular rings, which are rings where for every $x \in R$, there exists a $y \in R$ such that $xyx = x$. These are rings in which inverse need not exist, but they almost do, for example each field is Von Neumann regular rings because we can let $y = x^{-1}$. No integral domain is Von Neumann regular, but $M_n(F)$ is Von Neumann regular. There is a natural lattice structure that comes out of its idempotence that $K_0(R)$ is isomorphic to if R is a Von Neumann regular rings)

8. let $R = \varinjlim_i R_i$ where $\{R_i\}$ is a filtered system. Then every finitely generated projective R-module is of the form:
$$R \otimes_{R_i} P_i$$
where P_i is some finitely generated projective R_i-module. Any isomorphism:
$$R \otimes_{R_i} P_i \cong R \otimes_{R_i} P_i'$$
must set a finite set of generators to a finite set of generators, and so for j large enough this reduces to:
$$R_j \otimes_{R_i} P_i \cong R_j \otimes_{R_i} P_i'$$
But that means that $P(R)$ is then a filtered colimit of $P(R_i)$! We thus have the following wonderful result about colimits passing through K-groups:
$$K_0(R) = \varinjlim K_0(R_i)$$
If R is commutative, we can show that every R is the colimit of finitely generated subrings! As every finitely generated commutative ring is Neotherian with finite normalization (the integral closure is finitely generated over the ring), we may reduce studying $K_0(R)$ to the study of $K_0(R_i)$ where the R_i have nicer properties!

The following is very useful and allows for interesting geometric interpretations. For the following, we shall say that I is *complete* if every Cauchy sequence $\sum_i^\infty a_i$ where $a_i \in R/I^n$ converges to a unique element in I.

Theorem 6.2.4: K-group and Nilpotence

Let R be a ring and $I \subseteq R$ be nilpotent, or more generally complete (see definition above). Then:
$$K_0(R) \cong K_0(R/I)$$
And if R is commutative, then:
$$K_0(R) \cong K_0(R_{\text{red}})$$

***Proof*:**
We'll show $P(R) \cong P(R/I)$, which shall complete the result. We first need to build-up some important result:

lemma 1: Let P_1, P_2 be finitely generated projective R-modules and $I \subseteq R$ a radical idea. Then $P_1 \cong_R P_2$ if and only if $P_1/IP_1 \cong_R P_2/IP_2$

Proof. Certainly $P_1 \cong_R P_2$ implies $P_1/IP_1 \cong_R P_2/IP_2$, so we do the converse. There is a natural map $P_1 \to P_1/IP_1$, and then there is an isomorphism $\varphi : P_1/IP_1 \to P_2/IP_2$. Finally, there is a natural projection $\pi : P_2 \to P_2/IP_2$:

$$\begin{tikzcd} P_1 \arrow[d, "\pi"'] \arrow[dr, "\varphi\circ\pi"] & P_2 \arrow[d, "\pi"] \\ P_1/IP_1 \arrow[r, "\varphi"'] & P_2/IP_2 \end{tikzcd}$$

As P_1 is projective, there is a natural lift making the following diagram commute:

$$\begin{tikzcd} P_1 \arrow[r, "\tilde{\varphi}"] \arrow[d, "\pi"'] & P_2 \arrow[d, "\pi"] \\ P_1/IP_1 \arrow[r, "\varphi"'] & P_2/IP_2 \end{tikzcd}$$

If we show $\widetilde{\varphi}$ is an isomorphism, we're done. Note that as φ is an isomorphism, in particular it is surjective, then as I is radical it is a consequence of Nakayama's lemma that $\varphi : P_1 \to P_2$ is surjective (see exercises in [5, Chapter 20.3] or checkout [12]). Let's thus show injectivity. To that end, let us localize at each $\mathfrak{p} \subseteq R$ and consider:

$$\begin{tikzcd} P_{1,\mathfrak{p}} \arrow[r, "\tilde{\varphi}_{\mathfrak{p}}"] \arrow[d, "\pi"'] & P_{2,\mathfrak{p}} \arrow[d, "\pi"] \\ P_{1,\mathfrak{p}}/I_{\mathfrak{p}}P_{1,\mathfrak{p}} \arrow[r, "\varphi_{\mathfrak{p}}"'] & P_{2,\mathfrak{p}}/I_{\mathfrak{p}}P_{2,\mathfrak{p}} \end{tikzcd}$$

As P is a finitely generated projective module, its localization at a prime ideal is free, hence each $P_{i,\mathfrak{p}}$ has a basis. The key is that the projection onto $P_{1,\mathfrak{p}}/I_{\mathfrak{p}}P_{1,\mathfrak{p}}$ is still a basis. If we show then as $\varphi_{\mathfrak{p}}$ is an isomorphism, by commutativity the image of the basis in $\widetilde{\varphi}_{\mathfrak{p}}$ must be linearly independent which shows the map is injective. To that end, if $\{e_1, ..., e_n\}$ is a basis for $P_{1,\mathfrak{p}}$, its projection spans so let's show it is linearly independent. Say

$$\sum_i r_i \bar{e}_i = 0$$

Then $\sum_i r_i e_i \in I_{\mathfrak{p}}P_{1,\mathfrak{p}}$. By properties of ideals, $r_i \in I_{\mathfrak{p}}$, but then we must have that in the quotient $r_i = 0$ for all i, which by our previous reasoning allows us to conclude injectivity.

Hence, $\widetilde{\varphi}_{\mathfrak{p}}$ is injective for each prime $\mathfrak{p} \subseteq R$, and so it is a classical algebra result that $\widetilde{\varphi}$ is injective (see [5, chapter 20.3]. □

Now, let us show that if I is a nilpotent ideal, then there is a bijection between the finitely generated projective R-modules and the finitely generated projective R/I-modules, which shall then complete the proof. Without loss of generality we can assume that $I^2 = 0$ (we can then use induction to for the result for $I^n = 0$ for $n \geq 2$)[a]. Let us first show that an idempotent $\bar{e} \in R/I$ lifts to an idempotent $e \in R$. If r is a lift of e, then we can verify that:

$$e = r + rxr + (1-r)y(1-r)$$

where $x, y \in I$. Verify that $e^2 = e$ and it indeed maps to $\bar{e}$. This lift is unique up to conjugation by an element $u \in 1 + I$. With this lift, we shall show that every finitely generated R/I-module $\overline{P}$ is isomorphic to P/IP for an appropriate finitely generated projective module P. The key is that idempotent matrices correspond to summands of free modules. In particular if $\overline{P}$ is a finitely generated projective R/I-module, then there exists an idempotent $\overline{e} \in M_n(R/I)$ such that $\operatorname{im}(\overline{e}) = \overline{P}$. Then this e can be lifted to $e \in M_n(R)$ where $e^2 = e$ and $e \equiv \overline{e} \bmod I$, so by taking $\operatorname{im}(e) = P$ we get:

$$\overline{P} \cong P/IP$$

Finally, we are at the stage where we reduced our problem of finitely generated projective R-modules to modules of the form P/IP. Then by our lemma, $P_1/IP_1 \cong P_2/IP_2$ if and only if $P_1 \cong P_2$, showing that the isomorphism classes are in bijective corresponds, completing the proof.

[a]This is where the completeness generalization also comes in, as we can consider $\varprojlim R/I^n$

Example 6.2.5: 0-dimensional Commutative Ring

Let R be a commutative ring and R_{red} be its reduction. Then it is an easy exercise to show that R_{red} is Artinian if and only if $\operatorname{Spec}(R)$ is finite and discrete. More generally (in the "non-finite case"), if a ring is von Neumann regular (for each $x \in R$, there exists a $y \in R$ such that $yxy = x$; this is a "weak inverse"), then R has Krull dimension 0 and is reduced (in fact, this is an equivalent characterization of von-Neumann regular rings, giving a good geometric characterization of such rings[a])

[a]Note that there are many non-commutative examples, such matrix rings, left principal ideals generated by idempotents, and $\operatorname{End}_k(V)$ for a vector space, and as they are closed under direct product that makes every semi-simple von-Neumann regular ring semi-simple

6.3 Connection to H_0

We start connecting the K_0 ring to (co)homological results, starting with the most simple one of the $H_0(R)$ group, which we shall take as $\operatorname{Hom}(\operatorname{Spec}(R), \mathbb{Z})$. Recall that for a ring R, there is a natural map $\operatorname{Spec}(R) \to \mathbb{Z}$ sending $\mathfrak{p} \mapsto \dim_{\kappa(p)} M \otimes_R \kappa(\mathfrak{p})$. To see this connection, consider the following warm-up result:

Theorem 6.3.1: Pierce's Theorem

Let R be a 0-dimensional commutative ring. Then:

$$K_0(R) = \operatorname{Hom}(\operatorname{Spec}(R), \mathbb{Z})$$

Note that no assumption was made on R be finitely generated, for example it may be an infinite product of fields.

***Proof*:**

Recall that as R is 0-dimensional all its prime ideals are maximal, and $\operatorname{Spec}(R)$ has the discrete topology. Then by generalizing the classification of Artinian rings to 0-dimensional rings (in particular, every finitely generated projective R-module will be a finite product of modules of the form $R/\mathfrak{m}$), we can consider the rank of each $R/\mathfrak{m}$ individually which gives us the exact information of the $K_0(R)$ group, completing the proof.

We shall thus push this one step further and see the relation for more general commutative rings:

Definition 6.3.2: H_0 Group

Let R be a commutative ring. Then:

$$H_0(R) = [\operatorname{Spec}(R), \mathbb{Z}] = \operatorname{Hom}(\operatorname{Spec}(R), \mathbb{Z})$$

As $\operatorname{Spec}(R)$ is (quasi)compact, there are at most finitely many components , and so $H_0(R)$ will always be a free abelian group (ex. it can't be $\mathbb{Q}$). If furthermore R is Noetherian, then:

$$H_0(R) \cong \mathbb{Z}^n$$

If on top of that R is an (integral) domain (more generally $\operatorname{Spec}(R)$ is connected), then:

$$H_0(R) \cong \mathbb{Z}$$

Lemma 6.3.3: H_0 direct summand of K-group

The group $H_0(R)$ is a direct summand of $K_0(R)$. If $\dim R = 0$, then $K_0(R) \cong H_0(R)$.

***Proof*:**
Take the monoid L of the componentwise free modules R^n. Then as we saw, L is cofinal, and $L \to P(R)$ is a semiring map which naturally gives a map $L^{-1}L \to K_0(R)$. Then as L is isomorphic to $\operatorname{Hom}(\operatorname{Spec}(R), \mathbb{N})$, we must have $L^{-1}L \cong H_0(R)$, and it a subgroup.

Now, there is a natural map $K_0(R) \to H_0(R)$ given by the rank, namely there is a natural map from rank : $P(R) \to [\operatorname{Spec}(R), \mathbb{N}]$, which is a semiring homomorphism. Thus, we get $H_0(R) \to K_0(R) \to H_0(R)$, is the identity, and so $H_0(R)$ is a direct summand of $K_0(R)$, as we sought to show.

For the special case of $\dim(R) = 0$, By Pierce's Theorem we have $K_0(R) \cong H_0(R)$.

As $K_0(R)$ splits with $H_0(R)$, the other component is the kernel of the rank map. We shall denote it $\widetilde{K}_0(R)$ so that:

$$K_0(R) \cong H_0(R) \oplus \widetilde{K}_0(R)$$

It shall be shown later that $\widetilde{K}_0(R)$ is a nilradical, and as $H_0(R)$ is reduced already we would have that it is the nilradical of $K_0(R)$.

Lemma 6.3.4: Limit and Kernel

Let R be a commutative, let $P_n(R)$ denote the subset of all modules in $P(R)$ consisting of projective modules of constant rank n. There is a map $P_n(R) \to K_0(R)$ sending

$$P \mapsto [P] - [R']$$

This map is compatible with the stabilization map $P_n(R) \to P_{n+1}(R)$ sending $P \mapsto P \oplus R$, and the induced map is an isomorphism:

$$\varinjlim_n P_n(R) \cong \widetilde{K}_0(R)$$

***Proof*:**
Follows from 6.1.3.

Though we could take the colimit, by the base cancellation theorem we know that the interesting "geometric" structure of a finitely generated projective R-module over a ring of rank d can only happen for modules of rank $\leq d$:

Corollary 6.3.5: Higher Dimension Nilradical

Let R be a commutative Noetherian ring where $\text{kdim}R = d$. Then for every $n > d$,

$$P_n(R) \cong \widetilde{K}_0(R)$$

***Proof*:**
Let P, Q be finitely generated projective modules of rank $> d$. Then by the Base cancellation Theorem (theorem 5.1.12), we get that:

$$[P] = [Q] \text{ if and only if } P \cong Q$$

giving the desired result.

The H_0 group is based off of the spectrum of R, and hence is dependent on its commutativity. Let us now try to bring this interpretation to non-commutative rings. To do this, another interpretation of the kernel that is rather nice is that it is the intersection of all the kernels of map $K_0(R) \to K_0(F)$ for all maps $R \to F$ into a field: as rank is a natural map, $\widetilde{K}_0(F) = 0$, and $H_0(R)$ captures the information at each prime ideals (i.e. each stalk), where we can get this information by going over all fields for which we can have $R \to F$.

We may thus take the non-commutative equivalent of a field: a simple Artinian ring. As maximal commutative subrings of a simple Artinian ring S is a finite product o 0-dimensional local rings, we get that this is the correct generalization. We can define $\widetilde{K}_0(R)$ as the intersection of the kernels of $K_0(R) \to K_0(S)$ where S is a simple Artinian ring. If no map $R \to S$ exists, then we say $\widetilde{K}_0(R) = K_0(R)$. We define:

$$H_0(R) = \frac{K_0(R)}{\widetilde{K}_0(R)}$$

We saw that $H_0(R)$ is a free-abelian group in the commutative case. In the non-commutative case it is at least still torsion-free. To see this, let X be the set of homomorphisms $R \to S_x$ through which any other $R \to S'$ factors (such a set exists by using Zorn's lemma). Then as $K_0(S_x) \to K_0(S')$ is an isomorphism. Hence, $\widetilde{K}_0(R)$ is the intersection of all kernels of maps $K_0(R) \to K_0(S_x)$. We also then have that $H_0(R)$ is is the image of $K_0(R)$ in the torsion free group:

$$\prod_{x \in X} K_0(S_x) \cong \prod_{xinX} \mathbb{Z}$$

(The author of the textbook I'm reading talks a bit about whitehead groups, which I am omitting for now)

6.3.1 Some Important Functionals

We are in the realm of representation theory, hence we have our usual trace and determinant map that shows up. In this section, we shall see how these maps give us some interesting information on our K_0 and H_0 group.

Hattori-Stallings Trace Map

Let R be an associative ring, and let $[R, R] = \{rs - sr \mid r, s \in R\}$ be the commutator subgroup. Then the trace provides a natural map:

$$\mathrm{tr} : M_n(R) \to \frac{R}{[R, R]}$$

where we quotient by $[R, R]$ as tr should be be independent of conjugation, which shall require $a_{i\ell}b_{\ell i}$ to commute. Then it should be a simple computation to see that all traceless matrices are in $[M_n(R), M_n(R)]$ which gives us the isomorphism:

$$\frac{M_n(R)}{[M_n(R), M_n(R)]} \cong \frac{R}{[R, R]}$$

Naturally, not all elements of $\mathrm{End}_R(P)$ of a finitely generated projective R-module P is in matrix form, however it close enough to still define the trace: as P is finitely generated and projective choose Q such that $P \oplus Q \cong R^n$. Then we have an idempotent element $e \in M_n(R)$ such that $e(R^n) = P$, so:

$$\mathrm{End}_R(P) = eM_n(R)e$$

It is not too hard to verify that any other choices of idempotent will yield an e_1 which is idempotent and conjugate to e in some $M_m(R)$. Thus, we have that the trace map is independent of representative and we have a well-defined map:

$$\mathrm{tr} : \mathrm{End}_R(P) \to \frac{R}{[R, R]}$$

The trace of the identity map on P will be the trace of the idempotent element e; we call this particular value the *trace* of P.

Now, if P' is a finitely generated projective R-module represented by an idempotent f, then $P \oplus P'$ is represented by $\begin{pmatrix} e & 0 \\ 0 & f \end{pmatrix}$, and so we have its trace is $\mathrm{tr}(P) + \mathrm{tr}(P')$! Thus, the trace is an additive map on $P(R)$, and so we get a natural map:

Definition 6.3.6: Hattori-Stallings Trace Map

Let R be a (not necessarily commutative ring). Then by the above discussion, the natural map:

$$K_0(R) \to \frac{R}{[R, R]}$$

is called the *Hattori-Stallings Trace Map*.

In the case where R is commutative, we have that $R = \frac{R}{[R,R]}$. This gives a more explicit description of $K_0(R)$ and $H_0(R)$ (as we have $\mathrm{Spec}\,(R)$ well-defined). To see this, start by taking the trace map on $K_0(R) \to R$ and restrict to $H_0(R)$. Then notice that each $\mathrm{Spec}\,(R) \to \mathbb{Z}$ gives a decomposition of $\mathrm{Spec}\,(R)$ into components $R = R_1 \times \cdots \times R_c$ given by idempotents $e_1, ..., e_c$. As $\mathrm{tr}(e_i R) = e_i$, we have that for any $f \in R$,

$$\mathrm{tr}(f) = \sum_i f(i) e_i$$

And so we get:

$$\mathrm{tr}(fg) = \mathrm{tr}(f)\mathrm{tr}(g)$$

We thus get the following natural map:

Proposition 6.3.7: Hattori-stallings Map Factorization

Let R be *commutative*. Then the Hattori-Stallings trace map factors through:

$$K_0(R) \xrightarrow{\text{rank}} H_0(R) \to R$$

***Proof*:**
Consider the following diagram:

$$\begin{tikzcd} K_0(R) \arrow[r] \arrow[d, "\mathrm{tr}"'] & \prod_{\mathfrak{p}} K_0(R_{\mathfrak{p}}) \arrow[d, "\mathrm{tr}"] \\ R \arrow[r, hook, "\mathrm{diag}"] & \prod_{\mathfrak{p}} R_{\mathfrak{p}} \end{tikzcd}$$

As the kernel of the top arrow is $\widetilde{K}_0(R)$ by construction, we get the left arrows map to $H_0(R)$, which gives the desired factorization.

Example 6.3.8: Trace and Characters on Representations

1. Let k be commutative and $R = k[G]$ of a group G. Then if $g, h \in G$ are conjugate, then $g - h \in [R, R]$ (as $g = xhx^{-1}$ so $xhx^{-1} - h \in \left[xh, x^{-1}\right]$). Then we see that $\frac{R}{[R,R]} \cong \oplus k[g]$ for $g \in G/\sim$ (i.e. up to conjugacy class). Given such a basis, we then get:

$$\mathrm{tr}(P) = \sum_{g \in G} r_P(g)[g]$$

   where $r_P(g)$ are coefficients seen as functions of the set $G/\sim$, for each P.

2. In the special case where G is finite, then any finitely generated projective $k[G]$-module P is also a projective k-module, and so we can form the trace map $\mathrm{End}_k(P) \to k$! Hence, we get the character $\chi_P : G \to k$ given by the formula:

$$\chi_P(g) = \mathrm{tr}(g)$$

   It was proven by a mathematician called *Hattori* that we can write:

$$\chi_P(g) = |Z_G(g)| r_P(g^{-1})$$

which gives a very interesting connection between characters and the general theory of trace we've been building up! We shall not look into this more deeply, but it is an interesting result to point out.

One very interesting result we get out of this is that $\mathbb{Z}[G]$ does not split, i.e. it has no non-trivial idempotents:

Proof. Let $e \in \mathbb{Z}[G]$ be an idempotent element. Then $\chi_P(1)$ is the rank of the $\mathbb{Z}$-module $P = e\mathbb{Z}[G]$. Then by construction its rank is $\leq |G| = \text{rank}\mathbb{Z}[G]$. As $r_P(1) \in \mathbb{Z}$, this contradicts Hattori's formula as $\chi_P(1) = |G|r_P(1)$, completing the proof. □

3. If k is a field of characteristic 0 and G a finite group, then by Maschke's Theorem we have $k[G]$ is the simple product of k-algebras $S_1 \times \cdots \times S_c$ where c represents the conjugacy classes of G. Hence we get:
$$\frac{k[G]}{[k[G], k[G]]}$$
Then we saw in example 6.2.3 that
$$K_0(K[G]) \cong \mathbb{Z}^c$$
By Hattori's formula, we can see that the trace map from $K_0(K[G])$ is isomorphic to $\mathbb{Z}^c \subseteq k^c$.

Determinant

Recall definition 5.2.9 where we can define the determinant in a natural way if R is commutative on a finitely generated projective R-module. We saw in proposition 5.2.11 that we may classify all finitely generated projective R-modules over a 1-dimension ring, and in proposition 5.2.11 we saw that the determinant of a finitely generated projective R-module P can be seen as elements of $\text{Pic}\,(R)$. These results work when reducing to K-groups:

Proposition 6.3.9: K_0 Group and Determinant

Let R be a commutative ring. Then the determinant determines a natural map:
$$\det : K_0(R) \to \text{Pic}\,(R)$$

***Proof*:**
The key is that the non-top wedge products disappear. By the universal properties of monoids, it suffices to show that:
$$\det(P \oplus Q) \cong \det(P) \otimes \det(Q)$$
Without loss of generality, we may assume that P, Q have constant rank n, m respectively. Then $\bigwedge^{n+m} P \oplus Q$ is the sum over all i, j such that $i + j = n + m$ in $(\bigwedge^i P) \otimes (\bigwedge^j Q)$. Then if $i > n$ and $j > m$ then either $\bigwedge^i P = 0$ or $\bigwedge^j Q = 0$. But then:
$$\bigwedge^{n+m} P \oplus Q \cong (\bigwedge^{n} P) \otimes (\bigwedge^{m} Q)$$
as we sought to show.

Corollary 6.3.10: Rank and Determinant Represents K-group

Let R be a commutative ring. Then $H_0(R) \oplus \text{Pic}(R)$ is a ring with a square-zero ideal $\text{Pic}(R)$ and there is a surjective homomorphism:

$$K_0(R) \twoheadrightarrow H_0(R) \oplus \text{Pic}(R)$$

with kernel consists of equivalence classes representable by $[P] - [R^n]$ where P has constant rank n and $\bigwedge^n P \cong R$.

***Proof*:**
Let $SK_0(R)$ represent the kernel. Notice first that this is naturally the kernel of $\det : K_0(R) \to \text{Pic}(R)$ by the above proposition. Next, $SK_0(R)$ is an ideal of $K_0(R)$. After showing this, the rest is combining earlier results.

Corollary 6.3.11: Rank and Determinant: One Dimensional Case

Let R be a commutative Noetherian 1-dimensional ring. Then from the above we have:

$$K_0(R) \cong H_0(R) \oplus \text{Pic}(R)$$

***Proof*:**
This follows from the above corollary and proposition 5.2.10.

6.4 Morita Equivalence

An important invariant in the study of modules is the one given by the *Morita Equivalence*, which tells you when a ring action produces the same theory:

Definition 6.4.1: Morita Equivalence

Let R, S be (not necessarily commutative) rings. Then R-**Mod** and S-**Mod** are *Morita equivalent* if they are equivalent, namely there exists functors F, G such that $GF \cong \text{id}_R$ and $FG \cong \text{id}_S$ where id_R and id_S are the identity functors on their respective categories

We have in fact already seen many examples of Morita equivalence without invoking it, as we shall see in examples 6.4.4.

Note that F, G being inverses implies they preserve limits and colimits, hence the structure of in equivalence categories share many similarities (for example, direct sums and products pass through F, G).

We can in fact have a precise idea of what is the image of F, G. Let $F(R) = P$ and $G(S) = Q$. Then notice we have the natural maps:

$$R \cong \text{End}_R(R)^{\text{op}} \xrightarrow{F} \text{End}_S(P)^{\text{op}} \qquad \text{and} \qquad S \cong \text{End}_S(S)^{\text{op}} \xrightarrow{G} \text{End}_R(Q)^{\text{op}} \tag{6.1}$$

which make P into an S, R-bimodule and Q into a R, S-bimodule (if this is unfamiliar to you, see [5, chapter 23.2] for a review of the relation between R and its endomorphisms). Now, since F, G preserve limits/colimits, $F(R^{\oplus}) = P^{\oplus}$ and $G(S^{\oplus}) = Q^{\oplus}$, and any relation is preserved as equalizers are preserved, hence we have that for any $M \in \text{Obj}({}_R\mathbf{Mod}), N \in \text{Obj}({}_S\mathbf{Mod})$

$$F(M) \cong P \otimes_R M \qquad \text{and} \qquad G(N) \cong Q \otimes_S N$$

as bimodules, and:

$$GF(R) = G(P) = Q \otimes_S P \overset{!}{\cong} R \qquad \text{and} \qquad FG(S) = F(Q) = P \otimes_R Q \overset{!}{\cong} S$$

giving a very structured result for Morita equivalent categories! The $\overset{!}{\cong}$ comes from how we can think of the endomorphisms as being the "negative dimensional" lines bundles, or for more rigor you can checkout [5, chatper 24.2]. The following summarizes our above discussion and adds some useful insights the user may appreciate:

Theorem 6.4.2: Structure Theorem For Morita Equivalence

Let R, S be Morita Equivalent rings (represented by the functors F, G) and $F(R) = P$, $G(S) = Q$. Then:

1. P, Q are finitely generated projective both as R-modules and S-modules
2. $\text{End}_S(P) \cong R \cong \text{End}_S(Q)^{\text{op}}$ and $\text{End}_R(Q) \cong S \cong \text{End}_R(P)^{\text{op}}$
3. P, Q are dual to each other as S-modules:

   $$P \cong \text{Hom}_S(Q, S) \qquad \text{and} \qquad Q \cong \text{Hom}_S(P, S)$$

   The tensor gives us the R-tensor equivalent. Furthermore, $\text{Hom}_S(P, -)$ and $\text{Hom}_R(Q, -)$ (note it is over R) are faithful functors from **Mod** to **Ab** (for the respective rings).
4. The image of F, G are isomorphic (as bimodules) to:

   $$F(M) \cong P \otimes_R M \qquad \text{and} \qquad G(N) \cong Q \otimes_S N$$

***Proof*:**

1. This comes from $Q \otimes_S P \cong_R R$ and $P \otimes_R Q \cong_S S$
2. Follows from equation 6.1.
3. summarizes the above discussion
4. ibid

Corollary 6.4.3: Morita Equivalence and K Group

Let R, S be Morita Equivalent. Then

$$K_0(R) \cong K_0(S)$$

***Proof*:**
Since P, Q are finitely generated and projective, and F, G represent the equivalence, we get

$$P(R) \cong P(S)$$

giving the desired result

Naturally, having isomorphic K-groups does not imply Morita equivalence (think $K_0(\mathbb{Z})$ and $K_0(\mathbb{Q})$).

Example 6.4.4: Morita Equivalence
The prototypical example is R and $M_n(R)$ for $n \geq 1$. In this case, $F(R) = P$ is the bimodule R^n of "column vectors" while $Q(M_n(R)) = (R^n)^t$ is the bimodule of "row vectors".

More generally, by theorem 6.4.2(3), if P if a faithful finitely generated projective R-module, then $\mathrm{End}_R(P)$ is Morita Equivalent to R.

6.5 *Mayer-Vietoris Sequence

In algebraic topology, the Mayer-Vietoris sequence is a useful sequence for computing homotopic groups by gluing together pieces. We bring over this discussion for computing K groups when gluing user Milnor squres. As we saw via the Hattori-Stallings Trace Map allows for some very important decompositions:

$$R \to \frac{R}{[R, R]}$$

As we just saw, $M_n(R)$ produces equivalent categories. More generally, in representation theory we are interested in $\mathrm{GL}_n(R)$, the general linear group. Which n should we pick? Can we relate the n's in Chapter 7, we shall take the limit over all the n to define the group $\mathrm{GL}(R)$. For now, we relate these through the natural map:

$$g \mapsto \begin{pmatrix} g & 0 \\ 0 & 1 \end{pmatrix}$$

These groups shall be central in understanding K_1 in the next section. One important construction we shall use is patching together information using Milnor patching: consider the Milnor square:

$$\begin{array}{ccc} R & \xrightarrow{f} & S \\ \downarrow & & \downarrow \\ R/I & \xrightarrow{\bar{f}} & S/I \end{array}$$

Recall theorem 5.1.14(3) that any finitely generated projective R-module can derive from patching, hence define

$$\partial_n : \mathrm{GL}_n(S/I) \to K_0(R)$$

where $\partial_n(g) = [P] - [R^n]$. By theorem 5.1.14(4), we see that:

$$\partial_n(g) = \partial_{n+1} \begin{pmatrix} g & 0 \\ 0 & 1 \end{pmatrix}$$

Hence, taking the sequence of $\{\partial_n\}_{n=1}^{\infty}$ give a homomorphism:

$$\partial : \mathrm{GL}(S/I) \to K_0(R)$$

Then from Milnor-Patching

Theorem 6.5.1: Mayer-Vietoris

Given a Milnor square as above, the sequence

$$GL(S/I) \xrightarrow{\partial} K_0(R) \xrightarrow{\Delta} K_0(S) \oplus K_0(R/I) \xrightarrow{\pm} K_0(S/I)$$

is exact. The image of ∂ is the double coset space

$$GL(S)\backslash GL(S/I)/GL(R/I) = GL(S/I)/\sim$$

where $x \sim gxh$ for $x \in GL(S/I), g \in GL(S)$, and $h \in GL(R/I)$.

***Proof*:**
Follows from theorem 5.1.14(4) and our above discussion.

The Mayer-Vietoris sequence is a useful computation tool, so let's show how it can be used:

Example 6.5.2: Mayer-Vietoris
Consider the coordinate ring of an irreducible curve with a node (which can be shown is unique up to projective equivalence[a]). Let us compute K_0. As this is a 1-dimensional Noetherian ring, by corollary 6.3.11 we have:

$$K_0(R) \cong H_0(R) \oplus \operatorname{Pic}(R)$$

Then the result follows from the following exact sequence known as the *Units-Pic sequence*:
Let Δ denotes the diagonal map, $\pm$ the difference map sending $(s, \bar{r})$ to $\bar{s}f(\bar{r})^{-1}$, resp. (L', L) to $L' \otimes_S S/I \otimes_{R/I} L^{-1}$. Then given a Milnor square, the following is exact:

$$1 \to R^\times \xrightarrow{\Delta} S^\times \times (R/I)^\times \xrightarrow{\pm} (S/I)^\times \xrightarrow{\partial}$$
$$\mathrm{Pic}(R) \xrightarrow{\Delta} \mathrm{Pic}(S) \times \mathrm{Pic}(R/I) \xrightarrow{\pm} \mathrm{Pic}(S/I).$$

Proof. As R is the pullback of S and R/I, exactness at the first two places is immediate. Theorem 5.1.14 immediately gives exactness at the last two places, leaving only exactness at $(S/I)^\times$. Given $s \in S^\times$ and $\bar{r} \in (R/I)^\times$, set $\beta = \pm(s, \bar{r}) = \bar{s}f(\bar{r})^{-1}$, where $\bar{s}$ denotes the reduction of s modulo I. Then we see that $\lambda = (s, \bar{r}) \in L_\beta \subset S \times R/I$, and every element of L_β is a multiple of λ. It follows that $L_\beta \cong R$. Conversely, suppose given $\beta \in (S/I)^\times$ with $L_\beta \cong R$. If $\lambda = (s, \bar{r})$ is a generator of L_β, then I claim that s and $\bar{r}$ are units, which implies that $\beta = \bar{s}f(\bar{r})^{-1}$ which completes the proof. To that end, if $s' \in S$ maps to $\beta \in S/I$, then $(s', 1) \in L_\beta$, namely since $(s', 1) = (xs, x\bar{r})$ for some $x \in R$, this implies that $\bar{r} \in (R/I)^\times$. If $t \in S$ maps to $f(\bar{r})^{-1}\beta^{-1} \in S/I$, then $st \equiv 1$ modulo I. But then $I \subset sR$ because $I \times 0 \subset L_\beta$, so $st = 1 + sx$ for some $x \in R$. So $s(t - x) = 1$, and hence $s \in S^\times$ giving the desired result. □

Hence, we get that:

$$K_0(R) \cong \mathbb{Z} \oplus k^\times$$

[a] Equivalence after a linear isomorphism

7

K_1 and K_2 Group

The K_0 group let us classify the vector bundles over a ring or (paracompact) topological space. We saw in section 2.1 that for (say complex) vector bundles over X:

$$K_0(X) = [X, \mathbb{Z} \times BU]$$

Showing there is a relation between homotopy groups and $K_0(X)$. From the algebraic perspective, we found the isomorphism classes of the projective modules. It is now natural to ask for the automorphism classes. We shall focus on the algebraic perspective, but shall see that it will naturally tie with the topological by taking suspension (recall from [9, chapter 5] that there is a natural map $\pi_k(X) \to \pi_{k+1}(\Sigma X)$).

7.1 Defining K_1

Let R be a ring (with unit). Take the maps:

$$G_n(R) \to G_{n+1}(R) \qquad g \mapsto \begin{pmatrix} g & 0 \\ 0 & 1 \end{pmatrix}$$

The limit of these maps shall be denoted $\mathrm{GL}(R)$. Then:

Definition 7.1.1: K_1 Group

Let R be a ring (not necessarily commutative). Then:

$$K_1(R) = \frac{GL(R)}{[GL(R), GL(R)]}$$

Given a ring homomorphism $R \to S$, there is a naturally induced map $\mathrm{GL}(R) \to \mathrm{GL}(S)$ by mapping the elements of R to the diagonal, and as this respects the abelianization we have a natural map:

$$K_1(R) \to K_1(S)$$

And hence, $K_1(-)$ is a functor as well.

Before looking at examples, let us have a better grasp of the commutator. As it turns out, these shall always be generated by elementary matrices.

Definition 7.1.2: Elementary Matrices Group of Gl(R)

Let $E_n(R)$ denote the elemetnary matrices of $\mathrm{GL}_n(R)$. Then $E(R)$ denotes the limit of the stabilization morphisms.

Lemma 7.1.3: Elementary Matrix Group is Perfect

For $n \geq 3$, $E_n(R)$ is a prefect group, that is $E_n(R) = [E_n(R), E_n(R)]$

For $n = 2$, we get $\mathrm{SL}_2(\mathbb{F}_2) \cong S_2$ (symmetric group of order 2).

***Proof*:**
For distinct i, j, k, we get:

$$[e_{ij}(r), e_{k\ell}(s)] = \begin{cases} 1 & \text{if } j \neq k \text{ and } i \neq \ell, \\ e_{i\ell}(rs) & \text{if } j = k \text{ and } i \neq \ell, \\ e_{kj}(-sr) & \text{if } j \neq k \text{ and } i = \ell. \end{cases}$$

and so:

$$e_{ij}(r) = [e_{ik}(r), e_{kj}(1)]$$

Theorem 7.1.4: Whitehead's Lemma

$$K_1(R) = \frac{\mathrm{GL}(R)}{E(R)}$$

***Proof*:**
The commutator subgroup contains $E(R)$ by lemma 7.1.3. Conversely, every commutator in $GL_n(R)$ can be expressed as a product in $GL_{2n}(R)$:

$$[g, h] = \begin{pmatrix} g & 0 \\ 0 & g^{-1} \end{pmatrix} \begin{pmatrix} h & 0 \\ 0 & h^{-1} \end{pmatrix} \begin{pmatrix} (hg)^{-1} & 0 \\ 0 & hg \end{pmatrix}. \tag{7.1}$$

And certainly each of these terms is in $E_{2n}(R)$, giving the desired equivalence.

Example 7.1.5: K_1 Groups

1. Let R be commutative. Then we have a commutative group homomorphism:

$$\det : \mathrm{GL}(R) \to R^\times$$

   This induces a surjective map $\det : K_1(R) \to R^\times$ which whose kernel is written as $SK_1(R)$ (for reasons explained in a paragraph). Now, we also have a natural inclusion $R^\times \to \mathrm{GL}(R)$ mapping to the diagonals. We see that this map is a section map, i.e. it split by the determinant homomorphism. Taking $\mathrm{SL}(R) = \varprojlim_n SL_n(R)$, as we are working over groups we have the semi-direct product (see [5, chatper 5.3]):

$$\mathrm{GL}(R) \cong \mathrm{SL}(R) \rtimes R^\times$$

   Hence, when abelianizing we get:

$$K_1(R) \cong R^\times \oplus SK_1(R)$$

2. If we specifically let $R = k$ be a field where $k \notin \{\mathbb{F}_2, \mathbb{F}_3\}$, then we can show that $\mathrm{SL}(F) = [\mathrm{GL}(F), \mathrm{GL}(F)]$, and so we reduce to:

$$K_1(F) \cong F^\times$$

   To see this, we first certainly see that $[\mathrm{GL}(R), \mathrm{GL}(R)] \leqslant \mathrm{SL}(R)$ as each commutator $[A, B] = ABA^{-1}B^{-1}$ as determinant 1. For the other way around, we must show that each matrix in $\mathrm{SL}_n(R)$ can be written as a product of commutators. As we are working over a field, it suffices to show that all elementary matrices can be written as a product of commutators. But this is now a matter of computations, for example type I elementary matrices $A = I + tE_{ij}$ and $B = I + E_{jk}$ where $k \neq i, j$ (which is possible since $n \geq 3$), when taking their commutator we get:

$$[A, B] = ABA^{-1}B^{-1} = (I + tE_{ij})(I + E_{jk})(I - tE_{ij})(I - E_{jk}) = I + tE_{ik}$$

   notice how it was importance for $n \geq 3$ to insure the computations follow.

3. If $R = R' \times R''$ is a product of rings, then $\mathrm{GL}(R) = \mathrm{GL}(R') \times \mathrm{GL}(R'')$, and the commutator decompose too giving us that products are respected by the K_1

4. The K_1 group is invariant under Morita equivalence. First, as R and $M_n(R)$ are Morita equivalent, we have that:

$$M_{mn}(R) = \mathrm{End}_R(R^m \otimes R^n) \qquad M_m(M_n(R)) = \mathrm{End}_{M_n(R)}\left((M_n(R))^m\right)$$

   And so, we have a natural isomorphism:

$$\mathrm{GL}_{mn}(R) \cong \mathrm{GL}_m(\mathrm{GL}_n(R))$$

   which give the same "stabilization" morphism (the map between $\mathrm{GL}_n(R)$ and $\mathrm{GL}_{n+1}(R)$), and so:

$$K_1(R) \cong K_1(M_n(R))$$

5. Let $R = D$ be a division ring. Then for the same reason as field, every invertible matrix may be reduced to:
$$\mathrm{diag}(r, 1, ..., 1)$$
and so we have:
$$\frac{\mathrm{GL}_n(R)}{E_n(D)} \cong D^\times/N$$
By lemma 7.1.8, we can reduce to:
$$\frac{\mathrm{GL}_n(R)}{E_n(D)} \cong D^\times/[D^\times, D^\times]$$
for all $n > 1$.

6. Let $F/\mathbb{Q}$ be a finite field extension and $R \subseteq F$ the integrally closed subring (i.e. the algebraic integers). Then it is a result by Bass and Milnor that
$$SK_1(R) = 1$$
and so $K_1(R) = R^\times$. Then by Dirichlet Unit theorem (see [5, chapter 22]), we have that $R^\times$ is isomorphic to:
$$\mu(F) \oplus \mathbb{Z}^{s-1}$$
where $\mu(F)$ is the cyclic group of all roots of unity in F and s is the places at infinity of R, i.e. the number of real embeddings.

Let us take a moment to look at the K_1 group of semi-local rings due to their importance in representation theory (they are the "local rings" in the non-commutative case).

Definition 7.1.6: Semi-local Ring

Let R be a ring. Then R is *semilocal* if $R/J(R)$ is semisimple where $J(R)$ is the (left) Jacobian ideal.

Note that all division rings are semi-local, hence the discussion about semi-local rings generalizes the above discussion.

We shall find a representation of $[R^\times, R^\times]$ for semilocal rings. If $r, s \in R$ are such that $1 + rs$ is a unit, then so is $1 + sr$ because $(1 + sr)(1 - s(1 + rs)^{-1}r) = 1$. Then define subgroup $W(R) \leqslant R^\times$ generated by the $(1 + rs)(1 + sr)^{-1}$. It is not too hard to show that this subgroup belongs to $E_2(R)$. For $R = M_2(\mathbb{F}_2)$, $W(R) = R^\times \cong \Sigma_3$ and $K_1(R) = 1$ but $R^\times \neq [R^\times, R^\times]$. If T is the subring of upper triangular matrices in $M_2(\mathbb{F}_2)$, its group of units is abelian ($T^\times \cong \mathbb{Z}/2$), but $K_1(T) = 1$ since
$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = (1 + rs)(1 + sr)^{-1} \text{ for } r = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \text{ and } s = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Now, a mathematician called *Vaserstein* showed that that $W(R) = [R^\times, R^\times]$ if $\Lambda = R/\mathrm{rad}(R)$ is a product of matrix rings, none of which is $M_2(\mathbb{F}_2)$, and at most one of the factors in Λ is $\mathbb{F}_2$. In particular, $W(R) = [R^\times, R^\times]$ for every local ring R. The same applies for semilocal rings:

Proposition 7.1.7: K1 of Semilocal Rings

If R is a semilocal ring, then the natural inclusion of $R^\times = GL_1(R)$ into $GL(R)$ induces an isomorphism $K_1(R) \cong R^\times/W(R)$.
If R is a commutative semilocal ring, then

$$SK_1(R) = 1 \quad \text{and} \quad K_1(R) = R^\times.$$

***Proof*:**
By example 7.1.5 it suffices to prove that $R^\times$ maps onto $K_1(R)$. This will follow by induction after showing that $GL_n(R) = E_n(R)GL_{n-1}(R)$. Let J denote the Jacobson radical of R, so that R/J is a finite product of matrix algebras over division rings. We will show that $\mathrm{GL}_n(R) \twoheadrightarrow \mathrm{GL}_n(R/I)$. Indeed, if $I \subseteq R$ is an ideal, each homomorphism $E_n(R) \to E_n(R/I$ is onto as the generators $e_{ij}(r)$ of $E_n(R)$ map onto the generators of $E_n(R/I)$ as I is radical. Hence, $(R/J)^\times$ maps onto $K_1(R/J)$; every $\bar{g} \in GL_n(R/J)$ is a product $\bar{e}\bar{g}_1$, where $\bar{e} \in E_n(R/J)$ and $\bar{g}_1 \in GL_1(R/J)$.

Now, given $g \in GL_n(R)$, its reduction $\bar{g}$ in $GL_n(R/J)$ may be decomposed into $\bar{g} = \bar{e}\bar{g}_1$. Then we can lift $\bar{e}$ to an element $e \in E_n(R)$. The matrix $e^{-1}g$ is congruent to the diagonal matrix $\bar{g}_1$ modulo J, so its diagonal entries are all units and its off-diagonal entries lie in J. Using elementary row operations $e_{ij}(r)$ with $r \in J$, it is an it is a matter of computation to reduce $e^{-1}g$ to a diagonal matrix, say to $D = \mathrm{diag}(r_1, ..., r_n)$. But then the matrix matrix $\mathrm{diag}(1, ..., 1, r_n, r_n^{-1})$ is in $E_n(R)$. Multiplying D by this matrix yields a matrix in $GL_{n-1}(R)$, giving the induction and the proof.

Lemma 7.1.8: Determinant on semilocal Rings

Let R be a noncommutative semilocal ring ($R/J(R)$ is a semisimple ring). Then there exists a unique determinant map from

$$\det : GL_n(R) \twoheadrightarrow R^\times/W(R)$$

where:

1. $\det(e) = 1$ for every elementary matrix e,
2. if $\rho = \mathrm{diag}(r, 1, ..., 1)$ and $g \in GL_n(R)$, then $\det(\rho \cdot g) = r \cdot \det(g)$. Then show that det is a group homomorphism: $\det(gh) = \det(g)\det(h)$.

In particular, when reducing to $K_1(R)$:

$$K_1(R) \cong R^\times/W(R)$$

***Proof*:**
Since R is a semilocal ring, every matrix in $GL_n(R)$ can be expressed as a product of elementary matrices. Define the map $\det : GL_n(R) \to R^\times/W(R)$ as follows. For any elementary matrix e, we set $\det(e) = 1$. For a diagonal matrix $\rho = \mathrm{diag}(r_1, r_2, \ldots, r_n)$, define

$$\det(\rho) = r_1 r_2 \cdots r_n \cdot W(R)$$

For a general matrix $g \in GL_n(R)$, express g as a product of elementary matrices and diagonal matrices, and define $\det(g)$ as the product of the determinants of these factors. As any matrix can be written as product of elementary matrices, this map is the unique determinant map (if it satisfies the determinant properties). And indeed, let $\rho = \mathrm{diag}(r, 1, \ldots, 1)$ and $g \in GL_n(R)$. Express g as a product $g = e_1 e_2 \cdots e_k \cdot d$, where each e_i is an elementary matrix and d the diagonal matrix. Then $\rho \cdot g = \rho \cdot e_1 e_2 \cdots e_k \cdot d$. Then this is equivalent to multiplying the first row of g by r, which multiplies the determinant by r. Hence,

$$\det(\rho \cdot g) = r \cdot \det(g)$$

Next, we must show det is a group homomorphism

$$\det(gh) = \det(g)\det(h) \text{ for all } g, h \in GL_n(R)$$

It suffices to establish this for elementary and diagonal matrices. For elementary matrices e_1 and e_2, we have $\det(e_1 e_2) = 1 = \det(e_1)\det(e_2)$. For diagonal matrices $d_1 = \mathrm{diag}(r_1, \ldots, r_n)$ and $d_2 = \mathrm{diag}(s_1, \ldots, s_n)$, we have $\det(d_1 d_2) = \det(\mathrm{diag}(r_1 s_1, \ldots, r_n s_n)) = r_1 s_1 \cdots r_n s_n \cdot W(R) = (r_1 \cdots r_n \cdot W(R))(s_1 \cdots s_n \cdot W(R)) = \det(d_1)\det(d_2)$.

Finally, by definition $K_1(R)$ is the abelianization of $GL(R) = \varinjlim_n GL_n(R)$. The determinant map we constructed gives a homomorphism from $GL_n(R)$ to $R^\times/W(R)$ for each n, which are compatible under stabilization. Since $R^\times/W(R)$ is abelian and the determinant map captures exactly as we saw in example 7.1.5

$$K_1(R) \cong R^\times/W(R)$$

as we sought to show

With the basic properties of K_1 established on rings. Let us start by applying it to finitely generated projective R-modules. As $K_1(R)$ is defined as an amalgamation of all the "automorphisms" representable as matrices, one first thing we may want to find is a natural relation between a finitely generated projective R-module, say P, is the relation between $\mathrm{Aut}(P)$ and $K_1(R)$. To that end, as P is projective, we know there exits a finitely generated projective R-module Q such that

$$P \oplus Q = R^n$$

Then we use this to define a natural map $\mathrm{Aut}(P) \to \mathrm{GL}_n(R)$ where $\alpha \mapsto \alpha \oplus 1_Q$. This is the basis the map to $K_1(R)$:

Lemma 7.1.9: Automorphism Map

There exits a well-defined map $\mathrm{Aut}(P) \to \mathrm{GL}(R)$ up to inner automoprhism of $\mathrm{GL}(R)$, and hecne a well define homomorphism:

$$\mathrm{Aut}(P) \to K_1(R)$$

***Proof*:**
First, let's show it is defined up to inner automorphism. Fix Q and n. Then Two different isomorphisms between $P \oplus Q$ and R^n must differ by an automorphism of R^n, i.e., by an element

$g \in GL_n(R)$. Thus if $\alpha \in \mathrm{Aut}(P)$ maps to the matrices A and B, respectively, we have:

$$A = gBg^{-1}$$

Next we observe that we can stabilize; replacing Q by $Q \oplus R^m$ and $P \oplus Q \cong R^n$ by $P \oplus (Q \oplus R^m) \cong R^{n+m}$, then $GL_n(R) \to GL(R)$ factors through $GL_{n+m}(R)$. Finally, suppose given a second isomorphism $P \oplus Q' \cong R^m$. Since $Q \oplus R^m \cong R^n \oplus Q'$, we may stabilize both Q and Q' to make them isomorphic and invoke the above argument.

But then, we get a map $\mathrm{Aut}(P) \to \mathrm{GL}(R)$, and as its defined up to inner automorphism we have a well-defined map:

$$\mathrm{Aut}(P) \to K_1(R)$$

as we sought to show.

Corollary 7.1.10: Tensor and $K_1(S)$ as $K_0(R)$ module

Let R, S be rings. Then there is a natural tensor product:

$$K_0(R) \otimes K_1(S) \to K_1(R \times S)$$

If R is commutative and S an R-algebra, then we further have a natural map:

$$K_0(R) \otimes K - 1(S) \to K_1(S)$$

Hence, $K_1(S)$ is a module over the ring $K_0(R)$.

***Proof*:**
For each finitely generated projective R-module P and each m, lemma 7.1.9 gives us:

$$\mathrm{Aut}(P \otimes S^m) \to K_1(R \otimes S)$$

Next, for each $\beta \in GL_m(S)$, let $[P] \cdot \beta$ denote the image of the automorphism $1_P \otimes \beta$ of $P \otimes S^m$ under this map. Fixing β and m, the isomorphism

$$(P \oplus P') \otimes S^m \cong (P \otimes S^m) \oplus (P' \otimes S^m)$$

yields the identity

$$[P \oplus P'] \cdot \beta = [P] \cdot \beta + [P'] \cdot \beta$$

in $K_1(R \otimes S)$. Hence $P \mapsto [P] \cdot \beta$ is an additive function of $P \in \mathbf{P}(R)$, and so it factors through $K_0(R)$.

Now fix P. then the map $GL_m(S) \to K_1(R \otimes S)$ given by $\beta \mapsto [P] \cdot \beta$ is compatible with stabilization in m. Thus it factors through a map $GL(S) \to K_1(R \otimes S)$ and through a map $K_1(S) \to K_1(R \otimes S)$. Hence, we see the product is well-defined and bilinear.

When R is commutative, $K_0(R)$ is a ring (as we saw in chapter 6. If S is an R-algebra, there is a ring map $R \otimes S \to S$. Composing the external product with $K_1(R \otimes S) \to K_1(S)$ yields a natural product operation $K_0(R) \otimes K_1(S) \to K_1(S)$. But then we see that that

$$[P \otimes_R Q] \cdot \beta = [P] \cdot ([Q] \cdot \beta)$$

completing the proof.

Recall from section 6.3 that we connected K_0 with the H_0 homology. We now do the same with K_1. Recall first from [5, chapter 28] that $H_1(G;\mathbb{Z})$ is naturally isomorphic to $G/[G,G]$. This is great for us: if $G = \mathrm{GL}(R)$, we get:

$$K_1(R) = H - 1(\mathrm{GL}(R);\mathbb{Z}) = \lim_{n\to\infty} H - 1(\mathrm{GL}_n(R);\mathbb{Z})$$

By lemma 7.1.9, we get the composition:

$$H_1(\mathrm{Aut}(P);\mathbb{Z}) \to H_1(\mathrm{GL}_n(R);\mathbb{Z}) \to K_1(R)$$

Next, we want to show that $K_1(-)$ is Morita-invariant. To that end, we need to view the homological interpretation in a new light. Let $\mathbf{P}(R)$ be the category of finitely generated projective R-modules and let $t\mathbf{P}$ (the objects are the same, the morphisms between P and P' are the isomorphism classes of Q such that $P \oplus Q = P'$). Then $P \mapsto H - 1(\mathrm{Aut}(P) : \mathbb{Z})$ is a well-defined functor from $t\mathbf{P}$ to **Ab**. Thus, we can take filtered colimits of the functor (recall [5, chapter 26]! But then, by taking advantage of the fact that free modules are cofinal in $t\mathbf{P}$, we get:

Lemma 7.1.11: Morita Buildup (Bass)

$$K_1(R) \cong \lim_{P\in t\mathbf{P}} H_1(\mathrm{Aut}(P);\mathbb{Z})$$

***Proof*:**
outlined above

Now, as $\mathbf{P}(R)$ and $\mathbf{P}(S)$ are Morita equivalent, we apply the above corollary to get:

Proposition 7.1.12: Morita Invariance of K_1

Let R, S be Morita-equivalent rings. Then:

$$K_1(R) \cong K_1(S)$$

***Proof*:**
Looking at the usual case of R and $M_n(R)$ using the corollary 7.1.11, and that we have the Morita-equivalent categories to get:

$$K_1(R) \cong K_1(M_n(R))$$

Next, let us look at the structure given a ring homomorphism $f : R \to S$. We can define the same map as we've done with K_0, however we shall see that there is another arguably more interesting map that has better applications in cohomology. We shall show that whenever SS has a finite R-module resolution by finitely generated projective R-modules, then we can define:

$$f_* : K_1(S) \to K_1(R)$$

which is an interesting difference from the K_0 where we simply took $f \otimes S : \mathbf{P} \to \mathbf{S}$.

Lemma 7.1.13: Simple Transfer Functor

Let $T : \mathbf{S} \to \mathbf{R}$ be an additive functor. Then it induces a homomorphism

$$K_1(T) : K_1(S) \to K_1 9R)$$

And $T_1 \oplus T_2$ induces the sum $K_1(T_1) + K_1(T_2)$.

***Proof*:**
First, the functor T induces an evident functor $t\mathbf{P}(S) \to t\mathbf{P}(R)$. Next, if P is a finitely generated projective S-module, by cohomology theory T also induces a homomorphism $\mathrm{Aut}_S(P) \to \mathrm{Aut}_R(TP)$ and hence $H_1(\mathrm{Aut}_S(P); \mathbb{Z}) \to H_1(\mathrm{Aut}_R(TP); \mathbb{Z})$. As P varies, these assemble to give a natural transformation of functors from the translation category $t\mathbf{P}(S)$ to abelian groups. Since $K_1(S) = \varinjlim_{P \in \mathbf{P}(S)} H_1(\mathrm{Aut}_S(P); \mathbb{Z})$ by corollary 7.1.11 taking the direct limit over $t\mathbf{P}(S)$ yields:

$$K_1(S) \to \varinjlim_{P \in \mathbf{P}(S)} H_1(\mathrm{Aut}_R(P); \mathbb{Z}) \to \varinjlim_{Q \in \mathbf{P}(R)} H_1(\mathrm{Aut}_R(Q); \mathbb{Z}) = K_1(R)$$

as we sought to show.

Corollary 7.1.14: $K_1(-)$ Functor

Suppose that S is finitely generated projective as an R-module. Then the forgetful functor $\mathbf{P}(S) \to \mathbf{P}(R)$ induces a natural transfer homomorphism $f_* : K_1(S) \to K_1(R)$. If R is commutative, the composite

$$K_1(R) \xrightarrow{f^*} K_1(S) \xrightarrow{f_*} K_1(R)$$

is multiplication by $[S] \in K_0(R)$.

***Proof*:**

Let T be the forgetful map. Then the map $K_1(S) \xrightarrow{f_*} K_1(R)$ is the transfer map. Wecompute the composite $f_* \circ f^*$ by computing its effects on $\alpha \in GL_n(R)$. We see that the matrix $f^*(\alpha) = 1_S \otimes_R \alpha$ lies in $GL_n(S)$. To apply f_* we consider $1_S \otimes_R \alpha$ as an element of the group $\mathrm{Aut}_R(S^n) = \mathrm{Aut}_R(S \otimes_R R^n)$, which we then map into $GL(R)$. But this is just the product $[S] \cdot \alpha$, completing the proof.

7.1.1 Relative K-group: Connection to Homology

The next process should be familiar to any reader with some knowledge of relative homology, and will give us our first deep connection between the K_n groups and homology. Consider an ideal $I \subseteq R$. Then given a mp $R \to R/I$, we get a natural map $\mathrm{GL}(R) \to \mathrm{GL}(R/I)$. Let $\mathrm{GL}(I)$ be its kernel. Now, let $E(R, I)$ be the smallest normal subgroup containing the elementary matrices $e_{ij}(x)$ with $x \in I$. Note that $E(R, I) = \bigcup_n E_n(R, I)$. Then we have

Theorem 7.1.15: Whitehead's Lemma for Relative Groups

Let R be a ring and $I \subseteq R$ an ideal. Then $E(R, I) \trianglelefteq \mathrm{GL}(I)$ and contains the commutator subgroup of $\mathrm{GL}(I)$.

***Proof*:**
For any matrix $g = 1 + \alpha \in GL_n(I)$, the identity

$$\begin{pmatrix} g & 0 \\ 0 & g^{-1} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \alpha & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & g^{-1}\alpha \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -g\alpha & 1 \end{pmatrix} \tag{7.2}$$

shows that the matrix $\begin{pmatrix} g & 0 \\ 0 & g^{-1} \end{pmatrix}$ is in $E_{2n}(R, I)$. (The product of the first three matrices is in $E_{2n}(R, I)$.) Hence if $h \in E_n(R, I)$, then the conjugate

$$\begin{pmatrix} ghg^{-1} & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} g & 0 \\ 0 & g^{-1} \end{pmatrix} \begin{pmatrix} h & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} g^{-1} & 0 \\ 0 & g \end{pmatrix} \tag{7.3}$$

is in $E(R, I)$. Finally, if $g, h \in GL_n(I)$, then $[g, h]$ is in $E_{2n}(R, I)$ by equation (7.1).

Thus, the following is well-defined:

Definition 7.1.16: Relative K_1-group

Let R be a ring and $I \subseteq R$ an ideal. Then:

$$K_1(R, I) = \frac{\mathrm{GL}(I)}{E(R, I)}$$

Proposition 7.1.17: Snake Lemma For K-theory

Let R be a ring and $I \subseteq R$ an ideal. Then:

$$K_1(R, I) \hookrightarrow K_1(R) \to K_1(R/I) \xrightarrow{\partial} K_0(I) \to K_0(R) \to K_0(R/I)$$

***Proof*:**
First, we shall prove a simpler exact sequence:

$$1 \to GL(I) \to GL(R) \to GL(R/I) \xrightarrow{\partial} K_0(I) \to K_0(R) \to K_0(R/I). \tag{7.4}$$

First, consider the short exact sequence:

$$0 \to I \to R \to R/I \to 0 \tag{7.5}$$

where I is an ideal in R. This induces a sequence of group homomorphisms

$$1 \to GL(I) \to GL(R) \to GL(R/I) \tag{7.6}$$

where $GL(I)$ consists of matrices of the form $1+\alpha$ with α having entries in I. This sequence is exact since any matrix that maps to the identity in $GL(R/I)$ must have the form $1+\alpha$ with α having entries in I.

To extend this to the K-theory groups, we define the boundary map $\partial : GL(R/I) \to K_0(I)$ as follows. For any $\bar{g} \in GL(R/I)$, we can lift it to a matrix $g \in M_n(R)$ (not necessarily invertible). The matrix g induces an R-module homomorphism $R^n \to R^n$. Let C be the cokernel of this homomorphism. Since $\bar{g}$ is invertible, the reduction of C modulo I is zero, which means C is an I-module. We define $\partial(\bar{g}) = [C] \in K_0(I)$. Verifying exactness is now a standard procedure.

Thus, we have the complete exact sequence

$$1 \to GL(I) \to GL(R) \to GL(R/I) \xrightarrow{\partial} K_0(I) \to K_0(R) \to K_0(R/I). \tag{7.7}$$

Since the K_1 groups are quotients of the GL groups and $E(R)$ maps onto $E(R/I)$, this gives exactness except at $K_1(R)$. Suppose $g \in GL(R)$ maps to zero under $GL(R) \to K_1(R) \to K_1(R/I)$. Then the reduction $\bar{g}$ of g mod I is in $E(R/I)$. Since $E(R)$ maps onto $E(R/I)$, there is a matrix e in $E(R)$ mapping to $\bar{g}$, i.e., ge^{-1} is in the kernel $GL(I)$ of $GL(R) \to GL(R/I)$. Hence the class of ge^{-1} in $K_1(R, I)$ is defined and maps to the class of g in $K_1(R)$, giving exactness as we sought to show.

7.2 A word on K2 Groups

The K_0 and K_1 group we have covered so far had natural definitions motivated outside of homology. However, they both naturally where related to homology, namely they were tied to H_0 and H_1 of the group homology for matrix rings (where R can be seen as a 1-dimensional matrix ring). We may ask if we can further extend this and define higher K-groups via this method. We shall explore this in more detail in the next chapter, but in this chapter we shall cover more "naturally" how we may discover finding K_2 without simply saying $K_2(R) = H_2(E(R); \mathbb{Z})$. Given the group $E(R)$, a natural question to ask is what extensions we may have. Given we want to quotient by the centralizer, we are particularity interested in *central extensions*. Even better if we can find a *universal central extension*. Topologically, this would be associated to the *universal cover* of $BE(R)$ (similarly as to what we saw in topological K-theory). The group associated to this is called the *Steinberg Group*, and is often denoted $\mathrm{st}(R)$. To stay concrete, we shall define everything "low-brow" directly with elements instead of as the consequence of universal properties. The advantage of this approach is to make it easier to "play" with the objects and get a feeling to the information they store:

Definition 7.2.1: Steinberg Group

Let R be a ring. Then for $n \geq 3$ the *Steinberg group* $\mathrm{st}_n(R) \leqslant M_n(R)$ is given by the generators $x_{ir}(r)$ respecting the stein berg relation:

1. $$x_{ij}(r)x_{ij}(s) = x_{ij}(r+s)$$

2. $$[x_{ij}(r), x_{k\ell}(s)] = \begin{cases} 1 & j \neq k,\ i \neq \ell \\ x_{i\ell}(rs) & j = k,\ i \neq \ell \\ x_{kj}(rs) & j \neq k,\ i = \ell \end{cases}$$

Recall from lemma 7.1.3 that the elementary matrices of $E(R)$ satisfy the same relationships, hence there is a canonical surjection

$$\mathrm{st}_n(R) \to E_n(R) \qquad x_{ij}(r) \mapsto e_{ij}(r)$$

The Group $\mathrm{st}_n(R)$ naturally embeds in $\mathrm{st}_{n+1}(R)$ in the usual way, hence we may define $\mathrm{st}(R)$. By looking at the kernel of this map, we get the following:

Definition 7.2.2: K2 Group

Let R be a ring. Then the $K_2(R)$ is defined as the kernel of the map $\mathrm{st}(R) \to E(R)$.

Thus, it fits in the exact sequence:

$$1 \to K_2(R) \to \mathrm{st}(R) \to \mathrm{GL}(R) \to K_1(R) \to 1$$

We mentioned that $K_2(R)$ must come from the center of a group. We start by showing exaclty this

Theorem 7.2.3: K2 and Center of Steinberg

Let R be a ring. Then $K_2(R)$ is an abelian group and is the center of $\mathrm{st}(R)$.

***Proof*:**
If $x \in \mathrm{st}(R)$ commutes with every element of $\mathrm{st}(R)$, then $\varphi(x)$ must commute with all of $E(R)$. But the center of $E(R)$ is trivial (as it is a perfect group) so $\varphi(x) = 1$, that is $x \in K_2(R)$. Thus the center of $\mathrm{st}(R)$ is contained in $K_2(R)$.

Conversely, suppose that $y \in \mathrm{st}(R)$ satisfies $\varphi(y) = 1$. Then in $E(R)$ we have

$$\varphi([y,p]) = \varphi(y)\varphi(p)\varphi(y)^{-1}\varphi(p)^{-1} = \varphi(p)\varphi(p)^{-1} = 1 \tag{7.8}$$

for every $p \in \mathrm{st}(R)$. Choosing an integer n large enough so that y can be expressed as a word in the symbols $x_{ij}(r)$ with $i, j < n$, for each element $p = x_{kn}(s)$ with $k < n$ and $s \in R$, the Steinberg relations implies that the commutator $[y,p]$ is an element of the subgroup P_n of $\mathrm{st}(R)$ generated by the symbols $x_{in}(r)$ with $i < n$. On the other hand, we can see that that φ maps P_n injectively into $E(R)$. Since $\varphi([y,p]) = 1$, this implies that $[y,p] = 1$. Hence y commutes with every generator $x_{kn}(s)$ with $k < n$.

By symmetry, we get that y also commutes with every generator $x_{nk}(s)$ with $k < n$. Hence y commutes with all of $St_n(R)$, since it commutes with every $x_{kl}(s) = [x_{kn}(s), x_{nl}(1)]$ with $k, l < n$. Since n can be arbitrarily large, this proves that y is in the center of $\mathrm{st}(R)$ showing it is indeed the center, completing the proof.

Example 7.2.4: K_2 groups
The computation of K_2 group becomes a little non-trivial, however here are some interesting results:

1. Let $R = \mathbb{Q}$. Then it is somewhat remarkable that:

$$K_2(\mathbb{Q}) = (\mathbb{Z}/4\mathbb{Z})^* \times \prod_{p \text{ odd prime}} (\mathbb{Z}/p\mathbb{Z})^*$$

where this was proven following a proof similar to Gauss's proof of the Quadratic Reciprocity Law!

2. It was proven by R.K. Dennis and J. Sylvester that $K_2(\mathbb{F}[t]) = K_2(\mathbb{F})$, showing that $K_2(-)$ is invariant under extension by an algebraically independent element, and hence we can think of it as extracting "relational" or "number-theoretic" information.

3. More generally, Matsumoto's theorem (see [10]) states that:

$$K_2(k) \cong k^\times \otimes_\mathbb{Z} k^\times / \langle a \otimes (1-a) \big| a \in k^\times \setminus \{1\} \rangle$$

The proof is complicated,but due to our concrete definition given above we can see the outline of the proof. We shall draw on some results from section 7.2.1 to show that this group shall satisfy the same universal property of central extensions. For $a \in k^\times$, define the element $w_{ij}(a) \in \mathrm{st}(k)$ by:

$$w_{ij}(a) = x_{ij}(a)x_{ji}(-a^{-1})x_{ij}(a)$$

For $a, b \in k^\times$, define the Steinberg symbol $\{a, b\} \in K_2(k)$ as:

$$\{a, b\} = h_{ij}(a)h_{ij}(b)h_{ij}(ab)^{-1}$$

where $h_{ij}(a) = w_{ij}(a)w_{ij}(1)^{-1}$. Then by the linearity properties given in definition 7.2.1, $\{a, b\}$ satisfy the following bilinearity relations:

$$\{a_1a_2, b\} = \{a_1, b\}\{a_2, b\}$$
$$\{a, b_1b_2\} = \{a, b_1\}\{a, b_2\}$$

The crucial step is proving that: $\{a, 1-a\} = 1$ for all $a \in k^\times \setminus \{1\}$. To establish this, we work with explicit matrices in $\mathrm{SL}_3(k)$ and their lifts to the Steinberg group. Consider the elements:

$$u(a) = x_{12}(a)x_{23}(1)x_{12}(-a)x_{23}(-1)$$
$$v(a) = x_{13}(a(1-a))x_{23}(-a)x_{12}(-(1-a))$$

Through direct calculation in $\mathrm{st}(k)$, one can verify that $u(a)v(a)^{-1} \in K_2(k)$ and equals $\{a, 1-a\}^{-1}$. Furthermore, one can show that $u(a)v(a)^{-1} = 1$, establishing the Steinberg relation. Next, let $\mathcal{M}(k)$ denote the abstract group:

$$\mathcal{M}(k) = k^\times \otimes_{\mathbb{Z}} k^\times / \langle a \otimes (1-a) \big| a \in k^\times \setminus \{1\} \rangle$$

Define a homomorphism $\varphi : \mathcal{M}(k) \to K_2(k)$ by mapping $a \otimes b \mapsto \{a, b\}$. The bilinearity relations and the Stein berg relation ensure that φ is well-defined. If φ is an isomorphism, we're done. To that end, we construct a homomorphism $\psi : K_2(k) \to \mathcal{M}(k)$ and show that $\psi \circ \varphi = \mathrm{id}_{\mathcal{M}(k)}$ and $\varphi \circ \psi = \mathrm{id}_{K_2(k)}$.

The construction of ψ relies on the universal property of the Steinberg group as the universal central extension of $\mathrm{SL}(k)$. Specifically, we use the fact that any central extension of $\mathrm{SL}(k)$ factors uniquely through $\mathrm{St}(k)$ (see section 7.2.1). The proof would then be done by showing that $\mathcal{M}(k)$ satisfies the same universal property as $K_2(k)$ with respect to central extensions of $\mathrm{SL}(k)$. This involves constructing a central extension:

$$1 \to \mathcal{M}(k) \to E \to \mathrm{SL}(k) \to 1$$

and proving that any central extension of $\mathrm{SL}(k)$ factors uniquely through E (i.e. that it satisfies the universal property). By the uniqueness of universal central extensions, we conclude that

$$\mathcal{M}(k) \cong K_2(k)$$

and hence $K_2(k) \cong k^\times \otimes_{\mathbb{Z}} k^\times / \langle a \otimes (1-a) \big| a \in k^\times \setminus \{1\} \rangle$. We can verify this works by seeing that this matches our earlier claim of $K_2(\mathbb{Q})$!

4. Let $R = \mathbb{Z}$. Then $K_2(R) \cong \mathbb{Z}/2\mathbb{Z}$. Indeed, by Matsumoto's theorem, we have:

$$K_2(\mathbb{Q}) \cong \mathbb{Q}^\times \otimes_{\mathbb{Z}} \mathbb{Q}^\times / \langle a \otimes (1-a) \big| a \in \mathbb{Q}^\times \setminus \{1\} \rangle$$

Every element in $\mathbb{Q}^\times$ can be written as $\pm p_1^{n_1} \cdots p_k^{n_k}$ where p_i are distinct primes and $n_i \in \mathbb{Z}$. Thus, $K_2(\mathbb{Q})$ is generated by symbols of the form:

$$\{-1, -1\}, \{-1, p\}, \{p, q\}$$

where p, q are primes. For the next step, we shall combine theorem 7.2.10 which is a generalization of the snake lemma, and [11, chapter 5] from class field theory to get:

$$K_2(\mathcal{O}_F) \to K_2(F) \xrightarrow{\partial} \bigoplus_{\mathfrak{p}} \kappa(\mathfrak{p})^\times \to K_1(\mathcal{O}_F) \to K_1(F)$$

For $\mathbb{Z} \subset \mathbb{Q}$, this becomes:

$$K_2(\mathbb{Z}) \to K_2(\mathbb{Q}) \xrightarrow{\partial} \bigoplus_{p \text{ prime}} \mathbb{F}_p^\times \to K_1(\mathbb{Z}) \to K_1(\mathbb{Q})$$

where ∂ is the "generalized quadratic reciprocity":

$$\partial_p(\{a, b\}) = (-1)^{v_p(a)v_p(b)} \cdot \left(\frac{a^{v_p(b)}}{b^{v_p(a)}} \right) \bmod p$$

where v_p is the p-adic valuation. For the generators of $K_2(\mathbb{Q})$, we compute:

$$\partial_p(\{-1,-1\}) = 1 \text{ for all primes } p$$
$$\partial_p(\{-1,q\}) = \begin{cases} -1 \bmod p & \text{if } p = q \\ 1 \bmod p & \text{if } p \neq q \end{cases}$$
$$\partial_p(\{p,q\}) = \begin{cases} q \bmod p & \text{if } p = p \\ p^{-1} \bmod q & \text{if } p = q \\ 1 \bmod r & \text{if } p \neq p, q \end{cases}$$

By definition, $K_2(\mathbb{Z})$ is the kernel of ∂. An element $\{a,b\} \in K_2(\mathbb{Q})$ is in $K_2(\mathbb{Z})$ if and only if $\partial_p(\{a,b\}) = 1$ for all primes p.

Using the quadratic reciprocity law and properties, we see that the only non-trivial element in $K_2(\mathbb{Q})$ that maps to 1 under is $\{-1,-1\}$. To show its in $K_2(\mathbb{Z})$, we need to determine the order of $\{-1,-1\}$ in $K_2(\mathbb{Q})$. Using the properties of Steinberg symbols:

$$\begin{aligned} \{-1,-1\}^2 &= \{(-1)^2, -1\} \\ &= \{1,-1\} \\ &= 1 \end{aligned}$$

Thus, $\{-1,-1\}$ has order 2 in $K_2(\mathbb{Q})$ and hence in $K_2(\mathbb{Z})$.

From these examples, we also see how $K_2(-)$ is directly tied to number theory!

7.2.1 Universal Central Extensions

As described earlier, $\mathrm{st}(R)$ can naturally be seen as being the *universal central extension* of $E(R)$. Let us dive into what this means.

Let G be a group and A an abelian group. Then a *central extension* of G by A[1] if there exists a short exact sequence

$$1 \to A \to X \to G \to 1$$

such that A maps into the center of X. The extension splits if there is a section map $G \to X$, or equivalently that it is isomorphic to the extension:

$$1 \to A \to A \times G \xrightarrow{\pi_2} G \to 1$$

By theorem 7.2.3, we have a central extension:

$$1 \to K_2(R) \to \mathrm{st}(R) \to E(R) \to 1$$

If X, Y are extensions of G, they are said to be equivalent if there is an isomorphism $f : X \to Y$ which is the identity on A and induces the identity map on G. Then it is a known result that $H^2(G; A)$ classifies all the isomorphism classes (see [2] or [7]). A *universal central extension* of G would then be a central extension $X \to G$ such that for any other central extension $Y \to G$ it must

[1] remember that in group theory we usually reverse the order when we say "X" is extending "Y", hence why we say G is extended by A

factor through $X \to G$ uniquely. Of course, we have yet to show that a universal central extension always exists. We build-up to this result now:

Lemma 7.2.5: Universal Central Extension and Perfect Groups I

Let G have a universal central extension $X \to G$. Then G, X must be perfect groups (have no center).

***Proof*:**
Assume $B = \frac{X}{[X,X]}$ is nontrivial. Then there would be two at least homomorphisms over G from X to the central extension:

$$1 \to B \to B \times G \to G \to 1$$

namely $(0, \pi)$ and (pr, π) where $pr : X \to B$ is the natural projection, contradicting the uniqueness of universality.

Lemma 7.2.6: Universal Central Extension and Perfect Groups II

Let G have two central extension X, Y, and assume X is a perfect group. Then there is at most one homomorphism over G from X to Y.

***Proof*:**
If f and f' are two such homomorphisms, then for any x and x' in X we can write $f'(x) = f(x)c$, $f'(x') = f(x')c'$ for elements c and c' in the center of Y. Therefore $f'(xx'x^{-1}(x')^{-1}) = f(xx'x^{-1}(x')^{-1})$. Since the commutators $[x, x'] = xx'x^{-1}(x')^{-1}$ generate X, we must have $f' = f$, completing the proof.

Theorem 7.2.7: Universal Central Extensions of Perfect Groups

Let G be a perfect group. Then G has a universal central extension:

$$1 \to H_2(G; \mathbb{Z}) \to \frac{[F, F]}{[R, F]} \to G \to 1$$

where F is a free group mapping onto G and $R \subseteq F$ is the kernel of this map (i.e. the relations).

If X is any central extension of G, then the following are equivalent:

1. X is a universal central extension
2. X is perfect, and every central extension of X splits
3. $H_1(X; \mathbb{Z}) = H_2(X; \mathbb{Z}) = 0$

***Proof*:**
First, let $F \twoheadrightarrow G$ be a presentation of G with kernel R. Then $[R,F] \trianglelefteq F$, and so we have a short exact sequence:

$$1 \to \frac{R}{[R,F]} \to \frac{F}{[R,F]} \to G \to 1$$

By the universal properties of commutators with some diagram chasing, we obtain:

$$1 \to \frac{R \cap [F,F]}{[R,F]} \to \frac{[F,F]}{[R,F]} \to [G,G] \to 1$$

If G is perfect, $G = [G,G]$, and by homology theory(see [2] or [7]) we have $\frac{R\cap[F,F]}{[R,F]} \cong H_2(G;\mathbb{Z})$, giving us the exact sequence.

For the equivalences, (2) $\Rightarrow$ (1) is immediate as trivial 1st/2nd homology groups implies there are only split exact sequences and lemma 7.2.6 give us the perfect condition, Lemma 7.2.5 gives (1) $\Rightarrow$ (2). Hence we'll show (1) $\Leftrightarrow$ (3). It suffices to show that the above exact sequence is a universal central extension. Given any central extension X of G, the map $\varphi : F \to G$ lifts to a map $\Phi : F \to X$ (as F is projective, as its free). Then as $\Phi(R)$ is the center of X, $\Phi([R,F]) = 1$, and so by the universal property of commutators Φ induces a map $\frac{[F,F]}{[R,F]} \to X$ over G. This map is unique by lemma 7.2.6, and hence the above sequence is indeed universal giving the desired result.

Corollary 7.2.8: K2 Group and Homology

The Steinberg group $\mathrm{st}(R)$ is the universal central extension of $E(R)$, and hecne:

$$K_2(R) \cong H_2(E(R);\mathbb{Z})$$

***Proof*:**
We shall show that for $n \geq 5$, every central extension $Y \xrightarrow{\varphi} \mathrm{st}_n(R)$ splits, which by theorem 7.2.7 gives the desired result.

First, let's show that if $j \neq k$ and $l \neq i$, then every two elements $y, z \in Y$ with $\varphi(y) = x_{ij}(r)$ and $\varphi(z) = x_{kl}(s)$ must commute in Y. Pick t distinct from i, j, k, l and choose $y', y'' \in Y$ with $\varphi(y') = x_{it}(1)$ and $\varphi(y'') = x_{tj}(r)$. Then the Steinberg relations imply that both $[y', z]$ and $[y'', z]$ are in the center of Y, and since $\varphi(y) = \varphi([y', y''])$, this implies that z commutes with $[y', y'']$ and y.

Next, choose distinct indices i, j, k, l and elements $u, v, w \in Y$ with

$$\varphi(u) = x_{ij}(1), \quad \varphi(v) = x_{jk}(s), \quad \text{and} \quad \varphi(w) = x_{kl}(r). \tag{7.9}$$

If $G \leqslant Y$ is generated by u, v, w, then its commutator subgroup $[G,G]$ is generated by elements mapping under φ to $x_{ik}(s)$, $x_{jl}(sr)$, or $x_{il}(sr)$. From the first paragraph of this proof it follows that $[u, w] = 1$ and that $[G,G]$ is abelian. By group theory, we have $[[u,v],w] = [u,[v,w]]$, and so if $\varphi(y) = x_{ik}(s)$ and $\varphi(z) = x_{jl}(sr)$, we have $[y,w] = [u,z]$. Taking $s = 1$, this identity proves that the element

$$y_{il}(r) = [u,z], \quad \text{where } \varphi(u) = x_{ij}(1), \varphi(z) = x_{jl}(r), \tag{7.10}$$

does not depend upon the choice of j, nor upon the lifts u and z of $x_{ij}(1)$ and $x_{jl}(r)$.

Finally, I claim that the elements $y_{ij}(r)$ satisfy the Steinberg relations, so that there is a group homomorphism $St_n(R) \to Y$ sending $x_{ij}(r)$ to $y_{ij}(r)$. Such a homomorphism will provide the desired splitting of the extension φ. By the first paragraph of this proof, if $j \neq k$ and $l \neq i$, then $y_{ij}(r)$ and $y_{kl}(s)$ commute. The identity $[y, w] = [u, z]$ above may be rewritten as

$$[y_{ik}(r), y_{kl}(s)] = y_{il}(rs) \quad \text{for } i, k, l \text{ distinct.} \tag{7.11}$$

The final relation $y_{ij}(r)y_{ij}(s) = y_{ij}(r + s)$ follow similarly, giving us the desired result.

We shall not prove it here, but another important result is that $K_2(-)$ is stable under morita-equivalence (see [6] for the details). Instead, we shall finish off this section by extending the snake lemma for its use in example 7.2.4. First, for any ideal $I \subseteq R$, we may defined the augmented ring $R \oplus I$ via:

$$(r, x)(s, y) = (rs, ry + xs + xy)$$

There are two natural maps $\pi, + : R \oplus I \to R$ where $\pi(r, x) = r$ and $+(r, x) + r + x$. Let $\mathrm{st}'(R, I)$ be the normal subgroup of $\mathrm{st}(R \oplus I)$ generated by all $x_{ij}(0, v)$ where $v \in I$. Then there is certainly a map

$$\mathrm{st}'(R, I) \to E(R \oplus I, 0 \oplus I)$$

where $E(R \oplus I, 0 \oplus I) \leqslant \mathrm{GL}(R \oplus I)$. From this define:

Definition 7.2.9: Relative Steinberg Grouhatrp

Let R be a ring and $I \subseteq R$ an ideal. Then the *relative Steinberg group* $\mathrm{st}(R, I)$ is the quotinet of $\mathrm{st}'(R, I)$ by the normal subgroup generate by all cross-commutators:

$$[x_{ij}(0, u), x_{kl}(v, -v)] \qquad u, v \in I$$

We thus get:

Theorem 7.2.10: Snake Lemma For K-theory Extended

Let R be a ring and $I \subseteq R$ an ideal. Then:

$$K_2(R, I) \to K_2(R) \to K_2(R/I) \to K_1(R, I) \to K_1(R) \to \cdots \to K_0(R) \to K_0(R/I) \to K_0(I)$$

where there is no $\to 0$ at the end due to possible negative K-groups[a].

[a] We have not covered negative K-theory, but you can checkout [6] for more details

***Proof*:**
This follows by doing some diagram chasing on:

$$\begin{tikzcd}
K_2(R,I) \arrow[r] \arrow[d] & St(R,I) \arrow[r] \arrow[d, "+"] & GL(I) \arrow[r] \arrow[d, "\text{inj.}"] & K_1(R,I) \arrow[d] \\
K_2(R) \arrow[r] \arrow[d] & St(R) \arrow[r] \arrow[d, "\text{surj.}"] & GL(R) \arrow[r] \arrow[d] & K_1(R) \arrow[d] \\
K_2(R/I) \arrow[r] & St(R/I) \arrow[r] & GL(R/I) \arrow[r] & K_1(R/I).
\end{tikzcd}$$

and using snake's lemma (with proposition 7.1.17).

8

Higher K-Theory

As we have been discovering in the last 2 chapters, K-groups seem to be representable as the homology of certain groups. In the 1960s and 70s, a large amount of effort has been put in trying to find the right way to unify these results and give a single cohomology theory to unify these groups. As it turns out, the "right" approach was in fact to see each $K_n(R)$ as $\pi_n K(R)$ for a topological space $K(R)$, that is the *homotopy* groups rater than the homology groups. It turns out that in low dimensions, the homology and homotopy groups shall coincide, and in high dimensions the homotopy groups shall embed in the homology groups. The connection is less surprising when we see that the universal spaces given in section 2.1 were the homotopy groups of some universal spaces. In particular, there has been a push to represent the information many algebraic objects as "spaces" (in many ways, the current abstract definition of topology was made to accommodate the more odd geometric behaviour these "classifying" spaces had) and we extracted information from these spaces using algebraic topology. In this chapter, we shall go over the basics of how we may generalize the K-theory we have been developing into higher order K-theory, and give the beginning of how these results link to the K-groups we have studied earlier.

8.1 Defining the Space: BGL+

Let G be any group. In [9, section 2.2], we saw we can construct a connected topological space BT whose fundamental group $\pi_1(BG) \cong G$ and $\pi_n(G) = 0$ for $n > 1$. The homology of BG (with coefficients in a G-module M) coincides with the algebraic homology of G (with coefficients in M). For $G = \mathrm{GL}(R)$, this gives $\mathrm{BGL}(R)$.

Definition 8.1.1: $K_n(R)$

Let $\mathrm{BGL}(R)$ be the covering groups for $\mathrm{GL}(R)$, and let $\mathrm{BGL}(R)^+$ denote any CW complex X which has a distinguished map $\mathrm{BGL}(R) \to \mathrm{BGL}(R)^+$ such that the following holds:

1. $\pi_1(\mathrm{BGL}(R)^+) \cong K_1(R)$ and the natural map from $\mathrm{GL}(R) = \pi_1(\mathrm{BGL}(R))$ to $\pi_1(\mathrm{BGL}(R)^+)$ is surjective with kernel $E(R)$

2. $H_*(\mathrm{BGL}(R); M) \cong H_*(\mathrm{BGL}(R)^+; M)$ for every $K_1(R)$-module M

The space X is sometimes called the *model* for $\mathrm{BGL}(R)^+$.

For $n \geq 1$,

$$K_n(R) = \pi_n(\mathrm{BGL}(R)^+)$$

We shall show that any two models are homotopy equivalent, and hence the space $\mathrm{BGL}(R)^+$ is uniquely defined up to homotopy, and so the homotopy group $K_n(R)$ are well-defined up to a canonical isomorphism. We have not actually given a model X; there are in fact many such models, and we shall construct one of them soon.

Notice we have not included $K_0(R)$ in the definition. To amend this, we define the following:

Definition 8.1.2: $K(R)$

Let R be a ring, $K_0(R)$ the K-group given in definition 6.2.1. Then:

$$K(R) := K_0(R) \times \mathrm{BGL}(R)^+$$

Namely, $K(R)$ is the disjoint union of copies of the connected space $\mathrm{BGL}(R)^+$ with each element of $K_0(R)$. Then by construction:

$$K_0(R) = \pi_0(K(R)) \qquad \pi_n(K(R)) = \pi_n(\mathrm{BGL}(R)^+) = K_n(R),\ n \geq 1$$

recovering all the K-groups.

The maps $K_n(-)$ and $K(-)$ are functorial. The former is a functor from rings to abelian group, while the latter is a functor form rings to homotopy categories of topological spaces (a homotopy category is a category whose objects remain the same and the morphisms are weak homotopy equivalences with diagrams added to induce inverse morphisms; see [5, chapter 27.5]). Note that until an actual model X is concretely defined, we can only prove these results up to homotopy. To see this, let $R \to R'$ be a ring map. Then this induces a natural group map $\mathrm{GL}(R) \to \mathrm{GL}(R')$, which induces a natural map $\mathrm{BGL}(R) \to \mathrm{BGL}(R')$, and by construction we get a map $\mathrm{BGL}(R)^+ \to \mathrm{BGL}(R')^+$ up to homotopy (see theorem 8.1.7). Hence, we get that $K_n(R) \to K_n(R')$ is well-defined up to homotopy. A similar argument follows for composition, giving us that it is indeed a functor.

Let us now work towards a concrete model X known as *Volodin spaces*. This space shall be a special acyclic subspace[1] of $BGL(R)$, and so let us get a little more familiar with how these spaces relate to perfect groups

[1] Recal a pace is acyclic if it has trivial homology groups, i.e. $\tilde{H}_*(X; \mathbb{Z}) = 0$

Lemma 8.1.3: Acyclic and Perfect Groups

Let E be an acyclic space. Then E is connected, its fundamental group $G = \pi_1(E)$ is a perfect group, and $H_2(G;\mathbb{Z}) = 0$.

This proof shall involve the use of *Serre's Spectral Sequence* which is covered in [2] or [3].

***Proof*:**
First, the acyclic space E must be connected, as $H_0(E) = \mathbb{Z}$. Indeed, $G/[G,G] = H_1(E;\mathbb{Z}) = 0$, so we have $G = [G,G]$, i.e., G is a perfect group. To calculate $H_2(G)$, observe that the universal covering space $\tilde{E}$ has $H_1(\tilde{E};\mathbb{Z}) = 0$. Moreover, the homotopy fiber of the canonical map $E \to BG$ is homotopy equivalent to $\tilde{E}$. To see this, we shall construct a spectral sequence from fibers maps. If $E \to B$ is a continuous map, we can induce the maps $\pi_*E \to \pi_*B$. Then define the *homotopy fiber* $F(f)$ of f with respect to the base point $*_B$ to be the space of pairs (e,γ), $e \in E$ and $\gamma : [0,1] :\to B$ with $\gamma(0) = *_B$ and $\gamma(1) = f(e)$. Then a sequence of based spaces $F \to E \xrightarrow{f} B$ where $F \to B$is constant is called a *homotopy fibration sequence*. Then we get a long exact sequence of homotopy groups:

$$\cdots \to \pi_{n+1}(B) \xrightarrow{\partial} \pi_n(F(f)) \to \pi_n(E) \to \pi_n(B) \xrightarrow{\partial} \pi_{n-1}(F(f)) \to \cdots$$

Then the *Serre spectral sequence* for this homotopy fibration is $E^2_{pq} = H_p(G; H_q(\tilde{E};\mathbb{Z})) \Rightarrow H_{p+q}(E;\mathbb{Z})$ and the conclusion that $H_2(G;\mathbb{Z}) = 0$ follows from the associated exact sequence of low degree terms:

$$H_2(E;\mathbb{Z}) \to H_2(G;\mathbb{Z}) \xrightarrow{d^2} H_1(\tilde{E};\mathbb{Z})^G \to H_1(E;\mathbb{Z}) \to H_1(G;\mathbb{Z})$$

Definition 8.1.4: Volodin Space

Let R be a ring. Then the *Volodin space* $X(R)$ is an acyclic subspace of $\mathrm{BGL}(R)$ constructed as follows:

1. For each n ,let $T_n(R) \leqslant \mathrm{GL}_n(R)$ consist of upper triangular matrices with 1's on the diagonal. Let $T(R) = \varinjlim_{n\to\infty} T_n(R) \leqslant \mathrm{GL}(R)$
2. View the permutation group S_n as the subgroups of $\mathrm{GL}_n(R)$ by their representation as permutation matrices, and their union as S_∞.
3. For each $\sigma \in S_n$, let $T_n^\sigma(R)$ denote the subgroup of $\mathrm{GL}_n(R)$ obtained by conjugating $T_n(R)$ by σ. Then $BT_n(R)$ and $BT_n(R)^\sigma$ are subspaces of $\mathrm{BGL}_n(R)$, and hence of $\mathrm{BGL}(R)$
4. The *Volodin space* is defined to be:

$$X(R) = \bigcup_{n,\sigma} BT_n(R)^\sigma$$

We shall show that $\mathrm{BGL}(R)/X(R)$ is a model for $K(R)$. The whole proof is beyond the scope of this book, but we shall at least show that it matches with K_1 and K_2. First, the image of the map

$\pi_1(X(R)) \to \mathrm{GL}(R)$ is $E(R)$. Indeed, $\pi_1(X)$ is generated by the image of $\pi_1(BT_n(R)^\sigma)$, and the image of the composition

$$\pi_1(BT_n^\sigma(R)) \to \pi_1(X) \to \pi_1(\mathrm{BGL}(R)) = \mathrm{GL}(R)$$

is the subgroup $T_n^\sigma(R) \leqslant E(R)$, and every generator $e_{ij}(r)$ is contained in some $T_n^\sigma(R)$. This gives us that it matches K_1, and also the first condition for $K(R)$. We shall show π_2 of this group is K_2. This requires some more information about acyclic maps:

Definition 8.1.5: Acyclic Maps

Let X, Y be based connected CW complexes. Then a map $f : X \to Y$ is called *acyclic* if the homotopy fiber $F(f)$ of f is acyclic (i.e. Has trivial homology, the homology of a point).

Note that this implies that $F(f)$ is connected and $\pi_1(F(f))$ is a perfect group. Then from the exact sequence $\pi_1(F(f)) \to \pi_1 9X) \to \pi_1(Y) \to \pi_0(F(f))$ of homotopy groups, we get that if $X \to Y$ is acyclic, then $\pi_1(X) \twoheadrightarrow \pi_1(Y)$, and its kernel P is a perfect normal subgroup of $\pi_1(X)$. We shall give a name to such group

Definition 8.1.6: +-Construction

Let P be a perfect normal subgroup of $\pi_1(X)$ where X is a based connected CW complex. Then an acyclic map $f : X \to Y$ is called a *+-construction* on X (relative to P) if P is the kernel of $\pi_1(X) \to \pi_1(Y)$.

For any perfect normal subgroup of the fundamental group, there exists such a group.

Theorem 8.1.7: Existence of +-construction

Let $P \leqslant \pi_1(X)$ be a perfect normal subgroup. Then:

1. There exists a +-construction $f : X \to Y$ relative to P
2. Let $f : X \to Y$ be a +-construction relative to P, and let $g : X \to Z$ be a map st P vanishes in $\pi_1(Z)$. Then there is a map $h : Y \to Z$ such that $g = hf$ and h is unique up to homotopy
3. If g is another +-construction relative to P, then the map h in (2) is a homotopy equivalence, namely $h : Y \xrightarrow{\sim} Z$

***Proof*:**
See [4]

Let us now show why these are called acyclic maps:

Lemma 8.1.8: Acyclic Maps and Homology

Let X, Y be connected CW complexes. Then $f : X \to Y$ is acyclic if and only if

$$H_*(X; M) \cong H_*(Y; M)$$

for every $\pi_1(Y)$-module M

***Proof*:**
Suppose first that f is acyclic, with homotopy fiber $F(f)$. Then $\pi_1 F(f) \to \pi_1(Y)$ is trivial, and $\pi_1(F(f))$ acts trivially upon M. By the Universal Coefficient Theorem (see [9, chapter 4]), $H_q(F(f); M) = 0$ for $q \neq 0$ and $H_0(F(f); M) = M$.

Therefore $E^2_{pq} = 0$ for $q \neq 0$ in the Serre spectral sequence for f:

$$E^2_{pq} = H_p(Y; H_q(F(f); M)) \Rightarrow H_{p+q}(X; M)$$

The spectral sequence collapses, yielding $H_p(X; M) \cong H_p(Y; M)$ for all p.

Conversely, we suppose first that $\pi_1(Y) = 0$ and

$$H_*(X; \mathbb{Z}) \cong H_*(Y; \mathbb{Z})$$

By the Comparison Theorem for the Serre spectral sequences (see [3]) for $F(f) \to X \xrightarrow{f} Y$ and $* \to Y \xrightarrow{=} Y$, we have $\tilde{H}_*(F(f); \mathbb{Z}) = 0$. Hence $F(f)$ and f are acyclic.

The general case reduces to this by the following trick: Let $\tilde{Y}$ denote the universal covering space of Y, and let $\tilde{X} = X \times_Y \tilde{Y}$ denote the corresponding covering space of X. Then there are natural isomorphisms

$$H_*(\tilde{Y}; \mathbb{Z}) \cong H_*(Y; M) \qquad \text{and} \qquad H_*(\tilde{X}; \mathbb{Z}) \cong H_*(X; M)$$

where $M = \mathbb{Z}[\pi_1(Y)]$. The assumption that $H_*(X; M) \cong H_*(Y; M)$ implies that the map $\tilde{f} : \tilde{X} \to \tilde{Y}$ induces isomorphisms on integral homology. But $\pi_1(\tilde{Y}) = 0$, so by the special case above, the homotopy fiber $F(\tilde{f})$ of $\tilde{f}$ is an acyclic space. But by path lifting we have $F(\tilde{f}) \cong F(f)$, so $F(f)$ is acyclic. Thus f is an acyclic map.

Now, recall from section 7.2.1 that every perfect group P has a universal central extension $E \to P$ whose kernel is the abelian group $H_2(P; \mathbb{Z})$, namely theorem 7.2.7. From the above we now get:

Theorem 8.1.9: 2-homotopy group and K_2

let P be a perfect normal subgroup of G with corresponding $+$-construction $f : BG \to BG^+$. Then if $F(f)$ is the homotopy fiber of f, then $\pi_1(F(f))$ is the universal central extension of P, and

$$\pi_2(BG^+) \cong H_2(P; \mathbb{Z})$$

IF $G = \mathrm{GL}(R)$, we get:

$$K_2(R) \cong H_2(E(R); \mathbb{Z})$$

***Proof*:**
Consider the exact sequence from homotopy theory:

$$\pi_2(BG) \to \pi_2(BG^+) \to \pi_1(F(f)) \to G \to G/P \to 1$$

Then by algebraic topology we know we have $\pi_2(BG) = 0$ and $\pi_2(BG^+)$ is in the center of $\pi_1(F(f))$. Thus, $\pi_1(F(f))$ is a central extension of P with kernel $\pi_2(BG^+)$. As $F(f)$ is acyclic, $\pi_1(F(f))$ is prefect, so $H_2(F;\mathbb{Z}) = 0$, and so by theorem 7.2.7, we have that $\pi_1(F(f))$ is the universal central extension of P.

The last result comes from applying this to $G = \mathrm{GL}(R)$ so that $P = E(R)$, completing the proof.

Corollary 8.1.10: Fundamental Group of Volodin Space

The fundamental group of $\pi_1(X(R))$ is the Steinberg group $\mathrm{st}(R)$

Index

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